Finite energy foliations from broken books decompositions

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April, 2025

1 Local models and normal forms

In this section, we prove the existence of normal forms for the Reeb flow around binding orbits, which will be useful later on in order to construct the finite energy foliation, and to have a convenient asymptotic model for obstruction bundle gluing methods. These normal forms are designed to be compatible, in some sense, with adapted broken book decomposition.

Let α be a non-degenerate contact form, supported by a broken book decomposition $(\mathcal{K}, B)$. Throughout this whole section, we will be using tubular neighbourhood of Reeb orbits for α, which we will denote with coordinates (ρ, ϕ, θ), where ρ is the radial coordinate, and ∂_θ is parallel to R_α at $\rho = 0$. We will assume the binding orbit is non-degenerate, and will establish local models for both elliptic and hyperbolic orbits separately.

We will also use "partial blow-up" in order to construct our systems of coordinate. The idea is to replace the binding orbit by a 2-torus with coordinates (ϕ, θ) where ϕ and θ are both uniquely determined up to isotopy. Indeed, ϕ corresponds to the "limit" of the directions of the meridians of the tori $\{\rho = Cst\}$ for any system of coordinates, which are unique because they are the only isotopy classes of curves bounding a disc inside their respective torus. The θ-coordinate will also be well-defined as the direction given by the boundary of some radial pages. In this setting, the tubular neighborhood of the Reeb orbit becomes $\mathbb{T}^2 \times I$, with $I = [0, 1]$. By analogy with the usual blow-up, the tori $\mathbb{T}^2 \times \{0\}$ will be called the exceptional divisor.

We give the following result about this construction, which will simplify slightly the following proofs:

Lemma 1.1. If γ is a radial binding component, then it is possible to choose coordinates (ϕ, θ) on the exceptional divisor such that ∂_ϕ is in ξ and ∂_θ is tangent to the pages.

Proof. Choose a metric g on M, and for x in the image of γ, consider $D_x = \exp_x(t\mathbb{D}_x)$ where $\mathbb{D}_x$ is the unit disc in ξ_x and $t \in \mathbb{R}$. Up to choosing t small enough, this gives a 1-parameter family of disjoint discs, intersecting the binding once, and tangent to ξ at that intersection point. This gives a first parametrization $(\rho_0, \phi_0, \theta_0)$ of a tubular neighbourhood of γ. Consider the partial blow-up with respect to this parametrization. On the exceptional divisor, $\partial_\phi \in \xi$ by construction, while the intersection with the pages are loops intersecting each meridian curves once (by property of the pages in the radial model). Hence they are tangent to $\partial_\theta + n\partial_\phi$ for some n. Now a linear change of coordinates on the 2-torus gives the result. $\square$

We also give some constraints on local models for contact forms, as an auxiliary lemma which we will use later.

Lemma 1.2. Suppose γ is a Reeb orbit for α around which there are coordinates (ρ, ϕ, θ), such that the Reeb flow is always positively transverse to the surfaces $\{\phi = Cst\}$ (equivalently, $d\phi(R_\alpha) > 0$ for $\rho > 0$), and such that the "pages" $\{\phi = Cst\}$ have a positive end at γ. Suppose also that $\alpha = f(\rho, \phi, \theta)d\theta + g(\rho, \phi, \theta)d\phi$. Then, for ρ small enough :

1. $f > 0, g \geq 0$ pour $\rho \geq 0$.
2. $\partial_\rho f < 0$, $\partial_\rho g > 0$ for $\rho > 0$.
3. $g \sim_{\rho \to 0} C\rho^2$, and $\partial_\rho g \sim 2C\rho$
4. If $D := f\partial_\rho g - g\partial_\rho f$, then $D \sim C'\rho$

where the C, C' are functions of ϕ and θ independent of ρ.

Proof. Let us first prove the first two points.
One has :

$$d\alpha = \partial_\rho f(d\rho \wedge d\theta) + \partial_\rho g(d\rho \wedge d\phi) + (\partial_\phi f - \partial_\theta g)(d\phi \wedge d\theta)$$

and :

$$\alpha \wedge d\alpha = (f\partial_\rho g - g\partial_\rho f)(d\rho \wedge d\phi \wedge d\theta)$$

We know that $R_\alpha = \partial_\theta$ at $\rho = 0$, hence $f > 0$ for ρ small. Moreover, in order for α to be well-defined, g has to vanish at $\rho = 0$.

We know that γ is a positive end for the pages, hence by definition $d\alpha(\partial_\rho, \partial_\theta) < 0$ for $\rho > 0$. This implies $\partial_\rho f < 0$.
We also have : $d\alpha(\partial_\rho, R_\alpha) = \partial_\rho f d\theta(R_\alpha) + \partial_\rho g d\phi(R_\alpha) = 0$, and because $d\phi(R_\alpha) > 0$ and $d\theta(R_\alpha) > 0$, we get that $\partial_\rho f$ and $\partial_\rho g$ are of opposite signs and never vanishing for $\rho > 0$.

We now prove the last two items. Define $D := f\partial_\rho g - g\partial_\rho f$. It is positive by the contact condition for $\rho > 0$. We know by the construction of lemma **??** that $\alpha = fd\theta$ at $\rho = 0$, hence $g = o(\rho)$ and f is positive even at $\rho = 0$.

Moreover, because α is $\mathcal{C}^1$ at $\rho = 0$, $d\alpha$ is well defined and non-zero on this domain, which implies that $\partial_\rho g$ is a $O(\rho)$, while $\partial_\rho f$ is a $o(1)$. Moreover, D is a $O(\rho)$ at 0, so combining all this, we get that $\partial_\rho g$ is equivalent to $C\rho$ where C is some positive constant, hence $g \sim (C/2)\rho^2$ and $D \sim C'\rho$ where C' is another positive constant.

□

Finally, we state the following straightforward result.

Lemma 1.3. If $\alpha = f(\rho, \phi, \theta)d\phi + g(\rho, \phi, \theta)d\theta$ is a contact form, then its Reeb flow is given by :

$$R_\alpha = \frac{1}{D}((\partial_\phi f - \partial_\theta g)\partial_\rho - \partial_\rho f \partial_\phi + \partial_\rho g \partial_\theta)$$

where $D := f\partial_\rho g - g\partial_\rho f$ as usual.

Remark 1.4. In every computation that follows, we will always use the letter D to denote the function $f\partial_\rho g - g\partial_\rho f$, or $fg' - gf'$ when no confusion is possible.

We now build a first local model by studying locally what happens around an elliptic binding orbit, which is necessarily a radial binding component.

Lemma 1.5. Let γ be a radial binding component. Then there exists local coordinates (ρ, ϕ, θ) such that :

- $\xi = \mathrm{Ker}(d\theta + r^2 d\phi)$.

- The pages are $\{\phi = Cst\}$.

Proof. Start by taking the partial blow-up as in lemma **??**, with coordinates (ϕ, θ) on the exceptional divisor, with ϕ describing the characteristic foliation and θ following the boundary of the pages given by the radial model around the binding component. Define ∂_ρ on $\mathbb{T}^2 \times I$ as a vector field tangent to the characteristic foliation on the pages. This is possible as we know what this foliation looks like near a binding component (see the drawing below). Now consider the push-forward of ∂_ϕ and ∂_θ by ∂_ρ. This defines a new set of coordinates on $\mathbb{T}^2 \times I$ (in fact, it might be necessary to modify the neighbourhood near the boundary $\mathbb{T}^2 \times \{1\}$ but this does not change the construction). In this set of coordinates, the pages are given by $\{\phi = Cst\}$ as ∂_ρ and ∂_θ are tangent to them at all point.

We can now consider the set of coordinates (ρ, ϕ, θ) induced by the previous construction on the blow-down.
Because ∂_ρ is in ξ, the contact plane field is defined by an equation of the form : $fd\theta + gd\phi = 0$ with f and g functions of ρ, ϕ, θ. We know that f is non-vanishing up to taking the neighbourhood small enough, as R_α is parallel to ∂_θ and $\rho = 0$. Hence we can define $h = g/f$ and get : $\xi = \mathrm{Ker}(d\theta + hd\phi)$. Finally, consider the following change of variables :

$$p : (\rho, \phi, \theta) \mapsto (\sqrt{h(\rho, \phi, \theta)}, \phi, \theta)$$

To see that p is a diffeomorphism onto its image for ρ small enough, we compute :

$$2f\sqrt{fg}\,\partial_\rho(\sqrt{h}) = f\partial_\rho g - g\partial_\rho f$$

Now we use lemma **??** and get that $g/f \to 0$ and that $\frac{D}{2f\sqrt{fg}}$ converges to a positive constant when ρ goes to zero. This proves that p sends $\{\rho = 0\}$ to itself, and that $\partial_\rho\sqrt{h}$ is always well-defined and non-zero, which proves that p is indeed a diffeomorphism on the whole domain (i.e for fixed ϕ and θ, we can find a smooth inverse in ρ for $\sqrt{h}$).

A quick computation yields : $p_*\xi = \mathrm{Ker}(d\theta + \rho^2 d\phi)$. Moreover, the change of variables preserves the pages, hence we get a set of coordinates verifying the conditions we wanted.

□

The next lemma allow us to interpolate between any Reeb flow and a standard local model near a radial binding component. This model will be referred to as the "interpolation model" in the following parts.

Lemma 1.6. Take α and γ as before. Then there exists a contact form α' such that, in the set of coordinates given by the last lemma :

- $\alpha' = \alpha$ away from a neighbourhood of γ
- $R_{\alpha'}$ is supported by the broken book decomposition.
- $\alpha = f(\rho)d\theta + g(\rho)d\phi$ in a smaller neighbourhood around γ, where f and g are smooth maps verifying certain conditions. In particular, γ is elliptic for α'.

Moreover, for this new contact form, γ has an action greater than its initial action which can be chosen arbitrarily large, and we can arrange the model so that the Conley-Zehnder indices of an arbitrarily large number of iterates of γ is 1 (in the trivialization given naturally by the pages).

Remark 1.7. The conditions verified by f and g will be detailed in the proof ; they correspond to the usual conditions one imposes when constructing a contact form supported by a given open book decomposition (see for example [**wendl_open_2009**]).

Proof. Consider the coordinates (ρ, ϕ, θ) given y the previous lemma **??**, for $\rho \leq \delta$. In these coordinates, the contact form can be written $\alpha = h(\rho, \phi, \theta)(d\theta + \rho^2 d\phi)$ where h is positive. Denote $\overline{h}(\rho) := \max_{\phi,\theta} h(\rho, \phi, \theta)$. Take $0 < \delta'' < \delta' < \delta$ and take $\tau : [\delta'', \delta'] \to 1$ to be a strictly increasing, with value 0 in a neighborhood of δ" and value 1 in a neighborhood of δ'. Define $H(\rho, \phi, \theta) := \tau(\rho)h + (1 - \tau(\rho))\overline{h}$ and consider $\overline{\alpha} = H(d\theta + \rho^2 d\phi)$. Quick computations give the following :

$$\overline{\alpha} \wedge d\overline{\alpha} = 2\rho H^2 d\rho \wedge d\phi \wedge d\theta > 0$$

This shows that $\overline{\alpha} \wedge d\overline{\alpha}$ is positive, hence $\overline{\alpha}$ is a contact form.
Applied in our setting this gives :

$$d\phi(R_{\overline{\alpha}}) = -\partial_\rho H = -\partial_\rho \overline{h} - \tau'(h - \overline{h}) - \tau \partial_\rho (h - \overline{h})$$

We know that $\partial_\rho h < 0$ and by definition of $\overline{h}$: $-\partial_\rho \overline{h} > 0$, $h - \overline{h} < 0$ and $\partial_\rho(h - \overline{h}) < 0$. Combined with $\tau, \tau' \geq 0$, this proves that $d\phi(R_{\overline{\alpha}}) > 0$. We conclude that the Reeb flow is positively transverse to the pages, and $\overline{\alpha}$ is supported by the open book decomposition for $\rho \geq \delta''$.

Take smooth maps : $f, g : [0, \delta^{(3)}] \to [0, 1]$ where $\delta'' < \delta^{(3)} < \delta'$ and $\tau = 0$ on $[\delta'', \delta^{(3)}]$, such that :

- $f'(\rho) < 0,\ g'(\rho) > 0$.
- $f(0) = b > 0$, $g(0) = 0$.
- $f = \overline{h}$ et $g = \rho^2 \overline{h}$ pour $\rho \in [\delta'', \delta^{(3)}]$.
- f et g/ρ^2 are smooth at 0.
- $D := fg' - f'g > 0$.
- $-f'/g' = \eta \in \mathbb{R} \backslash \mathbb{Q}$, with η arbitrarily small and for $\rho \leq \rho_0$, with ρ_0 small enough.

An example of such maps f, g is drawn in figure **??**.
Finally, define $\alpha_0 := f(\rho)d\theta + g(\rho)d\phi$. We showed that the Reeb flow was supported by the pages of the open book for $\rho > \delta$", hence by lemma **??**, f and g are well-defined as the first condition is verified for $\rho \in [\delta'', \delta^{(3)}]$. The different conditions used here appeared in similar forms in **[wendl_open_2009]**,**[roux_cas_nodate]**, **[wendl_holomorphic_2010]**. They ensure that α_0 is a contact form, with a non-degenerate elliptic orbit γ at $\rho = 0$, with action b. Notice that b has to be greater than the initial action $\mathcal{A}(\gamma) + \epsilon$, where ϵ is a small number which can be chosen arbitrarily small by shrinking the neighborhood so that $\max_{\mathcal{U}} f < \mathcal{A}(\gamma) + \epsilon$ over this neighborhood. As long as b verifies this condition, it can be chosen arbitrarily large. Unlike the references mentioned above, the form chosen for f'/g' here is different and does not allow to control the action of other orbits in a neighborhood of γ, we just guaranty that γ is non-degenerate by asking $\eta \in \mathbb{R} \backslash \mathbb{Q}$.
We now show that the Conley-Zehnder index of γ and a certain number of iterates can be chosen to be 1. A short computation shows that the linearized Reeb flow along γ^r corresponds to a rotation of angle: $-\frac{t}{b}\frac{f'}{g'}$, so the first return map corresponds to a rotation of angle $r\eta$, which shows that up to choosing η small, the Conley-Zehnder index is 1.

Finally, the Reeb flow for $\rho \leq \delta''$ is $R_{\alpha_0} = \frac{1}{D}(g'\partial_\theta - f'\partial_\phi)$, and $-f'$ being positive implies that this flow is supported by the open book decomposition.

□

Remark 1.8. Note that this method allows one to transform hyperbolic orbits in the binding into elliptic ones. This process may create new periodic orbits, but it is not an issue as they will have greater action than the binding components.

Let us now build the model around broken binding components. Recall these orbits are always hyperbolic. We will also be using sometimes coordinates (x, y) instead of (ρ, ϕ) as they may be more convenient for hyperbolic models.
This model allows us to interpolate again in this setting, with particular constraints again on the action and Conley-Zehnder index.

Lemma 1.9. Take α and γ as before. Then there exists contact forms $\alpha'_i,\ i = 1, 2$ such that, in the set of coordinates given by lemma **??** :

- $\alpha'_i = \alpha$ away from a neighbourhood of γ
- $R_{\alpha'}$ is positively transverse to the rigid pages $\{\phi = 0\}$ and $\{\phi = \pi\}$, and negatively transverse to the pages $\{\phi = \pi/2\}$ and $\{\phi = -\pi/2\}$.
- $\alpha'_1 = (b + \epsilon(x^2 - y^2))d\theta + \frac{1}{2}(xdy - ydx)$ where b is the action of γ (with respect to α), and ϵ is a small constant, and $\alpha'_2 = be^{\epsilon H}(d\theta + \rho^2 d\phi)$ where $H(x, y, \theta) = x^2 - y^2$.

In particular, γ is hyperbolic, and its action in the new model is the same than in the initial one.

Remark 1.10. The Reeb vector field $R_{\alpha'}$ may not be transverse to the pages of the broken book near the binding anymore. This will not be an issue as we only need it to be transverse to the rigid pages. In fact, the arguments from intersection theory that we will use later will show that up to isotopy of the pages, we can still assume that the Reeb vector field for α' is supported by the broken book decomposition. The fact we had a stronger result in the radial case will simply mean that there is no need to isotope the pages near radial binding components, namely the projections of the curves of the finite energy foliation we will obtain will coincide exactly with the pages of the initial broken book near the radial binding, and up to isotopy elsewhere.

Remark 1.11. We can also ask that $\alpha' = be^{\epsilon H}(d\theta + \sigma)$ where σ is a Liouville form on the disc $\mathbb{D}_{x,y}$, with an appropriate boundary form, for example $e^{\rho} d\phi$ like what we will use in part **??**.

Proof. We start by using the same construction as in lemmas **??**,**??**. Indeed, applying the same method on the rigid pages, we can find coordinates (ρ, ϕ, θ) such that the rigid pages with negative ends are exactly $\{\phi = 0\}$ and $\{\phi = \pi\}$ and the rigid pages with positive ends are exactly $\{\phi = \pm\frac{\pi}{2}\}$, and the contact structure is $\mathrm{Ker}(d\theta + \rho^2 d\phi)$. Hence α can be written as $f(\rho, \phi, \theta)(d\theta + \rho^2 d\phi)$. Moreover, we can assume that at $\rho = 0$, f is constant equal to b, by taking the coordinate θ so that $\partial_\theta = R_\alpha$ at $\rho = 0$.

We interpolate α with α'_i. Consider $\tau : [\delta'', \delta'] \to 1$ as in the proof of lemma **??**, let $\overline{f}(\rho, \phi, \theta) = b + \epsilon(\rho^2 \cos(2\phi))$, and $\alpha'_1 = F(d\theta + \rho^2 d\phi)$ with $F = \tau f + (1 - \tau)\overline{f}$. First, this is a well-defined contact form as up to taking a small enough neighborhood, f is positive on it, and up to choosing ϵ small enough compared to b, $\overline{f}$ is too. We only need to check that the Reeb flow remains transverse to the four rigid pages. Notice that for $\rho \leq \delta$", this is true as the ∂_ϕ component of the Reeb flow is $-\partial_\rho(\overline{f}) = -2\epsilon\rho\cos(2\phi)$. Now we have : $\partial_\rho F = (1 - \tau)\partial_\rho \overline{f} + \tau\partial_\rho f + \tau'(f - \overline{f})$. Consider first the case $\phi = 0$. We want this page to have a negative end, that means we want $d\phi(R_\alpha) < 0$. But we know that $\partial_\rho f > 0$ and $\partial_\rho \overline{f} \geq 0$, hence $\tau\partial_\rho f + (1 - \tau)\partial_\rho \overline{f} > 0$. It remains to show that $\tau'(f - \overline{f}) > 0$, which is equivalent to $f - \overline{f} > 0$. This can be achieved up to taking ϵ small enough compared to b, as we know that $\partial_\rho f > 0$ on negative pages, so $f > b$ on such a rigid page, and: $f - \overline{f} = f - b - \epsilon\rho^2$. This also works for the positive rigid pages, where the condition becomes $f - \overline{f} < 0$, and because we know that: $\partial_\rho f < 0$, we have: $f - \overline{f} = f - b + \epsilon\rho^2 < 0$ if ϵ is small enough.

We interpolate again in order to have the right form for α'_1. Take $\delta_1 < \delta_2 < \delta''$, and $\tau : [\delta_1, \delta_2] \to [0, 1]$ which is 0 near δ_1 and 1 near δ_2. Let $\alpha'_1 = \overline{f}d\theta + (b + \tau\rho^2 \cos(2\phi))\rho^2 d\phi$. Note that we do not change the map in front of the $d\theta$ term, hence if we show that we have a contact form, the transversality condition for the Reeb flow and the rigid pages will be automatically verified. Write $\overline{g} := (b + \tau\epsilon(x^2 - y^2))\rho^2$. In order to verify that we have a contact form, it suffices to show that $\overline{f}\partial_\rho \overline{g} - \overline{g}\partial_\rho \overline{f}$ is positive. Because we can choose ϵ to be as small as necessary compared to b, it suffices to look at the term which do not depend on ϵ. A quick computation yields that this term is $2\rho b^2$ which is indeed positive. This concludes on this step of the interpolation. We now have for, $\rho \leq \delta_1$, $\alpha_1' = \overline{f}d\theta + bd\phi$.

Finally, we do one last interpolation. Choose $\delta_3 < \delta_4 < \delta_1$, and $\tau : [\delta_3, \delta_4] \to \mathbb{R}$ such that τ is monotone, takes value $\frac{1}{2}$ near δ_3 and b near δ_4. More precisely, if $b \geq \frac{1}{2}$, we can assume $\tau' \geq 0$. By the same arguments than before, we only need to check the contact condition, which, keeping only the terms without ϵ, boils down to $(\tau\rho^2)' > 0$. This is indeed verified if $\tau' \geq 0$. Now if $b < \frac{1}{2}$, we take τ such that $\tau' \leq 0$. Rewriting the condition$(\tau\rho^2)' > 0$, we see that is equivalent to saying that $\frac{1}{\rho^2}$ decreases faster than $\ln(\tau)$. This can indeed be achieved, taking δ_3 sufficiently small. This concludes for α_1'.

Only the first part of this interpolation is necessary for α_2', and again by taking ϵ small enough the same method shows the expected result.

Notice that this construction does not allow γ to have arbitrary small or large action. Indeed, if it were the case, we would be able to construct rigid pages on which the integral of $d\alpha$ would be negative by Stoke's theorem, which is not possible as the Reeb flow is positively transverse to the pages (see also lemma **??**). □

In the following, when considering this interpolation model, we will assume every broken binding orbit has the same small action b.

We now build a different kind of local model, adapted to radial and broken binding components alike. It will allow us to lift the pages of the broken book as a finite energy foliation for a given almost complex structure compatible with a *fixed* supported contact form α verifying the right conditions on the Conley-Zehnder index near the radial binding components, without modifying it near the binding components. This model will be referred to as the "exact model" in the following parts.

Lemma 1.12. Let γ be a radial **or** broken binding component, and a Reeb orbit for α a contact form supported by the broken book decomposition. Choose a finite collection of distinct pages with **positive** ends at γ, $\{P_1, ..., P_n\}$ that we call "rigid pages" (in the case of a broken binding component, choose the four natural rigid pages). Then there exists local coordinates (ρ, ϕ, θ) around γ such that, for all i and $(\rho, \phi, \theta) \in P_i$:

- $d\phi(R_\alpha) > 0$ and $d\theta(R_\alpha) > 0$, and R_α is collinear to ∂_θ at $\rho = 0$.
- $\alpha(\partial_\phi) > 0$ for $\rho > 0$.
- $P_i = \{\phi = \phi_i\}$ for some ϕ_i fixed constant.
- $\alpha = fd\theta + gd\phi$, and $\partial_\theta f = 0$.
- $d\rho(R_\alpha) = 0$.

Remark 1.13. Note that all the previous constraints are verified only on the collection of rigid pages chosen. The last two conditions are the additional constraints compared to the models built before.

Proof. Start as in the beginning of the proof of lemma **??** in order to get coordinates $(\tilde{\rho}, \tilde{\phi}, \tilde{\theta})$ on the partial blow-up such that $\partial_\rho \in \xi$ and the pages are level sets of ϕ (in the radial case one can do this for all pages, in the broken case the construction can be adapted to work on the rigid pages only). The three first conditions are then automatically verified by the arguments already used before. In order to show the two last conditions, we apply again two modifications of the coordinates. We will denote by T the neigborhood and T_0 the exceptional divisor of the partial blow-up.

We modify the previous coordinates in order to remove the dependency of f in the θ-coordinate on P_i. In order to do that, consider again the blow-up neighborhood T. We look at the level sets of f on P_i. They are all compact by lemma **??**. Hence they are conjugated to tori around T_0. Note that on T_0, $f = 1$ so $T_0 \cap P_i$ is a level set of f. Consider the diffeomorphism :

$$\begin{cases} \Psi : & P_i \to \Psi(P_i) \subset P_i \\ & (\rho, \theta) \mapsto (1 - f(\rho, \theta), \theta) \end{cases}$$

By lemma **??**, this is a well-defined diffeomorphism which extends as the identity at $\{\rho = 0\}$. Now look at the pullback of ∂_ρ, which is $-\frac{1}{\partial_\rho f}\partial_\rho$. This with the pullback of ∂_θ define our new set of coordinates on P_i. We extend them to T by the applying the same process than before.

Now, we want the following property to be verified for all i :

$$\text{On } \ TP_i, R_\alpha \in T(\phi_t^{\partial_\rho}(T_0)) \text{ for all time } t, \text{where } \partial_\rho = h(\tilde{\rho})\partial_{\tilde{\rho}}, \text{ with } h > 0$$

In order to chose this function h, define $g = \frac{d\tilde{\rho}(R_\alpha)}{d\tilde{\phi}(R_\alpha)}$, and chose some function $y = y(\tilde{\rho}, \tilde{\phi}, \theta)$ such that :

- $y(\tilde{\rho}, \tilde{\phi}, \theta) = \tilde{\rho}$ for $\tilde{\phi} = 0$ or $\tilde{\phi}$ large enough (i.e on P_i or away from a small neighborhood of P_i).
- $\partial_{\tilde{\phi}} y_{|\tilde{\phi}=0} = g$ near P_i.
- $\partial_{\tilde{\rho}} y > 0$ everywhere.

Define $h = dy(\partial_{\tilde{\rho}})$. Note that $d\tilde{\phi}(R_\alpha) > 0$ on T because R_α is positively transverse to the pages $\{\tilde{\phi} = \text{Cst}\}$, and at $\tilde{\rho} = 0$, $d\tilde{\rho}(R_\alpha) = 0$ because the Reeb vector field is well defined at $\rho = 0$ in the initial model. This implies that h is well-defined, is 1 on T_0 and is always positive. Moreover, for fixed $\tilde{\phi}, \tilde{\theta}$, $z(t) = y(t, \tilde{\phi}, \tilde{\theta})$ is the solution of $z' = h(z, \tilde{\phi}, \tilde{\theta})$ with initial condition $z(0) = 0$, i.e $\phi_t^{h\partial_{\tilde{\rho}}}(0, \tilde{\phi}, \tilde{\theta}) = y(t, \tilde{\phi}, \tilde{\theta})$.

Define $\partial_\rho := h\partial_{\tilde{\rho}}$, $\partial_\phi(\tilde{\rho}, \tilde{\phi}, \tilde{\theta}) := d_{(0,\tilde{\phi},\tilde{\theta})}\phi_t^{\partial_\rho}(\partial_{\tilde{\phi}})$ and $\partial_\theta(\tilde{\rho}, \tilde{\phi}, \tilde{\theta}) := d_{(0,\tilde{\phi},\tilde{\theta})}\phi_t^{\partial_\rho}(\partial_{\tilde{\theta}})$ where t is such that $\phi_t^{\partial_\rho}(0, \tilde{\phi}, \tilde{\theta}) = (\tilde{\rho}, \tilde{\phi}, \tilde{\theta})$ (we extend the coordinates (ϕ, θ) defined on T_0 to T with the flow of ∂_ρ). This gives a set of coordinates (ρ, ϕ, θ) on T.

Now we see that on P_i, the following happens : $\partial_\rho = \partial_{\tilde{\rho}}$, $\partial_\theta = \partial_{\tilde{\theta}}$ and $\partial_\phi = g\partial_{\tilde{\rho}} + \partial_{\tilde{\phi}}$. To get the latter, notice that : $d\phi_t^{h\partial_\rho}(\partial_{\tilde{\phi}}) - \partial_{\tilde{\phi}} = \lim\limits_{\epsilon \to 0} \dfrac{1}{\epsilon}(\phi_t^{h\partial_{\tilde{\rho}}}((0, \epsilon, \tilde{\theta}) - \phi_t^{h\partial_{\tilde{\rho}}}((0, 0, \tilde{\theta}))) = \lim\limits_{\epsilon \to 0} \dfrac{1}{\epsilon}(y(t, \epsilon, \tilde{\theta}) - y(t, 0, \tilde{\theta}))\partial_{\tilde{\rho}} = \partial_\phi y(t, 0, \tilde{\theta})\partial_{\tilde{\rho}} = \dfrac{d\tilde{\rho}(R_\alpha)}{d\tilde{\phi}(R_\alpha)}\partial_{\tilde{\rho}}$, so : $\partial_\phi = g\partial_{\tilde{\rho}} + \partial_{\tilde{\phi}}$. We can write : $R_\alpha = d\tilde{\phi}(R_\alpha)(g\partial_{\tilde{\rho}} + \partial_{\tilde{\phi}}) + d\tilde{\theta}(R_\alpha)\partial_\theta = d\tilde{\phi}(R_\alpha)\partial_\phi + d\tilde{\theta}(R_\alpha)\partial_\theta \in \langle \partial_\phi, \partial_\theta \rangle$ so these coordinates verify the condition we wanted. Notice that we still have $\partial_\theta f = 0$ on P_i as $\partial_{\tilde{\theta}} = \partial_\theta$ on P_0, and α, $d\phi$ and $d\theta$ are unchanged.

□

Finally, we consider the case of planar broken books where we allow pages to have radial negative ends. As we will see, these kind of broken books can still be lifted as a finite energy foliation, even though it will not be suitable to study symplectic fillings anymore. We consider both situations previously studied.

Lemma 1.14. If γ is a negative radial binding component, we can find a contact form interpolating such that all the conditions of lemma **??** are verified, except for the fact that the maps f, g verify instead that $\partial_\rho f, \partial_\rho g$ are both positive, so $f > 0$ and $g \geq 0$. The other conditions on f and g are left unchanged. Moreover, the new action can be chosen arbitrarily small as long as it is smaller than the initial action, and the Conley-Zehnder index of a finite number of covers of γ can be chosen to be 1.

Proof. The proof is essentially the same than then proof of lemma **??**, reversing some signs. Notice that in the proof of lemma **??**, we used the fact that we had $d\alpha(\partial_\rho, \partial_\theta) < 0$ in order to get $\partial_\rho f < 0$. In the negative case, this translates as : $d\alpha(\partial_\theta, \partial_\rho) > 0$, hence $\partial_\rho f > 0$. We also ask that $d\phi(R_\alpha) < 0$ which implies $\partial_\rho g > 0$, $f > 0$ and $g \geq 0$ with strict inequality for $\rho \geq 0$. All the other arguments still apply in this setting, with the difference that this time we have an upper bound on the possible action, instead of a lower bound, which is coherent with energy considerations. Moreover, f' is this time positive, hence $-\frac{t}{b}\frac{f'}{g'} = -\frac{\eta t}{b}$, and the first return map of the linearized Reeb flow is a rotation of angle $-\eta r$, so we have a Conley-Zehnder equal to -1 if we choose η small. □

Lemma 1.15. Under the same assumptions than in lemma **??** but for a negative radial end, there exists coordinates such that on the rigid pages, α verifies all the same conditions than in lemma **??**, but with the condition $d\phi(R_\alpha) < 0$ instead.

Proof. Apply the same proof as for lemma **??** but replacing $d\phi(R_\alpha) < 0$ and $\alpha(\partial_\phi) < 0$ like in the proof lemma **??**. □

Remark 1.16. We list here the properties verified for $\rho > 0$, in the case of a positive end, for future reference.

- $\alpha(\partial_\phi) > 0$.
- $d\phi(R_\alpha) > 0$.
- f > 0.
- g > 0 .
- $\partial_\rho f < 0$.
- $\partial_\rho g > 0$.

In the case of a negative end, the only differences are : $d\phi(R_\alpha) < 0$ and $\partial_\rho f > 0$. The other conditions are left unchanged.

2 Almost complex structures and lift of rigid pages

In this section we build the compatible almost complex structures that we will use in order to lift the pages as a finite energy foliation, and then study the extension of this foliation in the case of a symplectic filling. We start by recalling the standard model for the almost complex structure near the ends with the coordinates built in part **??**. We then explain how to suitable almost complex structures away from the ends on rigid pages. We also explain how to extend this construction in dimension 5, and we give in the last part another model, derived from the so-called *L-simple structures* from [**bao_definition_2018**], [**bao_semi-global_2023**]. This new model will allow to apply the obstruction bundle gluing arguments necessary for the proof of the Weinstein conjecture in part **??**, and will also allow to lighten the hypothesis necessary for the method used in part **??**.

2.1 Almost complex structures and lift near the ends

The purpose of these two first subsections is to explain how to choose appropriate almost complex structures around rigid pages so that we can lift them as a translation invariant 1-parameter family in the symplectization.

First, in the case of a classic radial model around a binding orbit in an open book, we recall how to construct an suitable almost complex structure, following the construction in [**wendl_open_2009**]. Let γ be the binding orbit, coordinates (ρ, ϕ, θ) around it as before on a neighborhood $\mathcal{U}$ of γ, and take $\alpha = f(\rho)d\theta + g(\rho)d\phi$. Denote $D = fg' - gf' > 0$ as usual.

Definition 2.1. Define a *model almost complex structure* J on $\mathcal{U}$ by the following formula :

$$\begin{cases} J\partial_a = R_\alpha = \frac{1}{D}(g'\partial_\theta - f'\partial_\phi) \\ J\partial_\rho = \frac{1}{D}(-g\partial_\theta + f\partial_\phi) \end{cases} \tag{1}$$

As a consequence of this definition, we also have the following relations :

- $J\partial_\theta = -f\partial_a - f'\partial_\rho =: -V$
- $J\partial_\phi = -g'\partial_\rho - g\partial_a$

Remark 2.1. Usually, like in [**wendl_open_2009**], the almost complex structure is defined by $J\partial_\rho = \frac{1}{B}(-g\partial_\theta + f\partial_\phi)$, where $B = B(\rho)$ is a map equivalent to ρ near 0, and equal to one away from 0. In fact, we will use this model later in the 5-dimensional case. The reason why we do not use this here is that D is also equivalent to a constant times ρ near 0, hence the almost complex structure is well-defined, but this expression allow to simplify the formulas for $J\partial_\rho$ and $J\partial_\phi$, which will be necessary in what follows.

Remark 2.2. This model will be valid for positive and negative ends alike, as this only changes the sign of f'.

Now when one has an almost complex structure of this form, it is possible to lift the ends $\{\phi = Cst\}$ as holomorphic curves in the symplectization. There are two ways to do this. The first one is the method used in [**wendl_open_2009**].

Proposition 2.3. Ends of pages $\{\phi = \phi_0\}$ can be parametrized by maps u of the form :

$$\begin{aligned} u : &\mathbb{R}_+ \times \mathbb{S}^1 \to \mathbb{R} \times \mathcal{U} \\ &(s,t) \mapsto (a(s), \rho(s), 0, t) \end{aligned}$$

If a and ρ verify the equations :

$$\frac{da}{ds} = f > 0 \qquad \frac{d\rho}{ds} = f'(\rho) < 0$$

then u is a holomorphic parametrization of these surfaces.

Note that these equations are order 1 ODEs, hence we can always find solutions.

Another way to define a holomorphic parametrization of the ends, which will be more suitable to higher dimension, is to use the flow of the vector field V defined in definition **??**.

Proposition 2.4. Define the line $L = (a = a_0, \rho = \rho_0, \phi = \phi_0, \theta \in \mathbb{S}^1)$ and consider the surface : $S = (\phi_t^V(L))_{t \in R_+}$.
Then S is a smooth holomorphic parametrization of the end $\{\phi = \phi_0\}$, and is asymptotic to γ.

Remark 2.5. Here we see with the two methods what happens if one uses the formula $J\partial_\rho = \frac{1}{B}(-g\partial_\theta + f\partial_\phi)$ as mentioned in remark **??**. In the Cauchy-Riemann equations, this leads to : $\frac{d\rho}{ds} = \frac{B}{D}f'$, and the vector fields V becomes : $V = f\partial_a + B\frac{f'}{D}\partial_\rho$. This becomes an issue when working with functions f, g which depends on θ (which implies that D also depends on θ). In this case, the equations cannot be solved as simply as before, and the flow method cannot be applied as the Lie bracket $[V, \partial_\theta]$ is not 0 anymore. Note that by taking $B = D$, requiring $\partial_\theta f = 0$ is enough is enough in order to apply the previous methods (as it also implies $\partial_\theta(\partial_\rho f) = 0$ on a fixed page.

Remark 2.6. In both cases, this gives asymptotically cylindrical (open) finite energy curves. The fact that they have finite energy can be seen by applying Stoke's theorem as the curves are topologically asymptotic to γ is the completion $\overline{R} \times M$.
It can also be checked by hand that the Hofer energy is finite: take $\varphi : \mathbb{R} \to]-1, 1[$ strictly increasing and look at :

$$E_\varphi(u) = \int_{\mathbb{R}_+ \times \mathbb{S}^1} u^*\omega_\varphi = \int_{\mathbb{R}_+ \times \mathbb{S}^1} \varphi'(a \circ u) e^{\varphi(a \circ u)} u^*(da \wedge \alpha) + \int_{\mathbb{R}_+ \times \mathbb{S}^1} e^{\varphi(a \circ u)} u^* d\alpha$$

Using the same parametrization of the cylindrical end $(s,t) \in \mathbb{R}_+ \times \mathbb{S}^1$ as before, we have : $u^*(da \wedge \alpha) = (f \circ u)^2 ds \wedge dt$, and $(f \circ u)^2 \leq C < +\infty$ with C some positive constant. Hence the first integral is finite. For the second one, $u^*d\alpha = (\partial_\rho f)^2(\rho(s))ds \wedge dt = (\frac{d\rho}{ds}\partial_\rho f \circ \rho)ds \wedge dt = \partial_s(f \circ \rho)$, and indeed $\displaystyle\int_{R_+} \partial_s(f \circ \rho)ds < +\infty$ as $f \to 1$ when $\rho \to 0$ i.e $s \to +\infty$. This shows the second integral is also finite, and that $E_\varphi(u)$ can be bounded by some constant independent on φ, which shows that the Hofer energy is finite.

Let us now use this model almost complex structure and the local models of part **??** in order to lift the ends of our rigid pages. Once again we will do this for both the interpolation and the exact models. Consider a rigid page P with an end (positive or negative) asymptotic to γ. We denote $P_\infty = P \cap \mathcal{U}$

Lemma 2.7. There exists an almost complex structure J_∞ on P_∞ such that J_∞ is α_i'-compatible in the interpolation case, or α-compatible in the exact case, and such that there is a lift of P_∞ as a 1-parameter family of holomorphic curves in the symplectization, whose projection onto M is P_∞. More precisely, on P_∞, α (or α_i') can be written as $fd\theta + gd\phi$, and the almost complex structure has the same form as in **??**, replacing f' by $\partial_\rho f$.

Proof. It suffices to notice that lifting methods used in the previous propositions **??**,**??** can still be used in more general setting than the one of definition **??**. Indeed, suppose without loss of generality that the page is $\{\phi = 0\}$. Then we only need the following **on** P_∞ **only**:

1. $\partial_\rho \in \xi$, so $\alpha = fd\theta + gd\phi$
2. R_α is transverse to P_∞.
3. $d\rho(R_\alpha) = 0$.
4. f does not depend on θ.

Conditions $1, 2$ and 3 ensure that the Reeb flow has the right form, and that the asymptotic behavior of f and g are as in lemma **??**. This implies that we can define the almost complex structure J on P_∞ with the same formulas the in definition **??**, replacing f' by $\partial_\rho f$. The last condition is the only additional constraints one needs in order to apply the lifting methods, as noticed in remark **??**. Hence we only need to check that these conditions are verified in the different models defined in part **??**.

Consider first the interpolation model, and the case of a radial binding component. The contact form given by lemma **??** has the form $f(\rho)d\theta + g(\rho)d\phi$ as in the exact model. Hence we can use the exact same definition as in **??** and get the wanted almost complex structure. Note that this also works in the case of a negative radial end as noticed before. Still in the interpolation model, but for a broken component, the contact form given by lemma **??** had a form which was either $(b+\epsilon(x^2-y^2))d\theta + \rho^2 d\phi = f(\rho,\phi)d\theta + g(\rho)d\phi$, or $be^{\epsilon H}(d\theta + \rho^2 d\phi) = f(\rho,\phi)d\theta + g(\rho,\phi)d\phi$. There is no dependency in θ, hence we can again define the same almost complex structures as before, replacing f' by $\partial_\rho f$. Notice that on the rigid pages, corresponding to $\phi = 0, \pi, \pm\frac{\pi}{2}$, we have : $f = b + \epsilon\rho^2$ or $f = be^{\epsilon\rho^2}$ and $g = \rho^2$ or $g = be^{\epsilon\rho^2}\rho^2$ so these maps obviously satisfy the same asymptotic estimates than in lemma **??**.

Consider now the exact model. It turns out that all necessary hypotheses were already verified in lemma **??**. Hence in the same way we can define a compatible almost complex structure on the ends. □

2.2 Almost complex structures on rigid pages

So far we have explained how to lift the ends of a rigid page. We now lift see how to lift a page away from the binding in a way such that everything can be attached smoothly. In fact, we will prove that we can lift any embedded surface with the right asymptotic behavior. This is expreseed by the following proposition.

Proposition 2.8.
Let (M^3, α) be a contact 3-manifold, and P be a smooth embedded compact surface with boundary, transverse to the Reeb flow on its interior, and with closed non-degenerate simple Reeb orbits as boundary components.
Then there exists $J \in \mathcal{J}(\alpha)$ such that P can be lifted (up to isotopy) to a translation invariant 1-parameter family of J-holomorphic curves in the symplectization. Moreover, we can require that $J \in \mathcal{J}(\alpha)$ is only fixed on P, and can be chosen arbitrarily elsewhere.

Proof. First notice that on any such surface, one can define the model from **??**, hence it suffices to consider the case where P is a page of some broken book decomposition (in fact, rigid or not, this does not change anything as we can lift the ends either way and the interior behaves the same).
Therefore, choose such a rigid page P, with boundary components Reeb orbits γ_i, and associated neighborhoods $\mathcal{U}_i$ such that on $P \cap \mathcal{U}_i$ we have a fixed almost complex structure which lift the ends of P. We need to define an almost complex structure on the interior of P, and to verify it can be glued smoothly to the one defined before.

We will use the notation $\dot{\Sigma} := \Sigma \backslash \Gamma$, where Σ is a compact Riemann surface, $\Gamma = \Gamma^+ \cup \Gamma^- = \{z_1^+, ...z_r^+\} \cup \{z_1^-, ..., z_s^-\}$ a collection of points of Σ, and $\dot{\Sigma}$ the associated punctured Riemann surface, such that topologically, $\dot{\Sigma} \simeq P$, and with the puncture in $1-1$ correspondance with the ends of P (with r the number of positive ends and s the number of negative ends). When denoting some puncture z_i, we refer to any puncture, positive or negative alike. We also take some initial complex structure j on $\dot{\Sigma}$.

Choose some function $h = (a_0, h_0) : \dot{\Sigma} \to \mathbb{R} \times M$ such that near each z_i, there is a cylindrical neighborhood which h sends in $\mathbb{R} \times \mathcal{U}_i$, and on which h coincides with the lifted holomorphic curve $(a(s), \rho(s), 0, \pm t)$ in local coordinates (ρ, ϕ, θ) as constructed before around the binding orbit γ_i associated to z_i. Extend this local parametrization smoothly outside of $R \times \mathcal{U}_i$, such that h_0 is an embedding and $\mathrm{Im}(h_0) = P$. On each $\mathbb{R} \times (P \cap \mathcal{U}_i)$, we built some α-compatible almost complex structure, which we can extend as a α-compatible almost complex structure J_0 on $\mathbb{R} \times P$ (because the fiber bundle of compatible almost complex structures is contractible). Consider the Liouville manifold $(\dot{\Sigma}, h^*\alpha, h^*d\alpha)$. Note that h is (j, J_0)-holomorphic near the punctures, where some vector fields (∂_s, ∂_t) associated to a set of cylindrical j-complex coordinates $(s,t) \in \mathbb{R}_\pm \times \mathbb{S}^1$ is sent to $(f\partial_a + (\partial_\rho f)\partial_\rho, \pm\partial_\theta)$ (the signs depend on the positivity of the end). We build an adapted Morse function ψ in the following way : we want $\psi \equiv a$ on the cylindrical ends, and we extend it elsewhere by asking that the vector field X such that $\alpha = \iota_X d\alpha$ is a pseudo-gradient for ψ, which is always possible up to an isotopy of P. Indeed, such a vector field generates the caracteristic foliation on P, and up to isotopy we can assume the only singularities are saddles and nodes, which are also the possible singularities of a Morse function. Note that a is Morse with pseudo-gradient X near the ends as we have : $-da \circ j = u^*\alpha$ by the Cauchy-Riemann equations, hence is even already a Stein potential near the ends. Hence we get $(\dot{\Sigma}, h^*\alpha, h^*d\alpha, \psi)$ a Weinstein manifold. We define a new complex structure j_0 on $\dot{\Sigma}$ by keeping j on $h^{-1}(\mathbb{R} \times \mathcal{U}_i)$, and extending it elsewhere as a $u^*d\alpha$-compatible almost complex structure, which is automatically integrable.

We now use the existence theorem from **[cieliebak_stein_2012]** (Theorem 13.8). There exists $h_1 : \dot{\Sigma} \to \dot{\Sigma}$ such that $(\dot{\Sigma}, h_1^* j_0, h_1^*(h^*\alpha), h_1^*(h^*d\alpha), \psi)$ is a Stein manifold, and with h_1 preserving the positive and negative ends. We define the following almost complex structure J on P :

$$\begin{cases} J\partial_a = R_\alpha \\ J(v - \alpha(v)R_\alpha) = (h_* j_0)v - \alpha((h_* j_0)v)R_\alpha \quad \forall v \in TP \end{cases}$$

Lemma 2.9.
This is a well-defined α-compatible almost complex structure along $P = (h \circ h_1)(\dot{\Sigma})$ which coincides on $\mathbb{R} \times \mathcal{U}_i$ with the previously defined almost complex structure.
Moreover, denoting by τ_a the translation by a in the $\mathbb{R}$-direction, $u := \tau_\psi(h \circ h_1)$ is $(h_1^* j_0, J)$-holomorphic, and coincides with the previous lift of P near the ends.

Proof. Denote $J_* := (h_* j_0)$.
J is well-defined because $v \mapsto v - \alpha(v)R_\alpha$ is an isomorphism between TP and ξ, and $J(J_*v - \alpha(J_*v)R_\alpha) = J_*^2 v - \alpha(J_*^2 v)R_\alpha$, but $J_*^2 = -Id$ hence J is a well-defined α-compatible almost complex structure. Now on $\mathbb{R} \times \mathcal{U}_i$, we have :

$$J(\partial_\rho) = J_*\partial_\rho - \alpha(J_*\partial_\rho)R_\alpha - \alpha(\partial_\rho)\partial_a$$

with the definition above. Hence :

$$J(\partial_\rho) = J_*\partial_\rho - \frac{1}{D}\alpha(J_*\partial_\rho)(g'\partial_\theta - f'\partial_\phi)$$

However, we have

$$J_*\partial_\rho = J_*(dh(\pm\frac{1}{\partial_\rho f}\partial_s)) = \frac{1}{\partial_\rho f}dh(\pm j_0\partial_s) = \frac{1}{\partial_\rho f}\partial_\theta$$

Hence :

$$J(\partial_\rho) = \frac{1}{\partial_\rho f}\partial_\theta - \frac{f}{D\partial_\rho f}(\partial_\rho g\partial_\theta - \partial_\rho f\partial_\phi) = \frac{1}{D}(f\partial_\phi - g\partial_\theta)$$

which is exactly the same formula than for the structure defined previously on $\mathcal{U}_i$. Hence J is an α-compatible almost complex structure well-defined on P.
Now, we know that $\psi = a$ on $(h \circ h_1)^{-1}(\mathbb{R} \times \mathcal{U}_i)$, h_1 is the identity on $(h \circ h_1)^{-1}(\mathbb{R} \times \mathcal{U}_i)$, and h coincides with the lifted holomorphic curve in the local model defined before. Hence $(\psi, h \circ h_1)$ is a smooth extension of the cylindrical curves defined in the local model.

Finally, let us show that the Cauchy-Riemann equations hold in local complex coordinates (s,t) :

$$du = (d\psi)\partial_a + d(h \circ h_1)$$

$$J \circ du(\partial_s) = d\psi(\partial_s)R_\alpha + J_*d(h \circ h_1)(\partial_s) - \alpha(J_*d(h \circ h_1)(\partial_s))R_\alpha - \alpha(d(h \circ h_1)(\partial_s))\partial_a$$

But :

$$J_*d(h \circ h_1)(\partial_s) = dh \circ j_0 \circ dh_1(\partial_s) = d(h \circ h_1)(h_1^* j_0)(\partial_s)$$

So :

$$\alpha(J_*d(h \circ h_1)(\partial_s)) = (h \circ h_1)^*\alpha(h_1^* j_0)(\partial_s) = -d\psi \circ (h_1^* j_0)(\partial_t) = d\psi(\partial_s)$$

by definition of a Stein manifold, and of the complex coordinates (s,t). Hence we have :

$$J \circ du(\partial_s) = d\psi(\partial_s)R_\alpha + d(h \circ h_1)(\partial_t) - d\psi(\partial_s)R_\alpha + d\psi(\partial_t)\partial_a$$

Now compute :

$$du \circ j(\partial_s) = du(\partial_t) = d\psi(\partial_t)\partial_a + d(h \circ h_1)(\partial_t) = J \circ du(\partial_s)$$

which proves that u verifies the Cauchy-Riemann equations.

□

This concludes the lifting of a single rigid page P. Now we can apply this process to define a α-compatible almost complex structure on all rigid pages at the same time, and then extend it (by contractibility of the bundle of compatible almost complex structure) as a smooth structure on the whole space $\mathbb{R} \times M$. The only thing one has to check before extending it is that it is actually smooth on an orbit asymptotic to several rigid pages (e.g a broken binding component).

Recall that on a rigid page in a local model, the almost complex structure is given by

$$\begin{cases} J\partial_a = R_\alpha = \frac{1}{D}(g'\partial_\theta - f'\partial_\phi) \\ J\partial_\rho = \frac{1}{D}(-g\partial_\theta + f\partial_\phi) \end{cases}$$

The first equation $J\partial_a = R_\alpha$ is indeed smooth on $\rho = 0$. For the second one, notice that f and g can be extended smoothly on $\mathcal{U}_i$ by writing $\alpha = fd\theta + gd\phi + hd\rho$ on the whole neighborhood ; moreoever (ρ, ϕ, θ) are smooth coordinates on $\mathcal{U}_i$, hence the almost complex structure is smooth at $\rho = 0$ as wanted.

The construction presented in this first part gives us a α-compatible almost complex structure $J \in \mathcal{J}(\alpha)$, or $J \in \mathcal{J}(\alpha'_i)$, such that the rigid pages (up to isotopy, which does not change the global structure of the broken book decomposition) are lifted to a 1-parameter translation invariant family of J-holomorphic curves.

Remark 2.10.
1) This proposition can be applied on the rigid pages, either for α directly in the case of the exact model, or for the different contact forms obtained by interpolation, as the construction was made so that the Reeb flow is always transverse to the rigid pages.
2) Note that in general we have no control on the Fredholm index of the holomorphic curves we construct with this method. This is why in the case of the exact model, we need additional assumptions on the Conley-Zehnder indices of the binding components for α, otherwise we have to interpolate to have the right asymptotic behavior for the Reeb flow.
3) We can also apply this proposition on regular pages. As we will see in the next section, this will give embedded index-2 regular curves.

□

2.3 Generalization in dimension 5

In this subsection we explain how to generalize the construction done before in the case of a 5-dimensional contact manifold supported by an open book decomposition. This will allow us to study the 5-dimensional Weinstein conjecture in part **??**. We will use a model very similar to the one used in dimension 3. This idea is that one can lift the 4-dimensional page of a 5-dimensional open book as asymptotically cylindrical complex surfaces in the symplectization, such that the induced complex structure on these lift is the initial compatible almost complex structure on the Weinstein manifold. In this way, any holomorphic curve living in the initial 4-dimensional Weinstein domain will be lifted to a curve in the 6-dimensional symplectization. It will also allow us to apply arguments from higher dimensional intersection theory, as in [**roux_cas_nodate**], [**moreno_holomorphic_2019**]. The model that follows is essentially a summary of [**wendl_open_2009**] and its adaptation in dimension 5 in [**roux_cas_nodate**], part 3.3.

Let (M^5, ξ) be a closed 5-dimensional manifold with contact structure supported by an open book decomposition. We will denote by Γ for the binding, W for a page, which corresponds to some Weinstein domain with an exact symplectic form $\omega = d\beta$, and is also a strong symplectic filling of (Γ, ξ_Γ). We will use coordinates a, ϕ for the symplectization direction and the fibration over $\mathbb{S}^1$ of the open book respectively. Finally, the monodromy corresponds to the data of an exact symplectomorphism of $(W, \omega = d\beta)$ which we denote by Ψ. Consider a neighborhood $\Gamma \times \mathbb{D}^2$ of Γ with coordinates (x, ρ, ϕ). We denote by λ a contact form on Γ such that $\xi_\Gamma = \mathrm{Ker}(\lambda)$.

Definition 2.2. We define a stable hamiltonian structure on M, (α_0, ω_0) by the following :

$$\begin{cases} d\phi \text{ for } \rho \geq 1 - \delta \\ f\lambda + gd\phi \text{ for } \rho \leq 1 - \delta \end{cases} \tag{2}$$

where $\delta > 0$ is a small constant, and with f, g verifying similar conditions to those in lemma **??** with the last condition replaced by:

- There exists $\eta > 0$ such that: $-\frac{g'(\rho)}{f'(\rho)} \leq \frac{1}{b} + \eta \in \mathbb{R}\backslash\mathbb{Q}$ for $\rho \leq 1 - \delta$, with equality for $\rho \leq \frac{2}{3}$.
- For $\rho \geq \frac{2}{3}$, we have: $g(\rho) \geq \frac{2}{3}$ and $f(\rho) \leq \frac{b}{3}$.

. The taming 2-form ω_0 is defined by:

$$\begin{cases} d\alpha_0 \text{ for } \rho \geq 1-\delta' \\ d(\tilde{f}\lambda) \text{ for } \rho \in [1-\delta', 1+\delta] \\ \epsilon_0 \omega \text{ for } \rho > 1+\delta \end{cases} \tag{3}$$

with $\delta' < \delta$, $\epsilon_0 > 0$ small constants, and $\tilde{f}$ a decreasing map which coincides with f for $\rho \leq 1-\delta'$ and with $\epsilon_0 e^{1-\delta-\rho}$ for $\rho \geq 1-\delta$.

These new conditions on f and g allow this time to ensure that the orbits in $\Gamma \times \{0\}$ are non-degenerate, and that they have smaller action than the other orbits created by this model, while keeping a control on Conley-Zehnder indices as we will see in part **??**.

This stable hamiltonian structure is contact for $\rho \leq 1-\delta$, and will allow us to define easily an almost complex structure such that the holomorphic curves living in the completion $\widehat{W}$ can be lifted to the 6-dimensional symplectization. Now we need to perturb this stable hamiltonian structure to a contact structure. We will write $\Psi^*\beta = \beta + dF$ by definition of an exact symplectomorphism, and we can require F to be equal to a constant C for $\rho \leq 1+\delta$.

Definition 2.3. We define a family of perturbations of (α_0, ω_0) which we denote $(\alpha_\epsilon, \omega_\epsilon)$, such that α_ϵ is contact for all $\epsilon > 0$.

$$\begin{cases} \epsilon[\beta + d(F\phi)] + d\phi \text{ for } \rho \geq 1-\delta \\ f_\epsilon \lambda + g_\epsilon d\phi \text{ for } \rho \leq 1-\delta \end{cases} \tag{4}$$

where $f_\epsilon = \frac{\epsilon}{\epsilon_0}\tilde{f}(\rho) + (1-\frac{\epsilon}{\epsilon_0})f(\rho)$, $g_\epsilon = g(\rho) + \epsilon C\tau(\rho)$, with τ increasing, being 0 near $1-\delta''$ and 1 near $1-\delta'$, $\delta'' < \delta'$.
The taming 2-form ω_ϵ is defined by:

$$\begin{cases} d\alpha_\epsilon \text{ for } \rho \geq 1-\delta' \\ d(\tilde{f}\lambda) \text{ for } \rho \in [1-\delta', 1+\delta] \\ \epsilon_0 \omega \text{ for } \rho > 1+\delta \end{cases} \tag{5}$$

Finally, by Giroux-Mohsen correspondance, the contacts structures $\xi_\epsilon = \mathrm{Ker}(\alpha_\epsilon)$ are isotopic to the original one on M.

Consider an almost complex structure on $\widehat{W}$, which is λ-compatible near the end, and ω-comatible everywhere. We can define an explicit almost complex structure adapted to (α_0, ω_0), esentially in the same way as before, which will allow us to lift J_0-holomorphic curves as J-holomorphic curves in dimension 6.

Definition 2.4. Let J be the almost complex structure on M defined by :
For $\rho \geq 1+\delta$:

$$\begin{cases} J_{|TW} = J_0 \\ J\partial_a = \partial_\phi \end{cases} \tag{6}$$

For $\rho \leq 1+\delta$:

$$\begin{cases} J\partial_a = R_\alpha = \frac{1}{D}(g'R_\lambda - f'\partial_\phi) \\ J\partial_\rho = \frac{1}{B}(-gR_\lambda + f\partial_\phi) \\ J = J_0 \text{ on } \xi_\Gamma \end{cases} \tag{7}$$

where, as mentioned before, B is such that $B \sim \rho$ near $\rho = 0$, and $B \equiv 1$ away from $\rho = 0$.

A crucial point of this construction is that, as we did in dimension 3, we will be able to lift the hypersurfaces $\{\phi = Cst\}$ as holomorphic hypersurfaces in $\mathbb{R} \times M$. Indeed, we have the following, which can be proved using the exact same method as in proposition **??**.

Proposition 2.11. For any $\phi \in [0,1]$, we can construct a 1-parameter family of holomorphic hypersurfaces in $\mathbb{R} \times M$, denoted by $\hat{P}^a_\phi$ in the following way: for $\rho \geq 1+\delta$, $\hat{P}^a_\phi$ is the page corresponding to ϕ and height a in the symplectization. Then extend this using the flow of $V := f\partial_a + B\frac{f'}{D}\partial_\rho$ on $\hat{P}^a_\phi \cap \{\rho = 1+\delta\}$, where $V = -JR_\lambda$ by definition.

We sum up in the next proposition the results we will need on these holomorphic hypersurfaces. This is based on [**roux_cas_nodate**] and [**moreno_holomorphic_2019**] ; see also [**dugast_sutured_2026**] for other details.

Proposition 2.12. The family of hypersurfaces $\hat{P}^a_\phi$ verifies the following properties:

1. $(\hat{P}^a_\phi, J_{|\hat{P}^a_\phi})$ is biholomorphic to $(\widehat{W}, J_0)$; in particular the coordinate of the Liouville flow in $\widehat{W}$ coincides with the coordinate given by the flow of V is the previous construction.
2. If $u : \dot{\Sigma} \to R \times M$ is an asymptotically cylincrical J-holomorphic map, with only **positive** ends and all asymptotic orbits in (Γ, λ), then u is entirely contained in a $\hat{P}^a_\phi$ for some a, ϕ.

We will use this model in part **??** to prove the Weinstein conjecture in dimension 5. To complete what we just did, it will also be necessary to have a more explicit description of the Cauchy-Riemann operator associated to J-holomorphic curves, and to compute the index of such curves. In the next subsection we adapt this model so that the almost complex structure J takes a *simple* form, which will be important to get a precise asymptotic control on the end of the holomorphic curves we will consider.

2.4 Simple tamed almost complex structures

We give in this section a different type of almost complex structure which has a simpler form near the end, which will allow us to control precisely the asymptotic behavior of the lifted rigid pages. This is very close to the notion of L-simple almost complex structures used by Bao-Honda [**bao_definition_2018**],[**bao_semi-g** with the same purpose to have simple asymptotic expressions.

First we start by changing the local form of the almost complex structure near broken binding components in the 3-dimensional manifold M. We recall first the definitions from [**bao_definition_2018**],[**bao_semi-global_** that we will need.

Definition 2.5. An almost complex structure J on $\mathbb{R} \times M$ is α-tamed if the following is verified :

1. J is $\mathbb{R}$-invariant.
2. $J\partial_a = h(x)R_\alpha$ where h is a smooth positive function on M.
3. $J\xi' = \xi'$ for ξ' oriented plane field on which $d\alpha_{|\xi'}$ is symplectic, and $d\alpha(v, Jv) > 0$ for $v \in \xi'$.

The interesting feature of these structures is that one does not need to take $\xi' = \xi$, and can in fact take ξ' to be integrable for instance; while a compactness result still holds for holomorphic curves associated with tame almost complex structures.
One crucial fact that makes these structures convenient to use is the following :

Theorem 2.13. *([**bao_definition_2018**])*
*Suppose J is α-tame. Then the SFT compactness result of [**bourgeois_compactness_2003**] still holds.*

Now in general the usual theory of moduli spaces of such holomorphic curves may fail because of the lack of control of the asymptotics of such holomorphic curves. Namely, we need the linearized Cauchy-Riemann operator to be asymptotic to an asymptotic operator with "good properties" on the ends.

This is solved by Bao and Hoda by introducing the notion of "L-simple" structure, which ensures a sufficiently simple form for the $\overline{\partial}$-operator so that we can compute explicitly the associated asymptotic operators, and still have the desired properties. In fact, in their setup, the Cauchy-Riemann operator becomes linear near the ends, hence one can easily read the asymptotic operator from this. Our five-dimensional setup will be slightly different and not strictly speaking "L-simple", but will be close enough so that the method still works. We start by reviewing explicitly our structure in dimension 3, which corresponds to a L-simple setting.

Definition 2.6. Consider γ an hyperbolic orbit corresponding to a broken binding component. We will only consider the interpolation model here and suppose that α can be written as $(b+\epsilon(x^2-y^2))d\theta+\frac{1}{2}(xdy-ydx)$ near γ, and the rigid pages are $\phi\in\{0,\pi/2,\pi,-\pi/2\}$. Define J_0 such that :

$$\begin{cases} J_0\partial_a=\partial_\theta-2\epsilon(y\partial_x+x\partial_y) \\ J_0\partial_x=\partial_y \end{cases}$$

Here we took $h=b$ in order to simplify the notations. We will explain in the proof of proposition **??** how to interpolate between the almost complex structure defined before and this new one on rigid pages, near hyperbolic orbits.

Proposition 2.14. In such a neighborhood, the Cauchy-Riemann equations are linear, and we can compute explicitly the asymptotic operator A, and its eigenvalues and eigenfunctions. In particular, any holomorphic curve with an end at γ admits a Fourier expansion in the basis of eigenfunctions for A.

Proof. Denote by j_0 the standard complex structure on $\mathbb{C}$. The Cauchy-Riemann equations in our model become: for $u(s,t)=(a(s,t),\theta(s,t),\eta(s,t))$:

$$\begin{cases} \overline{\partial}(a,\theta)=0 \\ \partial_s\eta+j_0\partial_t\eta+S\eta=0 \end{cases}$$

where the first equation allow us to assume that $a=s,\theta=t$ and with :

$$S=2\epsilon\begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix}$$

Hence the normal linearized Cauchy-Riemann operator near the end is :

$$\begin{aligned} D_u^N: \quad & W^{k,p}(N_u)\to W^{k-1,p}(\overline{Hom}(\dot{\Sigma},N_u)) \\ & \eta\mapsto\partial_s\eta+j_0\partial_t\eta+S\eta \end{aligned}$$

where we looked at the normal Cauchy-Riemann operator on the generalized normal bundle N_u, as we know that regularity of the full linearized Cauchy-Riemann operator is equivalent to the regularity of the linearized Cauchy-Riemann operator. Now this formula tells us that $D_u^N=\partial_s\eta-A\eta$ where $A=-j_0\partial_t-S$ is the asymptotic operator to which D_u^N is asymptotic.
We can actually compute the eigenvalues and eigenfunctions of A. We will only be interested in the extremal eigenvalues. A short computation, and taking into account periodic conditions, shows that: $\lambda_{\pm1}=\pm2\epsilon$ and : $f_1(t)=(x_0,0)$ and $f_{-1}(t)=(0,y_0)$.
Now we now that our Cauchy-Riemann operator is asymptotic to an explicit asymptotic operator, this is all that is necessary in order to use the usual analysis and show that it is Fredholm, and get the usual regularity results on moduli spaces. Moreover, the Fourier expansion we have for the ends is essentially all that is necessary in order to use Siefring's intersection theory [**siefring_intersection_2011**], hence we can make use of this theory as usual even for such models near the ends. See also [**zhou_unknottedness_2025**], part 2, for more details on this. □

Proposition 2.15. If P is a page with an end on an hyperbolic orbit γ around which we use a simple tame almost complex structure as defined above, then it is still possible to lift P as a 1-parameter family of holmorphic curves in the symplectization, such that the asymptotic representative of P at γ is an extremal eigenfunction of A (i.e associated with the eigenvalues 2ϵ or -2ϵ).

Proof. We first have to lift an end near a hyperbolic orbit with a model as defined above **??**, and then to show that the method used in the proof of proposition **??** in order to extend the lift to the rest of the page still works.

Consider a neighborhood $\mathcal{U}$ of γ with coordinates, contact form and simple almost complex structure as above. In order to lift the page, it suffices to notice that the Cauchy-Riemann equations in this model are already linear, and that we already exhibited special solutions which corresponds exactly to the rigid pages, namely $f_{\pm 1}$. More precisely, we have that :

$$\begin{aligned} u_- : \quad & \mathbb{R}_+ \times \mathbb{S}^1 \to \mathbb{R} \times \mathcal{U} \\ & (s,t) \mapsto (s,t,e^{2\epsilon s}x_0, 0) \end{aligned}$$

and

$$\begin{aligned} u_+ : \quad & \mathbb{R}_+ \times \mathbb{S}^1 \to \mathbb{R} \times \mathcal{U} \\ & (s,t) \mapsto (s,t,0,e^{-2\epsilon s}y_0) \end{aligned}$$

are holomorphic parametrization of the rigid negative ends and rigid positive ends respectively, with the choice of page in a pair given by the sign of x_0 or y_0. Indeed, one can verify directly that $\overline{\partial} u_\pm = 0$, or write for instance for u_- : $u_-(s,t) = \exp_{u_\gamma}(h(s,t))$ where u_γ is the trivial cylinder over γ, the exponential is the Riemannian exponential given by the euclidean metric in the coordinates chosen, and h is an *asymptotic representative* of u_- given by : $h(s,t) = e^{2\epsilon s}(x_0, 0)$. Here we recognize $(x_0, 0)$ as an eigenfunction for the eigenvalue -1 for A, and 2ϵ is the asymptotic decay rate of u_-. The same is true for u_+ which has asymptotic decay rate -2ϵ. Hence the rigid pages are lifted near the ends, and their asymptotic representative are extremal eigenfunctions of A as wanted. In particular, this implies that their Fourier decomposition in the basis of eigenfunctions of A is trivial.

In order to lift the rest of the page, we interpolate our simple model almost complex structure with the one defined in lemma **??** on rigid pages. Namely, we want to interpolate J given by : $J\partial_a = R_\alpha$, $J\partial_\rho = \frac{1}{b\rho}(-\frac{\rho^2}{2}\partial_\theta + (b + \epsilon(x^2 - y^2))\partial_\phi)$ and J_0 given by $J_0\partial_a = bR_\alpha$, $J_0\partial_\rho = \frac{1}{\rho}\partial_\phi$. The constraints on this interpolation is that the almost complex structure must remain α-tamed, and the holomorphic lifts of rigid pages must be well-defined. We see first that going from $J\partial_a = R_\alpha$ to $J\partial_a = bR_\alpha$ directly verifies this two conditions. Hence we only need to interpolate from $J\partial_\rho$ to $J_0\partial_\rho$ (this amounts to interpolate between ξ and $\langle \partial_x, \partial_y \rangle$).
In order to do this, notice that if we take $\tau = \tau(\rho)$ which varies from 0 to 1 on the interpolation domain as before and $\{J_\tau\}$ the almost complex structures interpolating, then if one has $J_\tau\partial_\rho = h_1\partial_\phi + h_2\partial_\theta$ and $\alpha = fd\theta + gd\phi$, then $d\alpha(\partial_\rho, J_\tau\partial_\rho) = (\partial_\rho f)h_2 + (\partial_\rho g)h_1$, hence we only need to verify that this quantity is always positive. Moreover, on rigid pages, we have : $\partial_\rho g = \rho$ and $\partial_\rho f = \pm 2\epsilon\rho$ so this condition is actually : $Q := \pm 2\epsilon\rho h_2 + \rho h_1 > 0$ where the sign is corresponds to the sign of the end of the rigid page.
For J_0, we have $h_1 = \frac{1}{\rho}$ and $h_2 = 0$ so this is indeed verified, and for $J_1 := J$, we have (on the rigid pages): $h_1 = \frac{b \pm \epsilon\rho^2}{b\rho}$ and $h_2 = \frac{-\rho}{2b}$, so $Q = 1$ is also positive. An easy way to interpolate is then to consider: $h_1^\tau = \frac{1}{\rho} \pm \frac{\epsilon\rho\tau}{b}$ and $h_2^\tau = -\frac{\rho\tau}{2b}$, which gives $Q = 1$ at all time.

We finally remark that during the interpolation, we can still lift the rigid pages. Indeed, after a short computation we see that : $J_\tau\partial_\theta = (-1 \mp \frac{\epsilon\rho^2\tau}{b})\partial_a \mp 2\epsilon\rho\partial_\rho$. This expression implies that we can use the method of proposition **??** in order to lift the positive/negative end of rigid pages. It coincides with the lift studied above by continuation principle, as ∂_θ is in the tangent space of all the curves defined. This proves that we can interpolate the almost complex structure on the pages, hence that we can lift the rigid pages as we wanted. □

Remark 2.16. This model also gives us a collection of holomorphic discs corresponding to the coordinates (x, y) for all values of a and θ. This may prove useful in part **??** for intersection arguments.

Now that we've established the 3-dimensional model, we can use this to get a simple 5-dimensional model. Consider as before (M^5, ξ) supported by an open book decomposition. In the previous subsection, we defined**??** a suitable stable hamiltonian structure (α_0, ω_0) and a compatible almost complex structure J. The purpose here is to simplify this almost complex structure in the setting of simple almost complex structures. We will use coordinates (x_1, y_1) or (ρ_1, ϕ_1) for a neighborhood of a binding component γ inside Γ, and coordinates (x_2, y_2) or (ρ_2, ϕ_2) for the coordinates of the 5-dimensional open book.

Lemma 2.17. We can define an α_0-tame almost complex structure J' as follows: for $\rho \geq 1-\delta$, take J' to be J. For $\rho \leq 1-\delta'$, with $\delta' > \delta$, take J' to be such that :

$$\begin{cases} J'\partial_{x_2} = \partial_{y_2} \\ J'_{|\xi_\Gamma} = J_{0|\xi_\Gamma} \\ J'\partial_a = R_\lambda - \frac{f'}{g'}\partial_{\phi_2} \end{cases} \tag{8}$$

Finally, we interpolate on $[1-\delta', 1-\delta]$.

Remark 2.18. Note that near broken binding components, J_0 does **not** preserve ξ_Γ.

Remark 2.19. The results about tame almost complex structures mentioned before still hold in this context as (α_0, ω_0) is a contact structure near Γ.

Remark 2.20. Near $\Gamma \times \{0\}$, $\alpha_0 = f((1+Q_1)d\theta + \frac{1}{2}(x_1dy_1 - y_1dx_1)) + gd\phi_2$, with Q_1 quadratic. By choosing $g = r^2/2$ near the binding (which can be done up to a few adaptations on the conditions on f and g), we can arrange that $\alpha_0 = f(1+Q_1)d\theta + f\beta_1 + \beta_2$ where $\beta_i = \frac{1}{2}(x_idy_i - y_idx_i)$. This does not correspond to the exact definition of a L-simple form, even though we will show in the following computations that everything works the same.

Proof. It is straightforward to check that for $\rho \leq 1-\delta'$, J is α_0-tamed.
The only thing that remains to be verified is that we can interpolate between J and J'. We use the same method we used in dimension 3. We have : $J\partial_{\rho_2} = \frac{1}{B}(-gR_\lambda + f\partial_\phi)$, and $J'\partial_{\rho_2} = \frac{1}{\rho_2}\partial_{\phi_2}$.
Similar than in the proof of proposition **??**, we have $\alpha_0 = f\lambda + gd\phi$, so $d\alpha(\partial_{\rho_2}, J_\tau\partial_{\rho_2}) = f'(\rho_2)h_2 + g'(\rho_2)h_1$ for an interpolating family $\{J_\tau\}$, and $J_\tau\partial_{\rho_2} = h_1\partial_{\phi_2} + h_2R_\lambda$. For $\rho \geq 1-\delta$, $h_1 = \frac{f}{B}$ and $h_2 = -\frac{g}{B}$, so $f'(\rho_2)h_2 + g'(\rho_2)h_1 = \frac{D}{B} > 0$. For $\rho \leq 1-\delta'$, we have : $f'(\rho_2)h_2 + g'(\rho_2)h_1 = \frac{g'(\rho_2)}{\rho_2} > 0$. Now interpolate by taking : $h_1(\rho_2) = \frac{1}{\rho_2} + \frac{f\tau}{B}(\rho_2)$ and $h_2(\rho_2) = -\frac{g\tau}{B}(\rho_2)$ with $\tau(1-\delta') = 0$ and $\tau(1-\delta) = 1$ as usual. We then get : $f'(\rho_2)h_2 + g'(\rho_2)h_1 = \frac{g'(\rho_2)}{\rho_2} + \tau(\rho_2)\frac{D}{B} > 0$. This concludes on the interpolation, and the definition of J. ☐

Remark 2.21. We can again define the complex hypersurfaces $\hat{P}^a_\phi$ by the following computation: $J'R_\lambda = -\partial_a - \frac{f'}{g'}\rho_2\partial_{\rho_2}$, which allows to define complex subspaces as f' and g' only depend on ρ_2. We will see that the results of **??** remain valid for this new almost complex structure.

It remains therefore to study what happens when $\rho_2 \to 0$ for the Cauchy-Riemann equations, namely to establish the properties for asymptotic operators and holomorphic curves similar to what we did in dimension three. Before doing this, we recall that in the setup we defined, if one has an holomorphic curve asymptotic to a broken binding component in Γ, then it is immersed asymptotically near its end, and its normal bundle can be decomposed as $N_1 \oplus N_2$ with $N_1 = \langle \partial_{x_1}, \partial_{y_1} \rangle$ and $N_2 = \langle \partial_{x_2}, \partial_{y_2} \rangle$.
We then have the following proposition, which is analogous to proposition **??**.

Proposition 2.22. Consider an end in $\{\rho_2 \leq 1-\delta'\} \cap \{\rho_1 \leq 1-\delta\}$ where (ρ_1, ϕ_1, θ) correspond to coordinates for a simple model around a broken binding component $\gamma \subset \Gamma \times \{0\}$, with action b. Then the Cauchy-Riemann equations for such an end are linear, and the normal Cauchy-Riemann operator is asymptotic to

$$A = -j_0\partial_t - \begin{pmatrix} -2\epsilon/b & 0 & 0 & 0 \\ 0 & 2\epsilon/b & 0 & 0 \\ 0 & 0 & f'/g' & 0 \\ 0 & 0 & 0 & f'/g' \end{pmatrix}$$

In particular, the ends of rigid curves in $\Gamma \times \{0\}$ corresponds to extremal eigenfunctions for A_γ.

Proof. We proceed as in proposition **??** and start by rewriting the Cauchy-Riemann equations for an end of the form : $u(s,t) = (a(s,t), \theta(s,t), \eta_1(s,t), \eta_2(s,t))$ where $\eta_1 = (x_1, y_1)$ and $\eta_2 = (x_2, y_2)$. Near $\{\rho_2 = 0\} \cap \{\rho_1 = 0\}$, the almost complex structure is:

$$\begin{cases} J'\partial_{x_i} = \partial_{y_i} \text{ i = 1,2} \\ J'\partial_a = \frac{1}{b}\partial_\theta - \frac{2\epsilon}{b}(y_1\partial_{x_1} + x_1\partial_{y_1}) - \frac{f'}{g'}\partial_{\phi_2} \end{cases}$$

hence the Cauchy-Riemann equations become:

$$\begin{cases} \overline{\partial}(a, b\theta) = 0 \\ \partial_s\eta_1 + j_0\partial_t\eta_1 + S_1\eta_1 = 0 \\ \partial_s\eta_2 + j_0\partial_t\eta_2 + S_2\eta_2 = 0 \end{cases}$$

with $S_1 = \begin{pmatrix} -2\epsilon/b & 0 \\ 0 & 2\epsilon/b \end{pmatrix}$ and $S_2 = \begin{pmatrix} f'/g' & 0 \\ 0 & f'/g' \end{pmatrix}$. Hence here again we can take $(a,\theta) = (s,t)$, and the normal Cauchy-Riemann operator is linear, hence $D^N = \begin{pmatrix} D_1 & 0 \\ 0 & D_2 \end{pmatrix}$ with $D_i = \partial_s - A_i$, and $A_i = -j_0\partial_t - S_i$. Hence D^N is asymptotic to $A = \begin{pmatrix} S_1 & 0 \\ 0 & S_2 \end{pmatrix}$. An important point here is that we took $\frac{f'}{g'}$ to be constant for ρ_2 small enough, otherwise these equations would not be linear. Recall that we chose: $\frac{f'}{g'} = -\frac{b_0}{1+\eta b_0}$ for ρ_2 small enough, with η and b_0 small positive constants. In what follows, we will denote $c := -\frac{b_0}{1+\eta b_0}$. We already discussed about the eigenfunctions and eigenvalues of A_1, and one could also do this for A_2, leading to a Fourier expansion for η_i, $i = 1, 2$. In particular, the largest negative eigenvalue is c, and the associated eigenspace is two-dimensional and consist of all constants maps. Now we are interested in the asymptotic of the curves in $\Gamma \times \{0\}$. For the latter, we have that $\eta_2 = 0$, hence by the computations already done in the proof of proposition **??**, we get that negative rigid curves in $\Gamma \times \{0\}$ corresponds to an eigenfunction of A, and that a pair of negative rigid curves corresponds to opposite eigenfunctions. ☐

Finally, the following proposition is the counterpart of proposition **??** in the simple case.

Proposition 2.23. If u has only positive asymptotic orbits, all inside $\Gamma \times \{0\}$, then u is contained inside a single leaf $\hat{P}^a_\phi$ for some (a, ϕ).

Proof. This is essentially just a consequence of the method used in [**zhou_unknottedness_2025**], part 2, combined with the same reasoning than in [**roux_cas_nodate**], of which we give the outline here.
It is essentially based on two observations. First, if u is as above, then we can look at its normal component $u_\perp$ with respect to H, which is some page $\hat{P}^a_\phi$. This corresponds to the η_2 component we defined in the previous proof. Then if u is not contained in H but intersects H, then we can define the intersection number $u * H = \delta(u, H) + \delta_\infty(u, H)$ where δ is the count of intersections, and δ_∞ is the count of intersections "hidden at infinity". These two counts are non-negative, hence u is either contained in a page, or there is one such that $u * H > 0$. In the latter case, because the intersection product is invariant by homotopy of smooth maps, we can homotope u so that it coincides with a

trivial cylinder for a larger than some constant. Hence we have: $u * H = \sum_\gamma u_\gamma * H$ with u_γ is the trivial cylinder over γ, and the sum is over the asymptotic orbits of u. Now to compute the intersection product, it suffices to notice that by definition (see definition 2.5 of [**zhou_unknottedness_2025**]), we have that $u_\gamma * H$ is the count of intersection of u_γ pushed in the direction of an eigenfunction for A_2 associated with the largest negative eigenvalue, with H. However, A_2 is elliptic so such an eigenfunction is constant as explained before. Now the constant can be chosen so that the pushed version of u_γ does not intersect H. This proves that $u_\gamma * H = 0$, hence $u * H = 0$ and this proves that u has its full image inside a single page. □

Remark 2.24. Note that this also proves that the only J'-holomorphic curves are the lifted J_0 holomorphic curves. Hence if we determine all the curves appearing inside a Weinstein filling of Γ, we also get all the curves in dimension 6. This will be one of the purpose of part **??**.

This concludes the definitions of the necessary almost complex structures. We are now ready to recollect the full finite energy foliation.

3 Finite energy foliations

In this section we finally use the work done in part **??** to lift the rigid pages as holomorphic curves, in order to get a full finite energy foliation on $\mathbb{R} \times M$. This is essentially based on a simple gluing argument, and some intersection theory showing that all curves are isotopic to pages. **From now on, we will assume that all pages have genus 0**. The following method works for both the exact and interpolation models. In what follows we may sometimes write "rigid pages" instead of "holomorphic lifts of rigid pages" to lighten the proofs.

3.1 Gluing rigid pages

The goal of this part is to recollect a sequence of regular pages from two rigid pages having a common asymptotic orbit, being negative for one and positive for the other. We will make use of Siefring's intersection theory in dimension 3, and refer to [**siefring_intersection_2011**] or [**wendl_contact_2019**] for reference. We will mostly use the notation of the latter, and will denote by $u * v$ the intersection product between u and v.

Lemma 3.1. Let u be a lifted rigid (positive or negative) page as given by proposition **??**. Then u is embedded, its Fredholm index is 1, and its linearized Cauchy-Riemann operator is surjective (we will say that u is Fredholm regular). Moreover, any two rigid curves u, v (potentially equal) verify : $u * v = 0$.

Proof. We start by computing the Fredholm indices of such curves. Let u be a rigid page, that we can assume without loss of generality to be positive (the computations are the same in the negative case). Then u has one positive end on an hyperbolic Reeb orbit corresponding to the broken binding component, and a certain number of ends on elliptic Reeb orbits corresponding to radial binding components.

We fix asymptotic trivialization given by the pages, i.e in the radial case the pages are $\phi = Cst$ and in the broken case the same is true for rigid pages only, which is enough to define a trivialization up to homotopy (which does not change the definition of the quantities below), which we will denote by τ.

Let us first compute the Conley-Zehnder index in this trivialization for the hyperbolic end γ. It is in fact always 0. In the case of the interpolation model, this can be computed directly as we have explicit formulas for everything. In general, consider an eigenvector of the first return map, which is hyperbolic. Then this eigenvector is in a quadrant delimited by the four rigid pages. Now notice that following the linearized flow of the Reeb vector field, the direction of the four rigid pages are constant by definition of the trivialization. Recall that the Conley-Zehnder index of the orbit is given by the number of half-turns of an eigenvector in the trivialization τ following the flow along the full Reeb orbit considered. Hence to conclude, it suffices to notice that the eigenvector cannot cross two successive directions given by rigid pages (say $\phi = \pi/2$ and $\phi = \pi$). However, the eigenvectors correspond to tangent vectors to stable and unstable manifolds of the hyperbolic orbit γ, and such tangent vectors cannot "cross" two consecutive directions given by rigid pages. If it were the case, looking close enough from γ, we would be able to find an embedded Reeb orbit inside the (un)stable manifold which we consider, crossing two successive rigid pages in the same sense, i.e for example verifying $d\phi(R_\alpha)$ on two successive rigid pages. This is not possible as the sign of rigid pages alternates. Hence the eigendirection cannot make a full half-turn, and the Conley-Zehnder index has to be zero.

In the case of an elliptic end (recall that it can be positive or negative), we already saw that the Conley-Zehnder index was ± 1 depending on the sign of the end. This was either a consequence of the hypotheses made in the theorem for the exact model, or a consequence of the choices made on the maps f and g in the interpolation in lemmas **??** and **??**.

We now show that the relative first Chern number of the normal bundle of u with respect to these trivializations is zero. This is rather straightforward as one can exhibit a non-vanishing vector field in the normal bundle. Indeed, away from binding components one can take R_α as a positive normal vector field, and near the binding components one can interpolate (by positive transverse vector fields) R_α with $-gR_\alpha + f\partial_\phi \in \xi$, given that the interpolation is made close enough from the orbit γ considered (so that g is small enough compared to f). $-gR_\alpha + f\partial_\phi$ is a transverse vector field and is constant "at infinity" (equal to $b\partial_\phi$ where $b = b(\gamma)$ is the action of the orbit) in the trivialization considered. This proves that $c_1^\tau(N_u) = 0$.
Hence we can compute the Fredholm index of our curves : denote by k_+ the number of radial positive ends, k_- the number of radial negative ends. Then the index is :

$$\mathrm{Ind}(u) = \chi(\dot{\Sigma}) + c_1^\tau(N_u) + (k_+ + k_- - 1) = (2 - k_+ - k_-) + (k_+ + k_- - 1) = 1$$

Hence the index is always 1 for this kind of curves. Note that positive and negative orbits play the same role in the computation of the index, because negative orbits have opposite Conley-Zehnder index than positive orbits.

Now we compute the intersection number $u * v$ for u, v two such (lifted) rigid curves, where u and v might be the same curve. We know that u and v only have simple ends, and we have :

$$u * v = u \bullet_\tau v - \sum_{(z,\zeta) \in \Gamma_u^\pm \times \Gamma_v^\pm} \Omega_\pm^\tau(\gamma_z, \gamma_\zeta)$$

where $u \bullet_\tau v = 0$ as it corresponds to the algebraic intersection number of u and a perturbation of v given by the trivialization τ, which gives 0 by definition of the trivialization τ. In our case, the quantities $\Omega_\pm^\tau(\gamma_z, \gamma_\zeta)$ are simply defined by:

$$\Omega_\pm^\tau(\gamma_z, \gamma_\zeta) := \alpha_\mp^\tau(\gamma)$$

if $\gamma = \gamma_z = \gamma_\zeta$ corresponds to a common end for both u and v with the same sign (i.e it corresponds to a positive or a negative ends for both); otherwise the quantity $\Omega_\pm^\tau(\gamma_z, \gamma_\zeta)$ is zero. Here the $\alpha_\pm^\tau(\gamma)$ are the extremal winding numbers in the considered trivialization. Now we know that the extremal winding numbers are always 0 in our chosen trivialization. Indeed, use the relation for any asymptotic orbit γ: $\mu_{CZ}^\tau(\gamma) = 2\alpha_\mp^\tau(\gamma) \pm p(\gamma)$ where $p(\gamma) \in \{0, 1\}$ is the parity of γ. For a hyperbolic orbit, $p(\gamma) = 0$, and $\mu_{cz}^\tau(\gamma) = 0$, hence the extremal winding numbers are both 0, and for an elliptic orbit, $p(\gamma) = 1$ and $\mu_{CZ}^\tau(\gamma) = \pm 1$, hence the extremal winding numbers are -1 and 0 for a negative end, and 0 and 1 for a positive end. In each case, the extremal winding number we are interested in for the computation of $\Omega^\tau_\pm(\gamma_z, \gamma_\zeta)$ is 0. This gives us the result we wanted, namely: $u * v = 0$.

We can now use the adjunction formula for asymptotically cylindrical curves in dimension 4, which reads as follows for our curves with only simple ends: $u * u = 2[\delta(u) + \delta_\infty(u)] + c_N(u)$, where $\delta(u)$ is the count of self-intersection and critical points, and $\delta_\infty(u)$ is the count of self-intersections "hidden at infinity". By the previous computations, this gives : $\delta(u) + \delta_\infty(u) = 0$, and because both quantities are non-negative, we get $\delta(u) = 0$, which implies that u is embedded.

Now we look at the normal first Chern number (see [**wendl_automatic_2009**]) $c_N(u)$, defined by: $2c_N(u) := \mathrm{ind}(u) - 2 + 2g + \#\Gamma_0(u)$ where g is the genus (here 0), and $\#\Gamma_0(u)$ is the number of even Reeb orbits in the asymptotics of u, here 1. Hence we get $c_N(u) = 0$.

Finally, we can show that u is Fredholm regular. We use the automatic transversality criterion from [**wendl_automatic_2009**]. We have: $\mathrm{ind}(u) = 1$, $c_N(u) = 0$ and $\delta(u) = 0$. Hence we get: $\mathrm{ind}(u) > c_N(u) + Z(du)$ where $Z(du)$ is the count of critical points of u, so the criterion is verified and u is regular. This concludes the proof of the theorem. □

Remark 3.2. The computation of the Conley-Zehnder index in the hyperbolic case shows in fact that the index of any over of the same orbit is also zero.

Remark 3.3. The interpretation of the Conley-Zehnder indices as some number of turns or half-turns for the first return map may not be valid *a priori* for the simple models detailed in subsection **??**. However, in these cases we can make explicit computations: namely, in the hyperbolic case, both extremal eigenfunctions are constant so they have the same winding number 0 which implies $\mu^\tau_{CZ}(\gamma) = \alpha^\tau_-(\gamma) + \alpha^\tau_+(\gamma) = 0$ where $\alpha^\tau_\pm(\gamma)$ corresponds to the extremal winding numbers. This is all we need for now, and we will make the remaining computations in part **??** for the indices of curves in the 6-dimensional symplectization.

Remark 3.4. As noticed at the end of the last part, we can apply exactly the same construction for regular pages, considering only asymptotic radial binding components. In that case, the very same index computation raises : $\mathrm{Ind}(u) = 2$. Hence our construction allows us in fact to lift an arbitrary finite number of curves of the broken book, rigid or regular, as a 1-parameter family of holomorphic curves in the symplectization with Fredholm index 1 or 2.

Lemma 3.5. For any pair of rigid pages (u_-, u_+), we can glue them and get a sequence u_n breaking onto (u_-, u_+). Moreover, any such sequence verifies that the curves u_n are embedded Fredholm regular index-2 curves.

Proof. The first part is just a consequence of the previous lemma, as both components of the building are Fredholm regular, which allow to "solve the gluing problem" by the Fredholm implicit function theorem. Note that by additivity of the Fredholm index, we indeed get a 1-parameter family of index-2 curves converging to (u_-, u_+).

The embeddedness follows from the adjunction formula for asymptotically cylindrical curves and the homotopy invariance of the different quantities used in this formula.
Indeed, choose some $u = u_n$ for some n, and write : $u * u = 2[\delta(u) + \delta_\infty(u)] + c_N(u)$ which is the adjunction formula for curves with simple ends, where $\delta(u)$ and $\delta_\infty(u)$ are again as before a non-negative count of self-intersections, critical points and intersections "hidden at infinity". Now we also have, by definition : $2c_N(u) = \mathrm{ind}(u) - 2 + 2g = 0$ if u has genus 0. Moreover, by continuity of the intersection product extended to holomorphic buildings (see the beginning of the next section), we have $u * u = 0$. This implies $\delta(u) = \delta_\infty(u) = 0$, hence u is embedded.

Finally, note that these curves are also Fredholm regular as we showed $c_N(u) = 0$, hence $\mathrm{ind}(u) > c_N(u)$ and again because u is embedded, the automatic transversality criterion[**wendl_automatic_2009**] gives us the Fredholm regularity of the curves. □

We finally give a last result which illustrates the control we have on sequences of curves degenerating to rigid pages.

Lemma 3.6. If a sequence of planar curves $\{u_n\}$ converges to a building $(u_-, ..., u_+)$ in the SFT topology, then it converges to (u_-, u_+).

Proof. Suppose we have such a sequence, then because the curves have genus 0 the intermediate levels can only be a succession of cylinders with extremities at the hyperbolic Reeb orbit. This is not possible except if they are all trivial cylinders, as the ends of u_- and u_+ are simple so they are not nontrivial branched covers of trivial cylinders, and all curves which are not covers of trivial cylinders have positive $d\alpha$-energy, which is not the case here by Stoke's theorem because the positive and negative ends are the same. □

Remark 3.7. This result also holds in the simple setting as we also have $u^* d\alpha > 0$ in that case.

Remark 3.8. The gluing method gives us at least one way to glue our rigid pages, but one could wonder if there is actually more than one. Using the fact that we have index 2 curves in symplectization, one could use arguments similar than [**hutchings_gluing_2007**], part 7, in order to show that the gluing map is bijective and hence that there is indeed a single way to glue. However, more straightforward intersection theoretic arguments in next part will allow us to get this result in a simpler way.

3.2 Recollecting the full foliation

In this last part we finally lift the full foliation. All the curves will be denoted with the notation $u = (a, f)$, with a the symplectization coordinate and f the projection on M.

First we recall some notions from [**wendl_contact_2019**], section C.5, on the intersection product of holomorphic buildings. We will denote in this context $u \odot v$ for the concatenation of two levels u and v with corresponding ends to produce a new holomorphic building.

Definition 3.1.
- If $u = u_1 \odot ... \odot u_N$ is a holomorphic building, an extension of u is a new building constructed from u by adding a certain number of intermediate levels with only trivial cylinders.
- If u, v are two holomorphic buildings, we define the intersection product $u*v$ by first taking extensions of u and v so that we can assume that $u = u_1 \odot ...u_N$, $v = v_1 \odot ... \odot v_N$ have the same number of levels, and then use the formula:

$$u * v := \sum_{i=1}^{N} u_i \bullet_\tau v_i - \sum_{(z,\zeta)\in\Gamma^+_{u_N}\times\Gamma^+_{v_N}} \Omega^\tau_+(\gamma_z, \gamma_\zeta) - \sum_{(z,\zeta)\in\Gamma^-_{u_1}\times\Gamma^-_{v_1}} \Omega^\tau_-(\gamma_z, \gamma_\zeta)$$

where the $\Omega^\tau_\pm$ are as in the proof of lemma **??**.

Proposition 3.9.
- The intersection product $u * v$ does not depend on the choice of extensions.
- It is continuous with respect to the SFT-topology.
- We have:

$$(u_1 \odot u_2) * (v_1 \odot v_2) = u_1 * v_1 + u_2 * v_2 + \sum_{(z,\zeta)\in\Gamma^+_{u_1}\times\Gamma^+_{v_1}} Br(\gamma_z, \gamma_\zeta)$$

where $Br(\gamma, \gamma') = \Omega^\tau_+(\gamma, \gamma') + \Omega^\tau_-(\gamma, \gamma')$ is the breaking contribution.

In particular, if $\gamma = \gamma'$ is a simple orbit and a binding component for a broken book decomposition, then: $Br(\gamma) = 1$ if γ is elliptic, and $Br(\gamma) = 0$ if γ is hyperbolic.
By the previous definition and proposition, it is interesting to understand the intersection number between our curves and trivial cylinders over binding components. The following lemma gives the expected answer in this direction.

Lemma 3.10. Suppose u is a lift of a rigid page as defined in the previous section, and γ is a closed Reeb orbit asymptotic to u. Let u_γ be the trivial cylinder over γ. Then $u * u_\gamma = 0$.

Proof. We simply use the definition of the intersection product as in the proof of lemma **??**. Suppose γ is a positive end for u (note that u could have several positive ends at γ if γ is elliptic, but cannot have positive and negative ends at the same time by construction of a broken book decomposition). We have:

$$u * u_\gamma = u \bullet_\tau u_\gamma - \alpha^\tau_\mp(\gamma)$$

Now $u \bullet_\tau u_\gamma$ is zero by construction, as one can push the trivial cylinder in a direction constant in the trivialization τ, such that u_γ and u do not intersect. Moreover, by the same arguments than in the proof of lemma **??**, all considered winding numbers are zero, which concludes. The same works for a negative end. □

Lemma 3.11. Take u to be the gluing of (u_-, u_+) as defined in lemma **??**, for gluing parameters large enough (i.e u is close enough from (u_-, u_+). Then the intersection number of u and any other rigid curve, or any trivial cylinder over binding orbits is zero.

Proof. Consider γ a binding orbit, and u_γ the trivial cylinder over it. Then by continuity of the intersection product, $u * u_\gamma = (u_- \odot u_+) * (\emptyset \odot u_\gamma) = u_+ * u_\gamma = 0$. The rest of the statement follows as in lemma 2.10 of [**SOBD2**], namely consider v a rigid page, and compute: $(w_1 \odot u) * (v \odot w_2)$ where w_1 and w_2 are collections of trivial cylinders over asymptotic orbits of u and v. By construction of a broken book decomposition, there cannot be negative orbits for u that are positive ends for v, hence the breaking contribution is 0 and: $u * v = u * w_2 + v * w_1$ which is zero by the first part of proof. □

Finally, we state for reference the following lemma, which is a direct computation from the definition or the adjunction formula.

Lemma 3.12. Let γ be a binding orbit, and u_γ the trivial cylinder over γ. Then if γ is radial, $u_\gamma * u_\gamma = -1$, and if γ is broken, $u_\gamma * u_\gamma = 0$.

In particular, because the breaking contribution of an radial binding component is 1, this means that we can ignore ignore intersections of trivial cylinders with themselves in the extensions we choose of the buildings u and v, as these intersections cancels with the breaking contributions of the orbits.

We now review certain properties of so-called *nicely embedded curves*, which will also be useful in part **??**. These are embedded curves verifying certain intersection theoretic conditions, expected to be verified by leaves of a finite energy foliations, which makes them particularly convenient to manipulate. This notion was first introduced in [**wendl_automatic_2009**] (with a slightly more restrictive definition), then in [**wendl_compactness_2008**],[**cioba_nicely_nodate**]. We also refer to [**SOBD2**], part 2.3, for a quick overview of the subject. The main definition is the following:

Definition 3.2. $u : \dot{\Sigma} \to \widehat{W}$ is said to be nicely embedded if it is somewhere injective, $u * u \leq 0$ and $\delta(u) = \delta_\infty(u) = 0$.

In this part, we will be particularly interested in the behavior of such curves in symplectizations. The main ingredient in that case will be an inequality on the following quantities, defined in [**hofer_properties_nodate**].

Definition 3.3. Let $\mathcal{H}$ is a stable hamiltonian structure and $J \in \mathcal{J}(\mathcal{H})$ in the symplectization, then for any J-holomorphic curve u define:

- $\mathrm{wind}_\pi(u)$ is the signed count of zeros of $\pi \circ Tu$ where $\pi : \mathbb{R} \times TM \to \xi$ is the projection parallel to the Reeb vector field and the $\mathbb{R}$-direction.
- $\mathrm{def}_\infty(u) := \sum_{z\in\Gamma} |\alpha^\tau_\mp(\gamma_z) - \mathrm{wind}^\tau(f_z)|$ where f_z is the eigenfunction appearing in the formula: $h(s,t) = e^{\lambda_z s}(f_z(t) + r(s,t))$ where λ_z is an eigenvalue for the asymptotic operator A_z, and h is an asymptotic representative of u at the puncture z.

Both terms are non-negative. Moreover, we have: $c_N(u) = \text{wind}_\pi(u) + \text{def}_\infty(u)$. In particular, $c_N(u) \geq 0$.

Remark 3.13. This result also holds in the simple setting. As all the algebraic computations work the same, the only thing to verify is that asymptotically, $\pi \circ Tu$ is non-vanishing for a holomorphic end in a simple neighborhood. This is indeed the case as we have an explicit Fourier expansion of the normal component. Hence the same arguments than in the compatible case are enough to conclude.

Using the adjunction formula and the previous notions, we obtain a reformulation of the definition of nicely embedded curves in the case of a symplectization:

Definition 3.4. $u : \dot{\Sigma} \to \mathbb{R} \times M$ is said to be nicely embedded if it is somewhere injective and: $u * u = 0$ or u is a trivial cylinder.

The following properties are direct consequences of the definitions and remarks we made, combined with automatic transversality.

Lemma 3.14. If u is a nicely embedded curve, then it verifies that: $c_N(u) \leq 0$ and $\text{ind}(u) \leq 2$. Moreover, if $u = (a, f)$ is nicely embedded inside a symplectization, then:

- $\text{wind}_\pi(u) = \text{def}_\infty(u) = 0$, hence f is an embedding, and the Reeb flow is transverse to the image of f
- $c_N(u) = 0$.
- If $\text{ind}(u) > 0$, u is Fredholm regular.
- $\text{ind}(u) \in \{0, 1, 2\}$. If u is regular, then $\text{ind}(u) = 0$ if and only if u is a trivial cylinder, and u has a single even end if it has index one and none otherwise. Moreover, in that case, u has genus 0.

In part **??**, we will also derive other constraints for certain type of nicely embedded curves in a strong symplectic filling.

Finally we give a few compactness results for moduli spaces of nicely embedded curves. These results are given in [**wendl_compactness_2008**].

Definition 3.5. A stable J-holomorphic building in $\mathbb{R} \times M$ is said to be nicely embedded if :

1. It has no nodes.
2. Any connected component is nicely embedded.
3. The projections on M of 2 distinct components are identical or disjoint.
4. Any breaking orbit is even and simply covered or even and a double cover of an odd hyperbolic orbit (in particular, in the latter case it is a bad orbit in the SFT terminology).

The following result works only in the setting of a symplectization and not in an arbitrary symplectic cobordism.

Theorem 3.15. *Suppose that (u_n) is a sequence of nicely embedded curves with $u_n : \dot{\Sigma} \to \mathbb{R} \times M$ and suppose that $u_n \to u$ where u is a stable J-holomorphic building. Then u is nicely embedded.*

Finally, we have the following fact, which is a straightforward consequence of the computations done before.

Lemma 3.16. The lifted rigid pages or the curves obtained by gluing them as in lemma **??** are nicely embedded curves.

Now that we have the necessary convergence results, we are ready for the proof of theorem **??**. First we introduce a bit of terminology for broken book decompositions. Consider (M, α) with supporting broken book decomposition. Then we will use the fact that M has a decomposition in several "sectors" delimited by 4 rigid pages. To define such a sector, take any regular page and notice than pushing it by a reparameterization of the Reeb flow allows to recollect the nearby pages, as long as they are transverse to the Reeb flow. If at some point the foliation cannot be obtained that way, it means that some part of the foliation is not transverse to the Reeb flow anymore, which can only happen at a broken binding component.
Hence if F is a regular page, we have 2 possibilities : either there is no problem of transversality whatsoever, and in this case F is a Birkhoff section, and we have a nice open book decomposition ; or we get a sector of M on which the fol.iation is $]0,1[\times M$, and which has boundary components consisting of 2 rigid pages, asymptotic to the same broken binding component with ends of opposite sign. Note that because of the local model around a broken binding component, the boundaries at $\{0\}$ and the boundaries at $\{1\}$ cannot be the same.
Finally, because any point in M belongs to a page, we get that M can be divided into a finite number of such sectors, "glued" to each other along rigid pages and binding components. In the following, we will use this notion of sectors even for the interpolation model, where the Reeb flow might not be transverse to the pages anymore near hyperbolic Reeb orbits. This is not an issue as it suffices to define the sectors using the initial Reeb flow before interpolation.

Proof. of theorem **??***:*
By what has been shown before, we will consider either the exact or interpolation model here without distinction, as in both cases (with additional hypotheses for the exact model), we can lift and glue rigid pages. When we will use arguments on pages using the flow of the Reeb vector field, it will sometimes be necessary, for the interpolation model, as above, to use instead the Reeb vector field of the initial contact form without interpolation.

First, if the broken book is in fact a genuine open book, then consider any regular page u, that we know that we can lift by remark **??**. We get a 2-parameter family of translation invariant, index-2 curves in the symplectization. As seen in the construction done before, we can choose our almost complex structure J to be generic except on u. Consider the moduli space $\mathcal{M}^u$ which corresponds to the connected component of u in the moduli space of curves with the same asymptotic and same relative homology class. All the curves inside this moduli space are nicely embedded, have index two and are transversally cut out. Consider a sequence (u_n) of curves inside this moduli space. Suppose it breaks on u_∞. Consider a breaking orbit γ. By theorem **??**, γ cannot be a binding orbit as it must be a cover of a hyperolic orbit. Hence it intersects u, so a component of u_∞ also intersects u or one of its translates, which is not possible. This implies that u_∞ has a single level which is a nicely embedded curve of index two, hence an element of $\mathcal{M}^u$, which proves that $\mathcal{M}^u/\mathbb{R}$ is a one-dimensional compact manifold, hence a circle. Moreover, the set of points of $\mathbb{R}\times(M\backslash B)$ by which passes a curve of $\mathcal{M}^u$ is open and closed, hence it is $\mathbb{R}\times M\backslash B$. Define the projection: $\pi : M\backslash B \to \mathcal{M}^u/\mathbb{R}$ that associates to each point the curve the goes through this point. This define an open book, which is isotopic to the initial one. Note that this gives in particular another proof of the result of [**wendl_open_2009**].

From now on we will consider that there is at least one broken component in the broken book decomposition, hence at least one rigid page. We make a choice of a sector S. We denote as F some regular page, and $F_-^0, F_+^0, F_-^1, F_+^1$, the four rigid pages delimiting the sector. These four pages can be lifted as $u_-^0, u_+^0, u_-^1, u_+^1$, which are J-holomorphic curves for some $J \in \mathcal{J}(\alpha)$, with a hyperbolic end γ_b^i. Denote $A = [F] = [u_-] + [u_+] \in H_2(M, \cup_i \gamma_i)$ where the γ_i are the radial binding components, which are also the asymptotic orbits of F.

Take u_-^0 and u_+^0 connected at a broken binding orbit γ_b^0; the previous part gives us (u_n) index-2 curves such that $[u_n] = A$ and $u_n \to u_\infty = (u_-^0, u_+^0)$ in the classical SFT-comptactness sense. We know that u_n are embedded curves in the symplectization, and for n large enough, we claim that their image belong to $\mathbb{R} \times S$. Indeed, suppose n is large enough and take $u := u_n$, then we can look at $u = (a, f)$ in the local hyperbolic model around γ_b. Denote as $\mathcal{U}_i$ the considered neighborhood. If $\mathrm{Im}(u) \cap (\mathbb{R} \times \mathcal{U}_i)$ is not entirely in $\mathbb{R} \times S$, then by definition of the local model, u has to intersect a lift of a rigid page, which contradicts the previous intersection considerations. Moreover, away from the breaking orbit, u has to stay "on the same side" of a rigid page, again because it cannot intersect it. This proves that f has its image inside S. Moreover, by lemma **??** that u is nicely embedded hence f is an embedding.

Now denote by $\mathcal{M}(A, J)$ the connected component of u in the moduli space of asymptotically cylindrical curves with homology class A modulo translation in the symplectization. This is a one-dimensional manifold by computation of the Fredholm index and regularity of the curves. Let us show that its boundary can be identified with the union of 2-level buildings : $(u_-^0, u_+^0) \cup (u_-^1, u_+^1)$.

First note that $\mathcal{M}(A, J)$ has non-empty boundary. If not, we could look at the evaluation map :

$$ev : \mathcal{M}_1(A, J) \to M$$

where $\mathcal{M}_1(A, J)$ is the moduli space of curves in $\mathcal{M}(A, J)$ with one marked point, modulo reparameterization. Notice that this space has compactification that we can write : $\overline{\mathcal{M}_1(A, J)} = \mathcal{M}_1(A, J) \cup (\bigcup \gamma_i)$ where we identified the asymptotic orbits of u with the trivial cylinders covering them. The evaluation map extends trivially on this space, and is smooth with degree one, hence by every point of M, we can find a curve in the homology class of u which passes through that point. This is a contradiction because u cannot cross any rigid page by intersection property. Hence $\mathcal{M}(A, J)$ is a connected 1-manifold with boundary $u_\infty^0 \cup u_\infty^1$. By the same intersection argument, any regular curve of $\mathcal{M}(A, J)$ has image inside $\mathbb{R} \times S$.

Suppose v_n is a sequence of $\mathcal{M}(A, J)$ converging to u_∞ (meaning u_∞^0 or u_∞^1). By genericity of the almost complex structure away from the rigid pages, we can assume that any somewhere injective curve is Fredholm regular, hence has positive Fredholm index if it is not a trivial cylinder. We know that v_n is a sequence of stable, nicely embedded curves because they are in the same moduli space connected component as u. We apply theorem **??**, which tells us at first that there can be no node in u_∞ (note that there cannot be bubbles either as we are in a symplectization).

Now suppose v_n breaks into a N-level holomorphic building, which is therefore nicely embedded. Then we claim that the only orbit that can be involved in the intermediate levels is γ_b^0 or γ_b^1. Indeed, v_n breaks on orbits in $\overline{S}$, which are all binding components. The previous theorem tells us that the sequence can only break on even Reeb orbits or double covers of odd hyperbolic orbits, however there are no odd hyperbolic orbits in the binding so it can only break on even Reeb orbits. Then because radial binding components are supposed elliptic and odd, we get that the breaking can only happen on γ_b^0 or γ_b^1.

If we now assume that (v_n) breaks on γ_b^0, then it cannot break on γ_b^1 as this would force u_∞ to cross u at some point which would contradict the self-intersection computation (as explained in [**wendl_contact_2019**], the intersection product extend to holomorphic buildings, and by proposition $C.6$, $0 = u * u = v_n * u \to u_\infty * u = 0$, which implies that no level intersects u). Hence (v_n) can only break on γ_b^0. We must show that the limit building is then (u_-^0, u_+^0). First look at the local model around γ_0^b. If a non-constant component of the limit building were distinct from u_+^0 or u_-^0, because it has to be in S, that means it crosses some curve u_n for n large enough. This is not possible, still by the same properties of the intersection product for holomorphic buildings. Indeed, the limit building has intersection 0 with u_n by continuity property of the intersection product, and by superadditivity this implies that the intersection product of u_n with all non-trivial components of the building is 0. Then for multiplicity reasons the only possible limit building is (u_-^0, u_+^0). The exact same reasoning holds if (v_n) breaks on γ_b^1.

Therefore we know that if the sequence (v_n) breaks, it has to be on γ_b^0 or γ_b^1 and the limit building has to be (u_-^0, u_+^0) or (u_-^1, u_+^1). Moreover it is easy to see that the 2 components of $\partial\overline{\mathcal{M}(A,J)}$ cannot be the same (S is divided in 2 parts by f, and each boundary component is in one of these parts so they cannot be the same) from which we deduce that $\partial\overline{\mathcal{M}(A,J)} = (u_-^0, u_+^0) \cup (u_-^1, u_+^1)$.

Looking at the moduli space of marked curves again, we get an evaluation map : $ev : \overline{\mathcal{M}_1(A,J)} \to \overline{S}$ which has degree one (here the degree is defined between compact manifolds with boundary). Hence it is surjective, and any point in S belong in the image of some curve (more precisely to the projection of a 1-parameter family of curves in the symplectization) of $\mathcal{M}(A,J)$. Moreover, 2 such curves are either disjoint or the same, hence exactly one curve passes through every point, and $S \simeq \mathrm{Im}(f) \times]0,1[$ (and even more : $\overline{S} \simeq \mathrm{Im}(f) \times [0,1]$ topologically, by considering $u_-^0 \# u_+^0$ and $u_-^1 \# u_+^1$).

Now we can apply this construction to all sectors in the manifold. We thus get a finite energy foliation of M as wanted. The last thing that remains to prove is that this foliation is isotopic to the initial broken book decomposition. This is fairly straightforward as we know the Reeb vector field is transverse to both foliations, except maybe in a small neighborhood around the broken binding components in the interpolation model. However, in the latter case we can look at this small neighborhood and we know that the pages and the curves are isotopic by construction, hence we can perturb slightly the Reeb vector field so that it remains transverse to both foliations on this neighborhood. Hence we will assume it is always transverse to the two foliations on a whole sector.

Chose some sector S, and take some Reeb orbit γ which is not an asymptotic orbit of any page in the sector (rigid or regular), with $\mathrm{Im}(\gamma) \subset \overline{S}$, and such that $\gamma(0)$ and $\gamma(1)$ belong to different rigid pages of $\overline{S}$. Denote as $\mathcal{F}$ the foliation of the broken book decomposition, and $\mathcal{F}'$ the foliation induced by the projection of the holomorphic curves. Then define :

$$\pi : S \to [0,1]$$

by associating to a point x the time t for which $\gamma(t) \in \mathcal{F}_x$ with $\mathcal{F}_x$ the page containing x. Because the Reeb vector field is transverse to $\mathcal{F}$, this is well defined and smooth. Consider $h : S \to \mathbb{R}$ such that $d_x\pi(R_\alpha) = h\partial_t$ with ∂_t corresponding to the $[0,1]$-coordinate. In the same way one can define π' and h' associated to $\mathcal{F}'$. Now consider the map :

$$\begin{aligned} F : \quad & S \times [0,1] \longrightarrow S \\ & (x,s) \longmapsto \phi_{\pi(x)}^{\frac{s}{h}R_\alpha} \circ \phi_{\pi'(x)}^{\frac{1-s}{h'}R_\alpha} \circ \phi_{\pi(x)}^{-\frac{R_\alpha}{h}}(x) \end{aligned}$$

This gives the isotopy we wanted. Because F is the identity on the rigid pages and on the binding orbits, we can compose all the isotopies of the different sectors to get a $\mathcal{C}^0$-isotopy between the projection of the finite energy foliation and the broken book decomposition. This isotopy can be chosen smooth away from the binding orbits up to reparametrizing the previous path γ, in such a way that π and π' are "flat" on the rigid pages. □

Remark 3.17. Note that with the terminology of [**wendl_strongly_2010**], this finite energy foliation could be neither stable nor asymptotically stable, as not all curves have index two (otherwise we are in the case of an open book) and some pages may have several ends on the same binding component.

Remark 3.18. By the same computations than in lemma **??**, we get that for any pair of lifted pages or trivial cylinders over binding orbits u, v, potentially equal, we have: $u * v = 0$ except if $u = v = u_\gamma$ with γ elliptic, in which case $u * v = -1$.

Remark 3.19. Here again the result is also true in the case of a simple almost complex structure as defined in definition **??**.

4 Spinal open books and broken book decompositions

The purpose of this section is to investigate the connection between spinal open book decompositions[**SOBD1**],[**SOBD2**], and open book decompositions. It turns out that spinal open book decompositions are exactly broken book decompositions whose negative rigid pages are all cylinders, and without negative radial components. With this correspondence at hand, we use methods coming from spinal open books to deduce certain properties for broken book decompositions.

4.1 Spinal open books

We start by recalling the basic definitions and properties of spinal open book decompositions from [**SOBD1**].

Definition 4.1. A spinal open book decomposition of M^3 is a decomposition $M = M_\Sigma \cup M_P$, the spine and the paper, with the following structure:

1. $\pi_\Sigma : M_\Sigma \to \Sigma$ with connected and oriented fibers, where Σ is a compact oriented surface with boundary, disconnected if there is more than one spine component.
2. $\pi_P : M_P \to \mathbb{S}^1$, whose fibers have connected components with non-empty boundary, such that $M_P \cap M_\Sigma$ consist of the boundaries of these fibers, which are also fibers of π_Σ.

We denote by $\boldsymbol{\pi}$ this data.

Definition 4.2. The multiplicity m_γ of π_P at a connected component of $M_P \cap M_\Sigma$ is the degree of the map:

$$\mathbf{m}_\gamma : \begin{array}{l} \gamma \subset \partial\Sigma \to \mathbb{S}^1 \\ \phi \mapsto \pi_P(\pi_\Sigma^{-1}(\phi)) \end{array}$$

This multiplicity counts the number of ends

Remark 4.1. We list here the different topological properties that are verified by open book decompositions, rational open book decomposition, spinal open book decompositions and broken book decompositions. We do not yet identify spinal open books with a subclass of broken book decompositions, hence the differences.

	OBD	ROBD	SOBD	BBD
Spine	Discs	Discs	Compact surface	Undefined
Ends	Positive	Positive	Positive	Positive or negative
Fibration over the circle	Yes	Yes	Yes (fibers may be disconnected)	No
All pages have the same topology	Yes	Yes	No	No
Several ends on the same boundary comp.	No	Yes	Yes	Yes
Several ends on different boundary comp. of the same spine comp.	No	No	Yes	-
Multiplicity for a single end	No	Yes	No	No

The three last rows correspond to three different notions of multiplicity that can be counted for ends of pages. The multiplicity of the previous definition corresponds to the first one (it counts the number of ends of a given page on the same boundary component of a given spine component).

Definition 4.3. Let $\boldsymbol{\pi}$ be a spinal open book on M. A contact form α is supported by $\boldsymbol{\pi}$ if:

1. $d\alpha$ is positive on the interior of the pages.
2. R_α is positively tangent to the fibers of $\pi_\Sigma : M_\Sigma \to \Sigma$

A contact structure ξ is supported by $\boldsymbol{\pi}$ if there exists a contact form supported by $\boldsymbol{\pi}$.

Proposition 4.2. Given a spinal open book $\boldsymbol{\pi}$ on M, there exists contact form supported by $\boldsymbol{\pi}$. Moreover, two contact structures supported by $\boldsymbol{\pi}$ are isotopic.

Adaptations of the proof of the previous proposition will be used in the next part to show similar results on broken book decompositions. It can be found in [**SOBD1**], part 2.3.

We will use the following coordinates: near the boundary of Σ, we have coordinates (s, ϕ), we parametrize a neighborhood of the boundary of M_Σ by coordinates (s, ϕ, θ), and a neighborhood of the boundary of M_P by (ϕ, t, θ). Here, ϕ corresponds to the coordinate defined by the fibration π_P and θ corresponds to the circle coordinate after identifying M_Σ with $\Sigma \times \mathbb{S}^1$.

Moreover, we define another region M_I as a neighborhood of ∂M_Σ in M_Σ and M_P. We also give here an explicit formula for a contact form supported by $\boldsymbol{\pi}$.

Definition 4.4.
Let:

- $H : \Sigma \to \mathbb{R}_+$ is a smooth map, Morse outside of $\partial\Sigma$, and near $\partial\Sigma$, H depends only on s, with $\partial_s H < 0$ except in a smaller neighborhood where it vanishes. We also ask that H has only index 1 and 2 critical points. We lift it on M_Σ by taking it constant along the fibers of π_Σ.
- σ is a Liouville form on Σ which is $me^s d\phi$ near the boundary, where m is the multiplicity defined in **??**. Moreover, we can ask that $\sigma = d(\frac{1}{H}) \circ j =$ away from the boundary, where j is a complex structure on Σ (see remark **??** below)
- $F_+ = F(\rho) + \epsilon H(F(\rho), \cdot)$, $G_+(\rho) = G(\rho) + \epsilon H(F(\rho), \cdot)$ where F, G are interpolating maps verifying certain conditions, see [**SOBD2**], 3.1.
- λ is a 1-form on M_P such that $d\lambda$ is positive on the fibers of π_P, and is $e^t d\theta$ near the boundary.
- K is a constant large enough.

Then we define the following contact form:

$$\alpha = \begin{cases} e^{\epsilon H}(\sigma + \frac{1}{K} d\theta) \text{ on } M_\Sigma \\ me^{F_+(\rho)} d\phi + \frac{1}{K} e^{G_+(\rho)} d\theta \text{ on } M_I \\ d\pi_P + \frac{1}{K}\lambda \text{ on } M_p \end{cases}$$

Remark 4.3. As stated above, we can choose $\sigma = -d\varphi \circ j$, with φ a subharmonic potential on Σ which coincides with $\frac{1}{H}$ away from $\partial\Sigma$, and is equal to e^s near the boundary. j is a complex structure which is defined near the boundary by: $j\partial_s = \frac{1}{m}\partial_\phi$. In order to find such a potential, start with a hamiltonian H as before, with the constraint that it is equal to e^{-s} near the boundary, before going to zero. Then post-compose it with a sufficiently convex function (see [**Latschev_2011**], lemma 4.1 for example). Note that this construction does not change the gradient flow lines of H, nor its critical points.

4.2 Correspondence and topological simplifications

In this section we explain how spinal open books can be identified with broken book decompositions whose negative rigid pages are all cylinders. We use this to derive a simplification result for the topological structure of broken book decompositions. A broken book will be called *simple* if all the regular pages only have positive ends, and all negative rigid pages are cylinders.

Definition 4.5. Let $\boldsymbol{\pi}$ be a spinal open book on M. Take a Hamiltonian H as in definition **??**.
Then one can extend the pages of M_P inside M_Σ by gluing it with the product of the fibers of π_Σ times the gradient flow lines of H.
The resulting foliation is a simple broken book decomposition, which we will denote by $B_H(\boldsymbol{\pi})$.

This result comes from the fact that pages of a broken book around a broken binding component behaves like the gradient flow line of a Morse function around an index-1 critical point, after taking the product with $\mathbb{S}^1$. Moreover, the negative rigid curves obtained that way are all cylinders as they are entirely contained inside the spine, and consist only of gradient flow lines by the condition on the critical point of H, hence the broken book is balanced.

Lemma 4.4. Let $\boldsymbol{\pi}$ be a spinal open book decomposition of M, and H as in the previous definition. If ξ is supported by $\boldsymbol{\pi}$, then up to isotopy, it is supported by $B_H(\pi)$.

Proof. First note that the notion of supported contact structure for a spinal open book as defined in [**SOBD1**] differs a bit from definition in the case of a broken book. Suppose ξ is supported by the spinal open book. Then by uniqueness, one can choose any hamiltonian H as in definition **??** and assume that ξ is defined by a contact form defined by the formulas :
It suffices to verify that the Reeb flow of this contact form is supported by the broken book decomposition, and by definition one just needs to check this on M_Σ and M_I. On M_Σ the Reeb flow is : $R_\alpha = e^{\epsilon \hat{H}}(1 + \epsilon\sigma(X_H)K\partial_\theta - \epsilon X_H)$ where ϵ is small and X_H is the Hamiltonian vector field for H with respect to the area form $d\sigma$ on Σ. Now this is indeed positively transverse to the pages of the broken book, or tangent to the binding : if $X_H = 0$, R_α is parallel to ∂_θ and thus follows a binding orbit, and if $X_H \neq 0$, by definition X_H is in a direction transverse to the one defined by the gradient flow line and the $\mathbb{S}^1$-coordinate, by definition of a Hamiltonian vector field. It is positively transverse because the induced area form is $dH \wedge d\theta$ on the page, and any page in this domain is positively asymptotic to a Reeb orbit induced by an index-2 critical point ; hence the natural orientation of the page is also $dH \wedge d\theta$. On M_I, the Reeb flow is collinear to:

$$-\frac{1}{m}e^{-F_+(\rho)}G'_+(\rho)\partial_\phi + Ke^{-G_+(\rho)}F'_+(\rho)\partial_\theta$$

Moreover, we have: $G'_+(\rho) = G'(\rho) + \epsilon F'(\rho)\partial_s H$, $G' \leq 0$, $F' \geq 0$, $\partial_s H \leq 0$, and we can ask that G' and $F'\partial_s H$ never vanish at the same time. This proves that $d\phi(R_\alpha)$ is always positive, so R_α is positively transverse to the pages on M_I. □

In the following, we may need the following assumption: "on negative rigid pages which are cylinders, the characteristic foliation is trivial". This corresponds more or less to the condition $\alpha(\partial_\theta) > 0$ in the definition of a Giroux form for a SOBD in [**SOBD1**], in section 2.3. Our condition is hopefully sufficient (?), one question is is it necessary ? i.e could one find another "exotic" contact structures supported by the broken book decomposition, with a different characteristic foliation on simple rigid pages? One could try first to work this out in the case of (simple) spinal open books, where the definition of Giroux form is relaxed, hence not allowing the uniqueness proof to work. It would be interesting to find another **tight** contact structure supported by the spinal/broken book.

Lemma 4.5. Let $\boldsymbol{\pi}$ be a spinal open book on M^3. Let α be a contact form such that R_α is supported by a broken book as defined above for a certain choice of hamiltonian H as before. Then $\xi = \mathrm{Ker}(\alpha)$ is supported by $\boldsymbol{\pi}$.

Proof. The proof follows roughly the method from **[SOBD1]**, lemma 2.8, used to prove the uniqueness of the contact structure supported by a spinal open book.
Take α a contact form as above, and take α_0 a Giroux form for the spinal open book decomposition. In particular, $R_{\alpha_0} = \partial_\theta$ on M_Σ. Moreover, one can arrange that $\alpha(\partial_\theta) > 0$. Indeed, this is true near binding components, and around the rigid negative pages, the characteristic foliation looks like straight lines going from one end of a cylinder to the other/ the caracteristic foliation can always be chosen to have no singularity A VERIFIER. Now taking ∂_θ always transverse to this foliation ensures that the condition is verified.
Take j and σ on Σ as in definition **??**, with $\sigma = -d\varphi \circ j$ and $\varphi = -H$ away from $\partial\Sigma$.
Define

$$\beta = \begin{cases} \sigma & \text{on } M_\Sigma \\ G d\phi & \text{on } M_P \end{cases}$$

where G is a interpolation map which is non-decreasing and goes from e^ρ to 2 on M_I (see part 2.3 of **[SOBD1]**).
Let $\gamma = \alpha + K\beta$ and $\gamma_0 = \alpha_0 + K\beta$. Then γ and γ_0 are contact forms. Indeed, we have $\gamma \wedge d\gamma = \alpha \wedge d\alpha + K(\alpha \wedge d\beta + \beta \wedge d\alpha)$ hence it suffices to show that $\alpha \wedge d\beta + \beta \wedge d\alpha$ is non-negative. This is the case for the following reasons :

- On M_P, $\beta = 2d\phi$ and $d\phi \wedge d\alpha > 0$, and $d\phi \wedge d\alpha_0 > 0$ by definition of being supported by a SOBD/BBD.
- On M_I, $\beta = G(\rho)d\phi$ and $\alpha \wedge d\beta = G'(\rho)\alpha \wedge d\rho \wedge d\phi > 0$ because $\alpha(\partial_\theta) > 0$. The same is true for α_0. We also have $\beta \wedge d\alpha > 0$ and $\beta \wedge d\alpha_0$ for the same reason as above.
- On M_Σ, $\alpha_0 \wedge d\beta > 0$ and $\alpha \wedge d\beta > 0$ for the same reason. Moreover, $\beta \wedge d\alpha_0 = 0$ because $\beta(R_{\alpha_0}) = \beta(\partial_\theta) = 0$. Finally, $\beta \wedge d\alpha > 0$. Indeed, on the interior of M_Σ, we can write: $H^2\beta(R_\alpha) = dH \circ j(R_\alpha) = d\sigma(X_H, -jR_\alpha) = d\sigma(\nabla H, (-j)(-j)R_\alpha)) = d\sigma(\nabla H, -R_\alpha) > 0$ because R_α is positively transverse to the pages of the broken book.

Now consider $\eta_t = t\gamma + (1-t)\gamma_0$. Denote $\alpha_t = t\alpha + (1-t)\alpha_0$. Then $\eta_t = \alpha_t + K\beta$ so : $d\eta_t = d\alpha_t + Kd\beta$, and : $\eta_t \wedge d\eta_t = \alpha_t \wedge d\alpha_t + K(\alpha_t \wedge d\beta + \beta \wedge d\alpha_t)$. Finally, we have that $(\alpha_t \wedge d\beta + \beta \wedge d\alpha_t) > 0$ everywhere, by the arguments given above. Hence we get an homotopy of contact structures, thus Gray's theorem allows us to conclude.

□

To obtain the inverse process, we define some structure than we will be useful later on, and can be seen as a generalization of the spine in the case of broken book decompositions.

Definition 4.6. Let $(\mathcal{K}, \mathcal{F})$ be a BBD. The *generalized spine*, denoted M_-, is a thickened compact neighbourhood of the union of the negative pages and binding components. In particular, it is taken in a way such that it is diffeomorphic to $\Sigma \times [-1, 1]$ around each negative page Σ, contains a (smooth) tubular neighbourhood of each boundary component, and the neighbourhood around two different pages which have common ends glue smoothly around this end.
A *generalized spine component* is defined by taking only certain pairs of negative pages or binding components. It is *maximal* if it is a connected component of the generalized spine.
A BBD is said to be *simple* if the generalized spine is $\mathbb{S}^1 \times \Sigma$ where Σ is some compact orientable surface with boundary, or equivalently if all negative rigid pages are cylinders. Similarly, a generalized spine component is simple if all the negative rigid pages it contains are cylinders.

In particular, if the broken book is just a classical open book decomposition, then M_- is just a union of tubular neighbourhood of the binding components, hence a union of $\mathbb{S}^1 \times \mathbb{D}^2$.

Lemma 4.6. Let $(\mathcal{K}, \mathcal{F})$ be a simple broken book decomposition. Then there exists a spinal open book decomposition $\boldsymbol{\pi}$, whose spine contains all negative rigid pages, and such that $B_H(\boldsymbol{\pi}) = (\mathcal{K}, \mathcal{F})$ for some map H as before.
In particular, there exists a unique contact structure supported by this broken book.

Proof. This simply follows that in this case, the generalized spine M_- turns out to be an actual spine. To see this, identify M_- with $[-1,1]_\phi \times [-1,1]_\rho \times \mathbb{S}^1_\theta$ around each negative page, and with $\mathbb{D}^2 \times \mathbb{S}^1_\theta$ where $\mathbb{D}^2$ has coordinates (ρ, ϕ), and such that the coordinates (ρ, θ, ϕ) defined in that way glue smoothly. Then consider the distribution spanned by : $\langle \partial_\phi, \partial_\rho \rangle$ outside of the binding orbits. This is integrable, and if we integrate it, and complete it with the binding orbits, it gives us a surface Σ with boundary. Note that it has no corners as all the ends of negative curves are on binding orbits. Now we can identify M_- with $\Sigma \times \mathbb{S}^1_\theta$. This corresponds to M_Σ for spinal open books. Finally, notice that the regular pages outside of M_- intersect ∂M_- on a union of tori, foliated by circles which can be chosen to follow ∂_θ, which therefore gives us a spinal open book decomposition $\boldsymbol{\pi}$ on $M = M_\Sigma \cup M_P$.
Now choose a hamiltonian on Σ such that H has index 2 critical points on regular binding components, and index 1 critical points at broken binding components, and such that the gradient flow lines of H follow the foliation given by the pages. This gives us that $B_H(\boldsymbol{\pi})$ is the initial broken book decomposition we started from. Hence by the previous lemmas, ☐

Remark 4.7. The surface Σ obtained with this construction is obtained by adding 1-handles between disks. Hence any compact surface with non-empty boundary can be obtained this way. Moreover, it can be described using a graph, and this graph is planar if and only if the surface is planar. Note that it is a disk if and only if the graph has no cycle. Finally, the topological type of the compact surfaces Σ obtained in that way is uniquely determined, as if Σ and Σ' are two compact surfaces with the same number of boundary components, $\Sigma \times \mathbb{S}^1 \simeq \Sigma' \times \mathbb{S}^1$ if and only if $\Sigma \simeq \Sigma'$.

Finally, we give a little result on the structure of broken books which allow for cancellation moves, in order to simplify the structure of a given broken book decomposition.

Proposition 4.8. Let $(\mathcal{K}, \mathcal{F})$ be a broken book decomposition supporting a contact structure ξ. Take $S \simeq \Sigma \times \mathbb{S}^1_\theta$ to be a simple generalized spine component. Take a hamiltonian $H : \Sigma \to \mathbb{R}_+$ as in definition **??**. We can define a new broken book decomposition $(\mathcal{K}', \mathcal{F}')$ using H. Then up to isotopy, $(\mathcal{K}', \mathcal{F}')$ supports ξ

Proof. Choose a hamiltonian $H_0 : \Sigma \to \mathbb{R}_+$ such that $(\mathcal{K}, \mathcal{F})$ is defined by this hamiltonian (i.e the pages are gradient flow lines of H_0). Let α_0 be a contact form supported by $(\mathcal{K}, \mathcal{F})$, with $\mathrm{Ker}(\alpha_0) = \xi$. We define a contact form α supported by $(\mathcal{K}', \mathcal{F}')$ by using the same construction as in definition **??**, and interpolating with α_0. Namely, we can define α on S so that near ∂S, it has the form: $md\phi + \frac{1}{K}e^{-\rho}d\theta$. We want to interpolate with α_0. In order to do this, it suffices to notice that the proof of lemma **??** can be adapted in our setting. More precisely, we choose the coordinates (ρ, ϕ, θ) near ∂S so that: $\alpha(\partial_\rho) = 0$, $\alpha(\partial_\phi) > 0$, $\alpha(\partial_\theta) > 0$, $d\theta(R_\alpha) > 0$, $d\phi(R_\alpha) > 0$. This allows to interpolate: in these coordinates, we can write our contact form as $\alpha_0 = f(\rho)d\theta + g(\rho)d\phi$, with f, g positive and respectively decreasing/increasing, and then extend f and g so that we can interpolate with α. Note that α can be chosen to have value: $ad\phi + be^{-\rho}d\theta$, with a small enough, and b large enough, so that we can interpolate. We now need to show that the contact structures defined by α and α_0 are isotopic. The proof of this relies on a variation of the standard Gray's stability theorem. Take δ so that for $\rho \geq \delta$, $\alpha = \alpha_0$, and so that $d\phi$ is well-defined on M_P for $\rho \leq \delta$ (recall that if we go too far from the simple generalized spinal component, the coordinate ϕ will stop being well-defined if there is a negative rigid page which is not a cylinder, which is why this proof differs a bit from the previous ones). Now for $\rho \leq \delta$ (and inside the generalized spine component), we can isotope both contact structures, using the same method as in the proof of lemma **??**. However, this isotopy do not extend to all of $\rho \geq \delta$ as we use the form $d\phi$ in the definition of β.

Starting of the standard proof of Gray's theorem using the Moser trick, see **[geiges_introduction_nodate]** theorem 2.2.2 for example, we find a suitable time-dependant vector field X_t whose flow will give us the desired isotopy. On $M_\Sigma \cup M_I$, we can find it using the standard method as we have a homotopy of contact structures. Now consider some area $\delta \leq \rho \leq \delta'$ on which we have $\alpha_0 = f(\rho, \phi, \theta)d\theta + g(\rho, \phi, \theta)d\phi$, and $\alpha = \alpha_0$. Then we can compute explicitly the vector field X_t induced by the homotopy on

$\rho \leq \delta$: the homotopy is: $\alpha + h(t)d\phi$ where $h : [-1,1] \to \mathbb{R}_+$ is an even function, increasing on $[-1,-\frac{1}{2}]$ and constant on $[-\frac{1}{2},0]$ (equal to some constant K large enough as before). Hence the derivative $\dot{\alpha}_t$ is given by: $\dot{\alpha}_t = h'(t)d\phi$. We have the following defining equation for $X_t \in \xi$: $\forall Y \in \xi, d\alpha_t(X_t,Y) = -\dot{\alpha}_t(Y) = -h'(t)d\phi(Y)$. Moreover, ξ is spanned by ∂_ρ and $v = -g\partial_\theta + f\partial_\phi$, so a short computation gives: $X_t = -\frac{h'f}{D}\partial_\rho$ where $D = f\partial_\rho g - g\partial_\rho f$ as before. So we can smoothly extend h as $\overline{h}(\rho,t)$ on $\delta' \leq \rho \leq \delta''$ such that: $\overline{h}(\rho,\cdot)$ is even for all ρ, and has the same monotonicity properties than h; $\overline{h}(\delta',t) = h(t)$ and $\overline{h}(\delta'',t) = 0$. Then extend X_t using this $\overline{h}$. Now let Ψ_t be the flow at time t of $\{X_t\}$. We verify that we have: $T\Psi_1(\xi_0) = \xi$. For $\rho \leq \delta'$, this is a consequence of the construction of X_t from the homotopy $\{\alpha_t\}$. For $\rho \geq \delta'$, it is a consequence of the parity of h, as Ψ_1 is in fact equal to the identity if $|h'(t)f/D|$ is small enough. This can be arranged up to multiplying the contact form by a small constant: indeed, in the proof of lemma **??**, we want $\alpha_t \wedge d\alpha_t + K(\alpha_t \wedge d\beta + \beta \wedge d\alpha_t)$ to be positive for K large enough. Hence taking $\epsilon\alpha$ instead of α allows to take $\epsilon^2 K$ instead of K, so: $|h'(t)f/D|$ becomes: $|(\epsilon^2 h')(\epsilon f)/(\epsilon^2 D)| = |\epsilon h' f/D|$, which can therefore be made as small as we want. This implies the desired contactomorphism, and concludes the proof.

□

Remark 4.9. One specific application is the following: suppose there is a pair of negative rigid pages which have the same negative end, and whose positive ends are different. Then we can simplify the broken book decomposition so that instead of the broken component and the two radial component, there is a single radial component and no rigid page (see figure **??**).
More generally, given a compact surface with non-empty boundary, one can always find a "minimal" Morse function with a single index 2 critical point, and exactly $2g+l-1$ index 1 critical points, where g is the genus and l the number of boundary components. This allows to simplify any broken book with a simple generalized spine component, by reducing the number of negative rigid pages in this component to $4g+2l-2$.

4.3 Completely connected broken books

In this subsection, using again the methods coming from spinal open book decompositions, we establish some results on the existence and uniqueness of a contact structure supported by a given broken book. Recall that a in a broken book decomposition, a "sector" is a connected component of the open subset of M foliated by regular pages; it can be parametrized by $[0,1] \times F$ where F is a regular page inside the sector.

Definition 4.7. Let $(K,\mathcal{F})$ be a broken book on a closed oriented smooth 3-manifold M. Recall K is divided in two subsets $K_r \cup K_b$ corresponding to radial and broken binding components.

- A broken book is said to be *completely connected* if the following condition is verified: for any sector S, surface F with the same topological type than the pages of the sector, and homology classes $\alpha, \beta \in H_1(F)$ corresponding to breaking orbits at both ends of the sector, then $\alpha \cdot \beta = 0$.
- We denote by M^N the union of the closure of negative rigid pages, and $M^\# = M \backslash M^N$. A connected component of $M^\#$ is foliated by pages, some of them rigid positive, whose topology can change. However, we can define a smooth surjective submersion : $\pi : M^\# \to \mathbb{S}^1$ such that the levelsets of π are exactly the pages. Note that this is a priori not a fibration as it is not proper (hence allowing the pages to change of topology). Now take $(M_1,\pi_1),(M_2,\pi_2)$ two connected components of $M^\#$. Suppose there exists some radial binding component γ such that γ is in the closure of both M_1 and M_2 in M. Then by definition of a radial binding orbit, there exists a tubular neighborhood of it and coordinates (ρ,ϕ,θ) such that the pages are given by $\{\phi = Cst\}$. Now look at $d\pi_i(\partial_\phi)$, $i=1,2$. If both have the same sign, we will say that π_1 and π_2 are **compatible at** γ. We will say that a broken book is **oriented** if for $M_1,...,M_r$ the connected components of $M^\#$, there exists a choice of π_i, $i=1,...,r$ such that for all $i \neq j$ and $\gamma \in \overline{M_i} \cap \overline{M_j} \cap K_r$, π_i and π_j are compatible at γ.

Remark 4.10. If the broken book supports an actual contact structure, it is indeed oriented, with the transverse orientation on each sector given by the Reeb flow.

Before actually building the contact and SHS structures, we start by defining coordinates around binding orbits which will be useful later. Take an oriented broken book on M, and let γ be a radial binding orbit. We identify a tubular neighbourhood of γ as $\psi(\mathbb{D}^2 \times \mathbb{S}^1_\theta)$ with $\mathbb{D}^2$ the unit disc, and ψ and embedding, such that the ∂_θ associated to the $\mathbb{S}^1$-coordinate is tangent to the pages and to the binding orbit. Define coordinates $\rho \in [-1, 1]$ and $\phi \in \mathbb{S}^1$ near the boundary, such that ∂_ρ is tangent to the pages and ∂_ϕ is transverse to the pages, with a direction compatible with the chosen orientation of the broken book. Define M_r to be the smaller neighbourhood composed of points with $\rho \leq 0$ (extending this coordinate up to the binding), and define $M_r^I = \{-1 \leq \rho \leq 1, \phi \in \mathbb{S}^1\}$ the boundary annulus.
Now take a broken binding orbit γ. Define Σ to be a hyperbolic octagon corresponding to a Morse chart around an index 1 critical point of a Morse function (see picture below). This can be described as the set of points : $\{(x, y) \in \mathbb{R}^2 | -1 \leq x^2 - y^2 \leq 1 \text{ and } |xy| \leq \cosh(1)\sinh(1)\}$ Now a neighbourhood of γ can be described as $\psi(\Sigma \times \mathbb{S}^1_\theta)$ with ψ a smooth embedding, in such a way that ∂_θ is tangent to the pages and to the binding orbit, and the intersection of page with a section $\{\theta = Cst\}$ is either empty, or it consists of a gradient flow line for the function $x^2 - y^2$ (this is possible by definition of a broken binding component in an broken book decomposition). Now we can identify Σ with a pair of pants with one negative boundary component and two positive in the following way : identify the sides of Σ corresponding to $\{x^2 - y^2 = -1\}$ and identify both corners of each side $\{x^2 - y^2 = 1\}$. Finally, glue the corresponding "oblique sides" so that they become preferred paths linking the three boundary components (see picture below). Now, away from a smaller octagon $\{(x, y) \in \mathbb{R}^2 \,|\, -\epsilon \leq x^2 - y^2 \leq \epsilon \text{ and } |xy| \leq \epsilon \cosh(1)\sinh(1)\}$, we can define coordinates (ρ, ϕ, θ) such that : ∂_ρ is tangent to the oblique boundary components, transverse to the others, and tangent to the pages, and ∂_ϕ is transverse to the oblique boundary components and to the pages, and is tangent to the vertical and horizontal boundary components. Namely, define the coordinates on some open set of the pair of pant defined before by noticing that on this subset, we can put coordinates (ρ, ϕ) where ρ is the height and ϕ the angular coordinate as on a cylinder. Finally, we ask that ρ is -1 on the vertical boundary, and 1 on the horizontal boundary.
A REFAIRE Define M_b to be the union of the subsets $\{(x, y) \in \mathbb{R}^2 \,|\, -\frac{9}{16} \leq x^2 - y^2 \leq \frac{9}{16} \text{ and } |xy| \leq \frac{9}{16}\cosh(1)\sinh(1)\}$ for all broken binding components in the BBD. We also define M_b^I to be the union of all the collars $\psi(\Sigma \times [0, 1]) \backslash \{(x, y) \in \mathbb{R}^2 \,|\, -\frac{1}{4} \leq x^2 - y^2 \leq \frac{1}{4} \text{ and } |xy| \leq \frac{1}{4}\cosh(1)\sinh(1)\}$ for all broken binding orbits.

Finally, define M_0 as $M \backslash (M_r \cup M_b)$ and $M^I = M_r^I \cup M_b^I$.

Lemma 4.11. *Let M^3 be a closed oriented smooth manifold and $(K, \mathcal{F})$ be an oriented broken book on M. Then there exists a vector field X on M_0 which coincides with ∂_ϕ in M^I, and such that the flow of X sends any oblique side of a octagon neighbourhood of a broken binding orbit to another oblique side associated to another broken binding orbit (possibly the same one). In particular, the trajectories of the flow of X are either defined on $\mathbb{R}$ or on a bounded interval.*

Proof. The main idea is a that there is a finite number of homotopy classes of embedded dividing loops on pages, and a finite subset of these that are "associated" to broken binding orbits. Let us clarify this a bit. On each rigid pages P_i, there exist a finite number of homotopy classes of loops which are separating curves. For each of these except of the homotopy class corresponding to the boundary component associated to the broken component, choose a representative $\gamma_{i,j}$, and a small tubular neighbourhood $\mathcal{U}_{i,j} \simeq \gamma_{i,j}(\mathbb{S}^1) \times [0, 1]$. Now consider a sector S. The orientation gives a pair of rigid pages at the "back end" of the sector (P_1, P_2) and a pair of rigid pages at the "front end", (P_3, P_4). We have O_1 and O_2 the 2 "oblique sides" of the octagonal neighbourhoods around the broken binding components, which are contained in the sector S. Consider $\pi_S : S \to [0, 1]$ the associated projection. Now consider a vector field X defined on $S \cap M_0$, which coincides with ∂_ϕ inside M^I, verifies : $d\pi_S(X) = 1$. Moreover, we ask the it sends any neighbourhood $\mathcal{U}_{i,j}$, $i = 1, 2$ on either a neighbourhood $\mathcal{U}_{k,l}$, $k = 3, 4$ or on O_2. We also ask that O_1 is also sent on the neighbourhood

corresponding to its homotopy class as a loop contained in a regular page. Now apply this construction to each sector. This is possible by the orientation assumption, and also asking that the vector field X constructed in that way is smooth on rigid pages, which is possible up to choosing well the projection π at each sector.
Now take any oblique side O, and look at where it is sent by the flow of X. If we assume it never reaches another oblique side, then it crosses an infinite number of times some rigid page P. But there is only a finite number of neighbourhood on such a page on which it can be sent by construction, hence it has to pass twice through the same neighbourhood $\mathcal{U}_{i,j}$. This implies that it is periodic, which is not possible as we assumed it never comes back to an oblique side. Hence all oblique sides are sent on another oblique side, and for any orbit of X, it is either periodic hence defined on $\mathbb{R}$, or it ends at an oblique side at some point, which means it is defined on an interval of $\mathbb{R}$ which is bounded at $\pm\infty$.

□

Now denote as H the union of the octagonal neighbourhoods and of the image of the sides by the flow of X. On $H \backslash M_b$ there are coordinates (ρ, ϕ, θ) which are well defined. Define M_H as $\{\rho \leq\} \subset H$, $M_H^I = \{\rho \leq\} \subset H$, and $M_0^H = M_0 \backslash M_H$. We obviously have : $M_b \subset M_H$, $M_b^I \subset M_H^I$ and $M_0^H \subset M_0$.

Lemma 4.12. Let M^3 be a closed oriented smooth manifold and $(K, \mathcal{F})$ be an oriented broken book on M. Chose $M_1 \sqcup ... \sqcup M_r = M^\#$ to be the connected components of $M^\#$, and $\{\pi_i\}_i$ compatible associated projections onto $\mathbb{S}^1$, which exist by the orientation assumption, and are compatible with the vector X constructed before, i.e $d\pi_i(X) = 1$.
Then there exists a 1 form λ on M_0^H and a constant $K > 0$ such that $\alpha = d\pi_i + \frac{1}{K}\lambda$ is a contact form on M_0, which also verifies the following boundary conditions :

$$\alpha = d\phi + \frac{1}{K} e^{-\rho} d\theta \qquad \text{on } M^I \cap M_0^H$$

Proof. First notice that even though the projections $\{\pi_i\}$ may not glue well, the 1-forms $\{d\pi_i\}$ glue as a global 1-form on M_0, which we will denote Π.
Notice that the connected components of M_0^H are actually of the form $\dot{\Sigma} \times \mathbb{S}^1$, where $\dot{\Sigma}$ is a compact surface with boundary. Moreover, this identification is given by taking any piece of page P^0 which is contained inside a connected component T of M_0^H, and taking :

$$\begin{aligned} \Psi :& \mathbb{R} \times \dot{\Sigma} \to T \\ & (t, x) \mapsto \phi_t^X(\varphi(x)) \end{aligned}$$

where φ is any identification of $\dot{\Sigma}$ with P^0. Note that by the hypotheses made on X, the flow sends a page on another page. Moreover, notice that the coordinates (ρ, ϕ, θ) are preserved by the flow of X on M^I as X coincides with ∂_ϕ on this domain.
Now choose a Liouville form on P^0 which verifies the right boundary conditions, namely it is $e^{-\rho}d\theta$ on M^I. Extend this form as a 1 form λ on T by pushing it forward with the flow of X, and by interpolation when it comes back to the initial page, as for classic open books (the set of Liouville forms with such fixed boundary conditions is convex). Note that this form also verifies the boundary conditions on each page because the coordinates are preserved by the flow of X.
Taking K large enough, the form $\alpha = \Pi + \frac{1}{K}\lambda$ is a contact form. Moreover, Π is $d\phi$ in M^I, hence the normal form of α in M^I.

□

We can now show the following :

Lemma 4.13. Let M^3 be a closed oriented smooth manifold and $(K, \mathcal{F})$ be an oriented broken book on M, such that rigid pages have at most one end asymptotic to a broken binding component. Then there exists a contact form α and a stable hamiltonian structure $(\Lambda, d\alpha)$ such that R_Λ is supported by the broken book (up to isotopy).

Proof. We consider Morse functions $H_r : \mathbb{D}^2 \to \mathbb{R}$ and $H_b : \Sigma_{0,3} \to \mathbb{R}$ where $\Sigma_{0,3}$ is the 3-punctured sphere, i.e a pair-of-pants. We impose the following conditions : H_r must have a single index 2 critical point, and in a neighbourhood of the boundary it must be at first strictly decreasing and then evenly equal to 0 in a smaller neighbourhood (roughly take $-(x^2+y^2)$ and extend it with a maps going flatly to 0). H_b has a single index 1 critical point, and must be such that it is constant on the boundary equal to ± 1, and equal to ρ away from a small neighbourhood of the critical point which we defined as $\{(x,y) \in \mathbb{R}^2 \mid -\epsilon \leq x^2 - y^2 \leq \epsilon \text{ and } |xy| \leq \epsilon \cosh(1)\sinh(1)\}$ in the octagon with the identification described before. Note that we can ask H_r and H_b to be $\phi-$invariant where the coordinate is well-defined.

Moreover, take Liouville forms σ_r and σ_b on $\mathbb{D}^2$ and $\Sigma_{0,3}$ such that σ_r is $e^\rho d\phi$ near the boundary and σ_b is $e^\rho d\phi$ in the same open set than the one described above (note that both Liouville forms can be described using subharmonic potentials ψ_r and ψ_b and a complex structure j_r and j_b and : $\sigma_{r,b} = -d\psi_{r,b} \circ j_{r,b}$). This will be useful later on to define compatible compatible almost complex structures.

Define "interpolation maps": $F, G, \beta, F_{r,b}, G_{r,b}$ with the following conditions (cf [**SOBD2**]) :
With these maps, we can define $\hat{H}_{r,b}$ as : $\hat{H}_{r,b}(x,\theta) = H_{r,b}(x)$ away from $M^I \cup M_0$ and $\hat{H}_{r,b}(\rho,\phi,\theta) = H(F(\rho),\phi)$ inside M^I.
Finally, define the following 1-forms :

$$\Lambda = \begin{cases} \Pi & \text{on } \tilde{M}_0 \\ e^{\epsilon \hat{H}_{r,b}}(\sigma_{r,b} + \frac{1}{K} d\theta) & \text{on } M_r \cup \tilde{M}_b \\ e^{F_{r,b}(\rho)} d\phi + \frac{1}{K} e^{G_{r,b}(\rho)} \beta(G_{r,b}(\rho)) d\theta & \text{on } \tilde{M}^I \end{cases}$$

Now take α_0 to be the contact one form on M_0 given by the previous lemma. We extend it to M by the following formulas :

$$\alpha = \begin{cases} \alpha_0 & \text{on } \tilde{M}_0 \\ \Lambda & \text{on } M_r \cup \tilde{M}_b \\ e^{F_{r,b}(\rho)} d\phi + \frac{1}{K} e^{G_{r,b}(\rho)} d\theta & \text{on } \tilde{M}^I \end{cases}$$

Because of the boundary conditions on the different Liouville forms used, these formulas define global 1-forms on M. Moreover, even though the 1-form Π may not be exact any more, it is still closed, hence we have a SHS on $\tilde{M}_0$. On $M_r \cup \tilde{M}_b$, both forms are contact forms by the same arguments that for α_0, and because adding a small hamiltonian perturbation does not change that. Finally, it remains to check that it is a SHS or a contact form on $\tilde{M}^I$. Around radial binding components, the computation is the same than in [**SOBD2**] hence the result. Even though it is very similar around $\tilde{M}_b$, let us detail a little bit what happens. The difference with the radial case is the fact the $\tilde{H}_b$ is not 0 near the boundary. In fact, it is equal to ρ in M^I. We must check that $\alpha \wedge d\alpha$ is a volume form, that $\mathrm{Ker}(d\alpha) \subset \mathrm{Ker}(d\Lambda)$ and that $\Lambda \wedge d\alpha > 0$. For the first one, it amounts to noticing that $\partial_\rho F_b - \partial_\rho G_b > 0$. However, this is true for $\partial_\rho F - \partial_\rho G$, hence up to choosing ϵ small enough, the same holds for the perturbed version. The same reasoning holds for $\Lambda \wedge d\alpha$. Now, in order to show that $\mathrm{Ker}(d\alpha) \subset \mathrm{Ker}(d\Lambda)$, it suffices to notice that because $F_b = F + H_b(F, \cdot)$, if $\partial_\rho F = 0$, then $\partial_\rho F_b = 0$, independently of the choice of H_b. This is the only ingredient necessary to conclude by construction of β, F and G.

□

The following lemma follows from a straightforward computation :

Lemma 4.14. The Reeb vector field R_Λ for $(\Lambda, d\alpha)$ is given by :

$$R_\Lambda = \begin{cases} X \text{ on } \tilde{M}_0 \\ \dfrac{1}{\beta(G_{r,b})F'_{r,b} - G'_{r,b}}(-e^{F_{r,b}(\rho)}G'_{r,b}(\rho)\partial_\phi + Ke^{-G_{r,b}(\rho)}F'_{r,b}(\rho)\partial_\theta) \text{ on } \tilde{M}^I \\ e^{-\epsilon\hat{H}_{r,b}}((1+\epsilon\sigma(X_{H_{r,b}}))K\partial_\theta - \epsilon X_{H_{r,b}}) \text{ on } M_r \cup \tilde{M}_b \end{cases}$$

with $X_{H_{r,b}}$ the hamiltonian vector field on $\Sigma_{0,3}$ given by $d\sigma_b$ and H_b.

Proof. The proof is exactly the same than in [**SOBD2**], part 3.5, despite the slight changes in $\tilde{M}_b$. □

Note that up to rescaling it is the same Reeb vector field than for $(\alpha, d\alpha)$.

We point out that the radial binding components of the BBD are elliptic closed Reeb orbits for R_Λ, and broken binding components are hyperbolic orbits. Moreover, we actually built this vector field by fixing the stable and unstable manifolds associated to these hyperbolic orbits. The following lemma is a straightforward consequence of the formulas for R_Λ.

Lemma 4.15. The Reeb vector field along the stable and unstable manifolds associated to a broken binding component is given by : $e^{-\epsilon\rho_0}(1+\epsilon)$
Hence these submanifolds have tangent space generated by ∂_θ and X.

Hence for any oriented broken book decomposition, we are able to build a contact structure supported by this broken book, with a Giroux form having the property that the stable and unstable manifolds associated to broken binding components form complete connections (see [**colin_existence_2022**]). We will see in the following that up to isotopy, there is only one such contact structure supported by a given oriented BBD. Let us finally note the following fact which will be useful for a later purpose :

Lemma 4.16. Let X and Y be 2 vector fields satisfying the same conditions, namely they coincide with ∂_ϕ, create only "connections" between broken binding components, and verify $d\pi_i(X) = d\pi'_i(Y) = 1$ for some families of projections $\{pi_i\}$ and $\{\pi'_i\}$ associated to the different oriented sectors. Then the contact structures built with the method above using X and Y are isotopic.

Proof. Consider the vector field $Z_t = tX + (1-t)Y$ for $t \in [0,1]$. Then Z_t also verifies all the conditions : it is ∂_ϕ in M^I, creates only connections and : $(t\Pi + (1-t)\Pi')(Z_t) = 1$. Hence we obtain a family of contact forms α_t, with associated contact structures ξ_t. Hence by Gray theorem, ξ_0 and ξ_1 are isotopic, which is the result we wanted. □

In this section we focus on the uniqueness of the contact structure supported by a broken book, and in the interpolation of a given contact form in order to have a normal form near the binding. We will denote by $(K, \mathcal{F})$ the considered BBD, and ξ_0 the contact structure constructed in the previous part supported by $(K, \mathcal{F})$.

Lemma 4.17. Suppose ξ is a contact structure supported by $(K, \mathcal{F})$. Suppose also that it admits a Giroux form α such that the stable and unstable manifolds associated to broken components of K form only connections. Then ξ is isotopic to ξ_0.

Remark 4.18. Note that we did not specify which vector field was used in order to build ξ_0, which is not important as we showed that all the different contact structures obtained in that way were isotopic. In fact, the beginning of the following proof starts by fixing such a contact structure.

Proof. This is essentially the same proof as in [**SOBD1**], part 2.3.

Consider α_0 the Giroux form defined in the previous part, such that $\text{Ker}(\alpha_0) = \xi_0$, where the stable and unstable manifolds of α_0 starting and ending at broken binding components are chosen so that they coincide with those of α. Let us precise this a little bit : choose projections $\{\pi_i\}$ for each sectors, an associated vector field X which is tangent to the stable and unstable manifolds of R_α, is positively transverse to the pages and verify $\Pi(X) = 1$ where Π is defined as before. Note that here the orientation of the broken book is given by the flow of R_α. Choose small enough neighbourhoods of the binding components and of the stable and unstable manifolds as before, with coordinates (ρ, ϕ, θ) chosen as in the first part, namely ∂_ρ and ∂_θ are tangent to the pages, ∂_ϕ is positively transverse, and ∂_θ is tangent to the binding components. We can moreover choose the coordinate θ such that $\alpha(\partial_\theta) > 0$ everywhere where it is defined. This is direct in $M_r \cup M_b$ up to choosing sufficiently small neighbourhoods because it is follows the Reeb vector field on the binding, and on the stable and unstable manifolds, it suffices to chose θ such that ∂_θ is always transverse to the characteristic foliation on the stable and unstable manifolds, and then to choose a small enough neighbourhood for M^I. Define $g = g(\rho) : [0, 1[\to [0, 2]$ such that $g(\rho) = e^\rho$ near 0, $g'(\rho) \geq 0$ for all ρ, and $g(\rho) = 2$ for $\rho \geq \delta$, with $\delta > 0$ fixed. Define $G : \tilde{M}_0 \to [0, 2]$ to be g on $\tilde{M}^I$ and 2 elsewhere. Define :

$$\beta = \begin{cases} \sigma_{b,r} \text{ on } \tilde{M}_b \cup M_r \\ G\Pi \text{ elsewhere} \end{cases}$$

Lemma 4.19. The forms $\alpha + K\beta$ and $\alpha_0 + K\beta$ are contact forms for $K \in \mathbb{R}_+$.

Proof. Consider first $\gamma = \alpha + K\beta$. Compute :

$$d\gamma = d\alpha + Kd\beta$$

$$\gamma \wedge d\gamma = \alpha \wedge d\alpha + K(\beta \wedge d\alpha + \alpha \wedge d\beta)$$

Note that the term $\beta \wedge d\beta$ cancels out as Π is closed and coincides with $d\phi$ when this coordinate is defined. Hence it suffices to show that $(\beta \wedge d\alpha + \alpha \wedge d\beta)$ is non-negative in order to conclude.

First, inside $\tilde{M}_b \cup M_r$, since $\alpha(\partial_\theta) > 0$, we have $\alpha \wedge d\beta > 0$. Moreover, $d\phi(R_\alpha) > 0$ everywhere where it is defined : indeed, the tangent space of the pages is spanned by $\{\partial_\rho, \partial_\theta\}$, and the Reeb flow is positively transverse to the pages hence $d\phi(R_\alpha) > 0$. Hence we have : $d\rho(-j_{r,b}R_\alpha) > 0$ and thus $-d\varphi_{r,b} \circ j_{r,b}(R_\alpha) \geq 0$ everywhere by definition, which amounts to $\sigma_{r,b}(R_\alpha) > 0$, and therefore $\beta \wedge d\alpha > 0$.

Now on $\tilde{M}_0$, we have that $\beta = \Pi$ is closed, and $\beta \wedge d\alpha$ is positive by construction of Π with a vector field X positively transverse to the pages.

On M^I, we have $\beta = G\Pi$, hence $\beta \wedge d\alpha$ is still non-negative ; moreover, $\alpha \wedge d\beta = G'(\rho)\alpha \wedge d\rho \wedge \Pi = g'(\rho)\alpha \wedge d\rho \wedge d\phi$. We know that $\alpha(\partial_\theta) > 0$ and $g' > 0$ which concludes.

The same reasoning applies to α_0, with an extra simplification as $d\alpha_0(\partial_\theta, \cdot) = 0$ inside $\tilde{M}_b \cup M_r$. $\square$

Finally, it remains to show that for K large enough, $t(\alpha + K\beta) + (1 - t)(\alpha_0 + K\beta)$ is a linear interpolation by contact forms of the two contact forms.

Write $\eta_t := t(\alpha + K\beta) + (1 - t)(\alpha_0 + K\beta)$ and $\gamma_t = t\alpha + (1 - t)\alpha_0$. Then $d\eta_t = d\gamma_t + Kd\beta$, and $\eta_t \wedge d\eta_t = \gamma_t \wedge d\gamma_t + K(\gamma_t \wedge d\beta + \beta \wedge d\gamma_t)$. It suffices to show that $(\gamma_t \wedge d\beta + \beta \wedge d\gamma_t) > 0$ in order to conclude. However, this follows directly from the arguments in the proof of the previous lemma, by noticing that with the same notations either $\beta \wedge d\alpha$ or $\alpha \wedge d\beta$ is in fact strictly positive. Hence for K large enough, η_t is a contact form for all t.

This concludes the uniqueness statement as we now have that ξ and ξ_0 are homotopic, hence isotopic by Gray's stability theorem.

$\square$

Remark 4.20. Note that it is a priori not true that any 2 contact structures satisfying the previous hypotheses are isotopic. Indeed, it may also depend on the orientation induced on the broken book ; it becomes true if we fix the orientation induced on the rigid pages.

Remark 4.21. Following the exact same method as in [**SOBD2**] part 3 and 4, one can use the stable hamiltonian structure defined above to construct an explicit almost complex structure, and explicit lifts of the pages. This gives yet another proof of theorem **??** in the case of an oriented completely connected broken book.

5 Extension of the foliation in symplectic fillings

We now show that the finite energy foliation built in the previous section foliates any strong symplectic filling of a weakly planar contact manifold M. This is a generalization of the same result for planar contact manifolds obtained by Wendl in [**wendl_strongly_2010**], and spinal open books [**SOBD2**]. For simplicity, all symplectic fillings here are considered to be strong fillings.

5.1 Sectorial nearly Lefschetz fibrations

In this subsection we give some definitions in order to describe the structure induced by a planar balanced broken book on a symplectic filling. This is a generalization of the structures of Lefschetz fibrations, bordered Lefschetz fibration, and nearly Lefschetz fibrations that appear on symplectic fillings of planar open books, planar Lefschetz-allowable spinal open books, and planar spinal open books respectively.

Let us recall quickly where these different objects come from. Suppose (M, ξ) is (strongly) filled by (W, ω), and that ξ is supported by a planar open book. Then, completing W with cylindrical ends, the curves from the finite energy foliation induced by the pages exist in this completion, and one can show that this foliation extends to the whole manifold. The only degeneracy that can occur is a simple nodal singularity, modelled on a Lefschetz-type singularity $z_1^2 + z_2^2 = 0$ ([**wendl_holomorphic_2018**], Appendix A). Hence the resulting foliation on W corresponds to a Lefschetz fibration over the disk.

If one now suppose (M, ξ) is supported by a planar spinal open book, one can use more elaborated methods, coming from intersection theory in dimension 4 and careful Fredholm analysis, to show that the resulting structure on W is a (nearly) bordered Lefschetz fibration. More precisely, the base of the fibration need not be a disk and can be any compact surface Σ_0 with an arbitrary number of boundary components, and the fibration restricted to the horizontal boundary of W factors through another surface Σ such that the fibers of this new fibration are circles. This is requested as pages can have multiple ends at the same boundary component, as seen in part **??**. Now $\Sigma \to \Sigma_0$ is *a priori* a branched cover. If this branched cover has no branching point, which corresponds to a condition called *Lefschetz-amenability*, the fibration is an actual true bordered Lefschetz fibration, with once again only Lefschetz-type singularities.

However, if the cover has branching points, then a new type of singularity may appear, namely so-called *exotic fibers*. In that case, the resulting structure was called a *nearly Lefschetz fibration* and was studied extensively in [**min_spinal_2024**], using asymptotic analysis techniques for holomorphic curves. We recall this model in the following definition:

Definition 5.1. Identify a neighborhood of an exotic fiber near $\partial_h W$ with $(\mathbb{C} \backslash B) \times \mathbb{C}$, where B the radius ϵ disc, with the radial direction of $\mathbb{C} \backslash B$ being the completion direction, the angular direction is the closed Reeb orbit associated with the exotic fiber, and the other $\mathbb{C}$ component accounts for a trivialization of the normal bundle of this orbit. Then in this neigborhood, the fibers are given by the levelsets of the map:

$$\begin{aligned}\pi_0 : \quad & (\mathbb{C}\backslash B) \times \mathbb{C} \to \mathbb{C} \\ & (z_1, z_2) \mapsto z_2^2 - z_1\end{aligned}$$

In the rest of this paragraph we introduce the generalized version of "Lefschetz fibration" we will need in order to study symplectic fillings of planar broken book decompositions. The motivations behind the introduction of this structure are given by the following facts:

- The negative rigid pages are in general not cylinders, and a pair of negative pages associated with a positive rigid page are not in general asymptotic to the same Reeb orbits.
- There is no "spine component" in general : the generalized spine is not equipped with a fibration over $\mathbb{S}^1$ (in fact a foliation by circles will always have singularities when there are non-cylinder negative rigid pages). Moreover, the non-existence of *holomorphic vertebrae* makes it difficult to use criterion such as Lefschetz-amenability to prevent exotic fibers from appearing. Hence in general we have to assume they do exist.

In order to take into account these differences, we introduce the following notions.

Definition 5.2. Let $\dot{\Sigma}$ be a compact surface with boundary, and b one fixed boundary component. Let $\widetilde{\Sigma}$ be the surface obtained by gluing a copy of $\dot{\Sigma}$ with $\dot{\Sigma}$ at b. Consider $D := \widetilde{\Sigma} \times [0, 1]$ a small thickening of $\widetilde{\Sigma}$. by Take a Morse function $f : \widetilde{\Sigma} \to [0, 1]$ with $2\#\partial\dot{\Sigma} - 1$ index 1 critical point, and 1 index 2 critical point, with at most one critical point in a given levelset and which is constant with no critical point along the boundary. Define a foliation on $\widetilde{\Sigma}$ by taking any metric on $\widetilde{\Sigma}$ and taking as integrable distribution the orthogonal of the gradient of f (see figure **??** below).
The *standard foliation* on D is a thickening of this foliation.

Definition 5.3. Let W be a compact 4-manifold with boundary and corners with $\partial W = \partial_h W \cup \partial_v W$ (horizontal and vertical boundary component), intersecting at a codimension 2 corner. A sectorial nearly Lefschetz fibration on W (SNLF) is the data of a division $\partial_h W = \partial_h^r W \cup \partial_h^b W$, which we call "regular horizontal boundary" and "broken horizontal boundary", and of a smooth map $\pi : W \to \Sigma$ where Σ is a compact connected oriented surface with boundary, where the following conditions are verified :

1. All connected components of $\partial_h^b W$ are thickened compact surfaces with an even number of boundary components, which must be at least 4.
2. $C_r := \partial\partial_h^r W$ is a disjoint union of tori, $C_b = \partial\partial_h^b W$ intersects C_r at a finite number of embedded annuli, which corresponds to thickened boundary components of the thickened compact surfaces composing $\partial_h^b W$.
3. $\partial_v W$ intersects $\partial_h W$ at $C_v = (C_r \cup C_b)\backslash(C_r \cap C_b)$.
4. For all $z \in \mathrm{Crit}(\pi)$, z is in the interior of W, and there exists local complex coordinates (z_1, z_2) around z such that : $\pi(z_1, z_2) = z_1^2 + z_2^2$.
5. $\pi_{|\partial_h^r W} : \partial_h^r W \to \Sigma$ is a smooth map with isolated critical points, called exotic. Around these singularities, there exists local coordinates in W, $(z_1, z_2) \in (\mathbb{C}\backslash B) \times \mathbb{C}$ as in definition **??** such that π is $\pi(z_1, z_2) = z_2^2 - z_1$.
6. $\pi_{|\partial_h^b W} : \partial_h^b W \to \Sigma$ is a smooth map whose fibers give a foliation on $\partial_h^b W$ isotopic to the standard foliation on $\partial_h^b W$.

7. $\pi_{|\partial_v W} : \partial_v W \to \partial\Sigma$ is a smooth surjective submersion.

8. $\pi^{-1}(\partial\Sigma) = \partial_v W$, and if $z \in \overset{\circ}{\Sigma}$, then $\pi^{-1}(\{z\})$ is connected and has boundary inside $\partial_h W$. Moreover, all fibers having their boundary entirely inside $\partial_h^r W$ have the same topology.

Remark 5.1. i) Let us give some elements to clarify the previous definition (see also remark **??** below). The broken horizontal boundary corresponds to thickened pairs of negative rigid pages of the broken book, which are not cylinders. The regular horizontal boundary corresponds to classic "spine components", i.e tubular neighborhood of radial binding components, and of negative rigid pages which are cylinders as in part **??**. $C_r \cap Cb$ corresponds to the intersection of (a thickening of) the negative rigid pages with a tubular neighborhood of their asymptotic radial binding components. The conditions for critical points and exotic fibers than in the spinal case. Finally, conditions 6 and 7 accounts for the change of topology of fibers in a component of M_0.
ii) In the case where $\partial_b^h W$ is empty, we get back to the nearly Lefschetz case.

iii) π only has non-degenerate critical points in the interior of W, $\pi_{|\partial_v W}$ does not have critical points but isn't proper (we cannot apply Ehresmann fibration theorem), and $\pi_{|\partial_h W}$ only has Morse-Bott critical points, and the families of critical points are either circles in $\partial_h^r W$ is the case of exotic fibers, or curves linking two boundary components in $\partial_h^b W$, corresponding to the standard foliation (i.e to "thickened index-1 critical points" in negative rigid pages).

In order to further justify this structure, we explain how to get a broken book decomposition on the boundary of a sectorial nearly Lefschetz fibration.

Lemma 5.2. Let W^4 be a SNLF. Then $M := \partial W$ admits a natural decomposition by a balanced broken book, with $M_0 \simeq \partial_v W$ and $M_- \simeq \partial_h W$.

Proof. It suffices to understand how to glue the different "parts" of the pages, as in a usual Lefschetz fibration.
Recall that in this latter case, the following happens: choose a fiber in each connected component of $\partial_h W$ which will correspond to the binding orbit, then a page can be obtained by taking a fiber inside $\partial_v W$, "glued" with a union of circles inside $\partial_h W$ (and also applying some smoothing operation which we will not describe here).
In the case of a bordered Lefschetz fibration ($\partial_h^b W = \emptyset$), one must first choose a collection of circles in each connected component of $\partial_h W$, which essentially amounts to choosing a hamiltonian on each spine component, with the chosen circles corresponding to index-1 and one index-2 critical points. Then as before, extend the fibers of $\partial_v W$ using unions of fibers in $\partial_h W$ to connect them to the chosen circles (one must also be careful to only link two pages with each index-1 critical point, and the rest to the index-2 critical point). This gives a broken book decomposition where each negative rigid page is a cylinder, which is another interpretation of the construction done in part **[p:IV]** to go from spinal open books to broken book decompositions.
In the general case, we may have $\partial_h^b W \neq \emptyset$. In that case, a little more care must be taken as the topology of fibers inside $\partial_v W$ may change. By definition, this happens at singularities of the standard foliation of $\partial_h^b W$, as they correspond to critical points of $\pi_{|\partial\partial_v W}$. First choose a collection of fibers in $\overset{\circ}{\partial_h^b W}$, corresponding to critical points of the Morse function used to get the standard foliation, hence we get $2\#\partial\dot{\Sigma} - 1$ fibers which are a wedge sum of two circles, corresponding to index-1 critical points of f, and one circle corresponding to the index-2 critical point. Moreover, consider the thickening of these preferred "index-1" fibers ; this gives for each component of $\partial_v^b W$ a division into different subset, one of which contains the "index-2" fiber. We will denote this subset $(\partial_h^b W)_0$.
From this, we can get the rigid pages: in order to get positive rigid pages, for each component of $\partial_h^b W$, choose a fiber of $\partial_v W$, such that it has an end in $(\partial_h^b W)_0$, and glue it with a union of fibers in $\partial_h^b W$ such that it goes to the chosen circle. Note that there are two connected components in $(\partial_h^b W)_0 \cap \partial_v W$, which allows to create a pair of positive rigid pages. In order to get negative rigid pages, just take a

collection of fibers between the chosen circles. Once again, we can take two of these by the condition that the number of ends is even, which allows to get the pair of negative rigid pages. Note that we also have to complete the pages by gluing a union of fibers in $\partial_h^r W$ so that the pages are asymptotic to the chosen fibers in $\partial_h^r W$. Finally, complete all the other pages by gluing fibers of $\partial_h^b W$

□

Remark 5.3. We can encode on Σ extra data which allows us to clarify the previous definition. We have three disjoints subsets inside Σ, which we will denote by $\Sigma^{crit}, \Sigma^{exot}$ and Σ^w. $\Sigma^{crit} \subset \mathring{\Sigma}$ is a finite union of disjoint points which are critical values of π. Σ^{exot} is a disjoint union of discs which corresponds to exotic fibers; note that because we are working with a compact domain we do not have a single exotic fiber but a family of these parametrized by a disc, all of which have a different topology than the regular pages. Finally, Σ^w corresponds to so-called "walls" that we will encounter again when working with holomorphic curves as 3-parameter family of rigid curves. Here, it can be described in the following way: $\Sigma^w \subset \Sigma$ consists of a union of embedded strips $[0,1] \times [0,1]$, with $\{0\} \times [0,1] \subset \partial\Sigma$ and $\{1\} \times [0,1] \subset \partial\Sigma$, with the property that if S is such a strip, and $z \in S$, then $\pi^{-1}(z)$ has a different topology than the topology of regular fibers. Note that because Σ^w is disjoints from $\Sigma^{crit} \cup \Sigma^{exot}$, this implies that $\pi^{-1}(z)$ has a boundary component in a component of $\partial_h^b W$. Note also that for $z \in ([0,1] \times \{0\}) \cup ([0,1] \times \{1\})$, the fiber $\pi^{-1}(z)$ is singular as it contains a critical point of $\pi_{|\partial_h^b W}$ (i.e on of its boundary components is a singular leaf of the standard foliation on a component of $\partial_h^b W$. Finally, we can ask that any two such strips intersect transversally, and that the intersection of any given triple of strips is always empty. The chosen terminology "sectorial" comes from the existence of these walls, which divides W into different sectors where the topology of the pages is the same, each pair of adjacent sectors being separated by such a "thick wall", in which the topology of the pages is different (namely the pages have strictly less boundary components).

We now define an extension of these notion to the symplectic/contact case.

Definition 5.4. A symplectic sectorial nearly Lefschetz fibration is (W, ω) strong symplectic filling of $(M = \partial W, \xi)$ such that $\omega = d\lambda$ near the boundary, with λ a contact form with $\xi = \mathrm{Ker}(\lambda)$. We also ask that: $d\lambda > 0$ on the fibers in $\partial_v W$, the fibers in $\partial_h^r W$ are closed Reeb orbits, and the Reeb vector field on $\partial_h^b W$ is transverse to the fibers away from the thickened boundary of the underlying surface (near the boundary, it is tangent as in $\partial_h^r W$ as it is close from C_r.

Lemma 5.4. The boundary of a strongly convex SSNLF is naturally equipped with a contact structure supported by the BBD defined before.

Proof. This is proved by the same method as in lemmas **??** and **??**, as the only thing that needs to be shown is that we can make the Reeb vector field transverse to the pages in $\partial_h^r W$. □

Remark 5.5. Our construction only allows to get supporting broken books for contact structures such that the negative rigid pages which are cylinders have trivial characteristic foliation (see part **??**).

Definition 5.5. A broken book decomposition is said to be *balanced* if any two negative rigid pages with their negative end at the binding component have the same topology, and if moreover no page have a negative end at a radial binding component.
A broken book decomposition is said to be *strongly balanced* if it it is balanced, and the condition is also verified for positive rigid pages.

Lemma 5.6. All pages of a strongly balanced broken book have the same topology.

Proof. First note that for a balanced broken book, all pages in a given connected component of M_0 have the same topology. Now suppose there exists two connected components M_0^1 and M_0^2 of M_0 such that the pages inside these components have different topologies.

We claim that we can assume that these two components are adjacent, meaning that they have a common negative rigid page in their boundary, and in particular a common broken binding component. This is because we can always find a sequence of connected components of M_0, $M_0^{i_1}, ..., M_0^{i_k}$ such that $M_0^{i_1} = M_0^1$, $M_0^{i_k} = M_0^2$ and $M_0^{i_j}$ is adjacent to $M_0^{i_{j+1}}$. Indeed, define an equivalence relation on $\pi_0(M_0)$ by $M^{(i)} \sim M^{(j)}$ if and only if there is such a sequence between $M^{(i)}$ and $M^{(j)}$. Then the union of the closures of the element of an equivalence class is open and closed in M by definition ($\pi_0(M_0)$ is finite, and the union is open by definition of being adjacent). Because M is connected, this proves the claim.

However, this contradicts the strongly balanced condition, as this means that topologically the pages are obtained by gluing a given negative rigid page, and one positive rigid page of a pair depending on the component, and we know that both rigid pages of the same pair have the same topology. □

The purpose of the following subsections will be to show the following theorem :

Theorem 5.7. *Let (M, ξ) be a weakly planar strongly balanced contact manifold such that a supporting strongly balanced planar broken book verifies the hypotheses of Theorem* **??**. *Suppose (M, ξ) admits a strong filling W. Then "the finite energy foliation extends".*
More precisely, denote as $\overline{\mathcal{M}_{0,1}}$ the compactified moduli spaces of marked unparametrized curves which have the same homology class than a regular page of the foliation defined before. Define an equivalence relation on $\overline{\mathcal{M}_{0,1}}$ by : $[u, z] \sim [v, z'] \Leftrightarrow u(z) = v(z')$, and let $\widetilde{\mathcal{M}_{0,1}} := \overline{\mathcal{M}_{0,1}}/\sim$. Then :

- *$\widetilde{\mathcal{M}_{0,1}}$ can be equipped with a smooth structure of manifold with boundary such that $\pi : \overline{\mathcal{M}_{0,1}} \to \widetilde{\mathcal{M}_{0,1}}$ is smooth and is a diffeomorphism if one restricts both spaces to index-2 regular curves.*
- *$ev : (\widetilde{\mathcal{M}_{0,1}}, \partial\widetilde{\mathcal{M}_{0,1}}) \to (\overline{W}, M)$ is well-defined, is a smooth map of degree 1, and is a diffeomorphism on its image if restricted to index-2 regular curves.*
- *This foliation induces a structure of symplectic sectorial nearly Lefschetz fibration on an enlarged filling obtained from W by attaching a trivial symplectic cobordism, and the induced broken book decomposition on M is isotopic to the initial one.*

5.2 Setup and strategy of proof

Let us start by explaining the setup and constructing the moduli spaces we will need in the proof. This is an adaptation of the proof done in [**SOBD2**], hence we will not give the details for all the arguments and refer to this paper when necessary. We will mostly explain how to generalize their arguments to our context. However, notice that because we are not considering the parametric version nor the case of weak symplectic fillings, many proofs get simpler in our setup.

First we define as usual the completion of the symplectic filling, where the curves of the original foliation live.
Let (M, ξ) and (W, ω) as in the hypothesis of the theorem **??**. Let Y be the Liouville vector field for ω near M. Suppose that the finite energy foliation is well-defined for an almost complex structure compatible with the contact form λ on M, and that $\iota_Y \omega = e^f \lambda$ on $M = \partial W$. Then, using the same method as in [**wendl_strongly_2010**], we can attach a symplectic cobordism $\mathcal{S}_f^R := \{(a, x) \in \mathbb{R} \times M \mid f(x) \leq a \leq R\}$ with symplectic form $d(e^a\lambda)$ and Liouville vector field ∂_a, and then another one $\mathcal{S}_R^{+\infty} := [R, +\infty[\times M$ with the same symplectic form, which is the positive end of the symplectisation of (M, λ). Denote as $\widehat{W}$ the resulting manifold. We will take J generic so that it is λ-compatible on $\mathcal{S}_R^{+\infty}$, ω-compatible elsewhere, and so that any somewhere injective curve is regular. This construction also allow us to ask that J is such that any closed, non-constant holomorphic curve is contained in the interior of W. Finally, we will denote as J_+ the λ-compatible almost complex structure on $\mathcal{S}_R^{+\infty}$, which is also well-defined on the symplectisation of λ and cylindrical.

Now as seen in part **??**, in a neighbourhood of infinity, we have a foliation $\mathcal{F}^+$ which comes from the balanced broken book decomposition. When it is defined, it is invariant by $\mathbb{R}$-translation. However the negative rigid curves are missing as well as the trivial cylinders, as they have negative ends ; hence a neighbourhood of infinity in $\widehat{W}$ minus a neighbourhood of negative rigid curves and binding orbits is foliated by regular holomorphic curves of index 1 or 2.

Before defining our moduli spaces, we need to make some additional assumptions on the Reeb flow and the broken book.

In the following, we will denote as n the number of ends of regular pages (recall it is the same for any regular page). The following lemma is a consequence of the constructions done in part **??**.

Lemma 5.8. We can choose a contact form α supported by the broken book such that, for $k = 1, ..., n$ and any radial binding orbit γ, we have : $\mu^{\tau}_{CZ}(\gamma^k) = 1$ where τ is the natural trivialization induced by the pages.

Proof. □

Now we define all the relevant moduli spaces that we will need in what follows.

In the symplectisation :
Let $\mathcal{M}^{\mathcal{F}_+}(J_+)$ be the space of J_+-holomorphic curves that compose the finite energy foliation $\mathcal{F}_+$. Denote as $\overline{\mathcal{M}}^{\mathcal{F}_+}(J_+)$ its closure, in the SFT sense. We consider $\mathcal{M}_i^{\mathcal{F}_+}(J_+)$ for $i = 0, 1, 2$ to be the subset of index i curves inside $\mathcal{M}^{\mathcal{F}_+}(J_+)$, with $\overline{\mathcal{M}}_i^{\mathcal{F}_+}(J_+)$ their compactification.
Define : $\widehat{\mathcal{M}}^{\mathcal{F}_+}(J_+) := \overline{\mathcal{M}}_2^{\mathcal{F}_+}(J_+)/\sim$ where 2 buildings are considered to be equivalent if they have the same ground level. We equip this moduli space with the quotient topology. Note that this is a disjoint union of circles, and using the notation of the previous part, each of these circles correspond to a connected component of M_+.

Remark 5.9. Here there is a little difference compared to the setup of [**SOBD2**] : we do not ask for the asymptotic orbits of the upper levels to be the same, as 2 negative rigid pages associated to a same hyperbolic orbit will not have the same asymptotics in general.

In $\widehat{W}$:
We define the spaces $\mathcal{M}(J)$, $\overline{\mathcal{M}}(J)$ and $\widehat{\mathcal{M}}(J) := \overline{\mathcal{M}}(J)/\sim$. The first one is the moduli space of unparametrized J-holomorphic curves with genus 0 in W which have asymptotics at binding orbits of the broken book, and has $d\alpha$-energy E. The second and third spaces are its compactification and the quotient for the same equivalence relation as before, respectively.
Up to choosing E high enough, we have an inclusion $\widehat{\mathcal{M}}^{\mathcal{F}_+}(J_+) \subset \widehat{\mathcal{M}}(J)$, so we can define $\widehat{M}^{\mathcal{F}}(J)$ to be the connected component of $\widehat{\mathcal{M}}(J)$ containing $\widehat{\mathcal{M}}^{\mathcal{F}_+}(J_+)$. This moduli space corresponds to the curves we can get by "extending the foliation in W", and we want to show good properties for curves in this moduli space, namely that they are regular, nicely embedded, foliate the whole space and have intersection number 0.

Nicely embedded curves :
In order to study the compactness properties of the moduli spaces, we need to restrict the class of curves we will be working with, and for which we will be able to control the behaviour.
Recall that a simple regular J-holomorphic curve u in a 4 manifold with cylindrical ends is said to be *nicely embedded* if it verifies :

- $\delta(u) = \delta_\infty(u) = 0$
- $u * u \leq 0$

We saw that all curves inside $\mathcal{M}^{\mathcal{F}_+}(J_+)$ were nicely embedded, which means that the same curves in the completion of $\widehat{W}$ are too. Moreover, this notion is invariant by homotopy, by properties of the intersection numbers. However, we will have to show that a limit of nicely embedded curves has good properties ; namely we will show that the main level is always nicely embedded.
We can define the moduli spaces : $\mathcal{M}^{nice}(J) \subset \mathcal{M}(J)$ the subset of nicely embedded curves, $\mathcal{M}_i^{nice}(J)$ the moduli spaces of nicely embedded curves of index i, and as before take their compactification.

Now, we define $\widehat{\mathcal{M}}_i^{\mathcal{F},nice}(J)$ to be the subset of $\widehat{\mathcal{M}}_i^{\mathcal{F}}(J)$ of buildings whose main components are curves of $\mathcal{M}^{nice}(J)$ of index i, except maybe if $i = 0$ in which case we allow such a curve to have 2 index-0 components inside $\mathcal{M}^{nice}(J)$, intersecting each other transversally at a singular node. Finally, take $\widehat{\mathcal{M}}^{\mathcal{F},nice}(J)$ to be the closure of $\widehat{\mathcal{M}}_2^{\mathcal{F},nice}(J)$ inside $\mathcal{M}^{\mathcal{F}}(J)\backslash\widehat{M}^{\mathcal{F}_+}(J_+)$.

Strategy of proof :
In order to show our theorem, we will need to show some properties on the moduli spaces defined above. The main steps of the proof are the following :

1) Show that the nicely embedded curves involved, and limits of sequences of such curves, have good properties from the point of view of regularity and intersection theory.

2) Use this to show that we can describe explicitly the elements of $\overline{\mathcal{M}}_i^{nice}(J)$ for $i = 0, 1, 2$.

3) Because we have enough information near the boundary elements of $\overline{\mathcal{M}}_i^{nice}(J)$, we can show that $\widehat{\mathcal{M}}^{\mathcal{F},nice}(J)$ is actually equal to $\mathcal{M}^{\mathcal{F}}(J)\backslash\widehat{M}^{\mathcal{F}_+}(J_+)$.

4) Now we know that all the curves that we consider are nicely embedded, we can use their properties to show that they foliate $\widehat{W}$.

5) Finally, $\widehat{\mathcal{M}}^{\mathcal{F},nice}(J)$ has a well defined smooth structure, so that there is a well defined smooth projection from a little completion of W to the moduli space. This gives us the desired structure on our space.

5.2.i Properties of nicely embedded curves:

In this section we recall the basic properties of nicely embedded curves, and write the lemmas we will need in the following. We refer to **[SOBD2]** for many proofs, except for the few ones where our situation differs from theirs.

Lemma 5.10. Consider u is a simple J-holomorphic curve in $\widehat{W}$, whose ends are covers of binding orbits. Then u is either a leaf of $\mathcal{F}_+$ inside the completion, or is intersecting the interior of W. In any case, u is regular, and in particular $\text{ind}(u) \geq 0$.

Proof. If u intersects the interior of W, then it intersects the domain where J is generic, and because u is somewhere injective, it is regular. Suppose now that u does not intersect the interior of W, then it can be seen as a curve inside $\mathbb{R} \times M$ which is J_+-holomorphic. Moreover, it only has positive ends. Because it is simple, it cannot be a multiple cover of a leaf. If it is not a leaf or a trivial cylinder, then it intersects a regular leaf at a finite number of points. But by intersection theory this is not possible, as by applying a homotopy of curves with cylindrical ends, this would amount to saying that the leaf has non-zero intersection number with the trivial cylinders appearing over its ends. However, we computed in the first part these intersection numbers, which were all zero. Hence u has to be a leaf, and in particular it is regular.

□

Remark 5.11. Using the same proof, we can show that any J_+- holomorphic curve in $\mathbb{R} \times M$ whose positive ends are all covers of binding orbits has to be a cover of a leaf. In particular, this applies for upper levels of buildings living in the completion $\widehat{W}$.

Lemma 5.12. Let $u^\infty \in \overline{\mathcal{M}}(J)$. Then any component in an upper level of u^∞ is a cover of a leaf of $\mathcal{F}_+$. Moreover, the main level of u^∞ is empty if and only if the u^∞ is a single regular page or a single positive rigid page.

Proof. The first part is a consequence of the previous remark, as the constraint on the energy of the curves in $\overline{\mathcal{M}}(J)$ implies that all positive ends are covers of binding orbits. The second part is a consequence of the fact that we chose all the binding orbits to have the same action, and that all the regular pages have the same number of ends : hence if a component of u^∞ is a regular page, then all other components of u^∞ have energy 0, so they are unbranched covers of trivial cylinders, which does not exist in W, so because u^∞ is connected and regular pages have simple ends, there is none. Similarly, if a component is a positive rigid page, then because of the asymptotic conditions, there need to be another level above this level in which u^∞ has a component which is a single negative rigid page. In that case, component formed with the positive rigid page and the negative rigid page has all the $d\alpha$-energy, hence we conclude in the same manner.
Conversely, if the main level of u^∞ is empty, by the previous lemma the components of u^∞ are covers of leafs of $\mathcal{F}_+$. However, the only leaves without negative ends are regular leaves or positive rigid leaves, hence a component of u^∞ is a regular leaf, which concludes. □

Remark 5.13. Note that there cannot be multiple covers of such leaves either, as this would contain too much $d\alpha$-energy.

This second lemma corresponds to lemma 6.12 in [**SOBD2**].

Lemma 5.14. For $u \in \overline{\mathcal{M}}^{nice}(J)$, all asymptotic orbits of u are at most doubly covered, $0 \leq \text{ind}(u) \leq 2$, $u * u = 0$ and :

$$2 = \text{ind}(u) + \#\Gamma_0 + 2\#\Gamma^2$$

where Γ_0 is the set of punctures asymptotic to even Reeb orbits, and Γ^2 corresponds to punctures asymptotic to doubly covered orbits. In particular, if u is a nicely embedded curve, $c_N(u) = -\#\Gamma^2$ hence u has positive index, it is regular.

Proof. First notice that these properties are invariant by homotopy, hence it suffices to show that it is true for a nicely embedded curve u.
By the previous lemma, we have : $\text{ind}(u) \geq 0$, and combining the definition of the normal Chern number with the adjunction formula yields :

$$\text{ind}(u) - 2 + \#\Gamma_0 = 2(u * u - [\overline{\sigma}(u) - \#\Gamma]) \leq -2[\overline{\sigma}(u) - \#\Gamma]$$

Hence for the same reason than in [**SOBD2**], the result will follow if we can show that for each asymptotic orbit γ^k of u with γ simple, there exists an asymptotic trivialisation τ for $\gamma^*\xi$, such that: $\alpha_-^\tau(\gamma^k) = 0$ (here we used the same notation τ for the trivialisation induced by the initial τ on $(\gamma^k)^*\xi$). However, because of the relation : $\mu_{CZ}^\tau(\gamma^k) = 2\alpha_-^\tau(\gamma^k) + p(\gamma)$, this amounts to asking that $\mu_{CZ}^\tau(\gamma^k) = 0$ if γ^k is even, and $\mu_{CZ}^\tau(\gamma^k) = 1$ if γ^k is odd. Now in our particular setup, we know that odd orbits are elliptic, and even orbits are positive hyperbolic. Moreover, we know that the energy of u is bounded from above by some constant E, hence k is also bounded from above. Recall that in the first part, we had natural trivialisations coming from the pages around elliptic and hyperbolic orbits. The condition we have here means the following :

- Around elliptic binding components, the Reeb vector field needs to turn "slow enough" so that the Conley-Zehnder index is small enough, so that in the natural trivialization given by the pages it is always 1 for any possible value of k.

- Around hyperbolic binding component, the trivialization follows the eigenspaces, hence the Conley-Zehnder index is always 0 for any cover, so there is nothing to check.

Now the first condition can always be achieved up to changing locally the contact form λ, without changing the action of the binding orbits, which gives us what we wanted.
This method only gives at first the equality :

$$2(u * u + 1) = \mathrm{ind}(u) + \#\Gamma_0 + 2\#\Gamma^2 \in \{0, 2\}$$

To get our result, notice that if we had : $u * u = -1$, the we would have : $\mathrm{ind}(u) + \#\Gamma_0 = 0$, hence : $\overline{\sigma}(u) - \#\Gamma = 1$. But we know that the previous argument tells us that $\overline{\sigma}(u) - \#\Gamma = \#\Gamma^2$, which is a contradiction because we also necessarily have $\#\Gamma^2 = 0$ in this situation. Hence $u * u = 0$.

To get the last statement, use the formula : $2c_N(u) = \mathrm{ind}(u) - 2 + \#\Gamma_0(u)$ which works for a nicely embedded curve. By the previous formula, $\mathrm{ind}(u) + \#\Gamma_0(u) = 2 - 2\#\Gamma^2(u)$ so $c_N(u) = -\#\Gamma^2(u)$. If $\mathrm{ind}(u) > 0$, in particular : $\mathrm{ind}(u) > c_N(u) + Z(du)$ as u is embedded, so the automatic transversality criterion is verified and u is regular.
Beaucoup de choses à préciser ici

□

The next lemma is a variation of lemma 4.9 of [**SOBD2**] in our setting.

Lemma 5.15. Suppose u is a connected stable holomorphic building in $\mathbb{R} \times M$, without nodes, with arithmetic genus 0, whose connected components are all covers of negative rigid leaves of $\mathcal{F}_+$, or of trivial cylinders over binding orbits. Denote by $\{\omega_i\}_{i \in I}$ the levels corresponding to branch covers over negative rigid leaves, and by p_i the associated leave. Then :

$$\mathrm{ind}(u) = -2 + \#\Gamma_0^+ + 2\#\Gamma_1^+ + \#\Gamma_0^- + 2\sum_{i \in I} k_i(1 - \#\Gamma_+(p_i))$$

Proof. The first part is a consequence of the index formula. Recall that if p_i is a rigid page, we can split $p_i^* TW$ as $T_{p_i} \oplus N_{p_i}$, with N_{p_i} the generalized normal bundle, and we have : $c_1^\tau(p_i) = c_1^\tau(T_{p_i}) + c_1^\tau(N_{p_i}) = \chi(p_i)$ because p_i is embedded, and in particular immersed. Now if one takes $\omega_i = p_i \circ \varphi_i$ to be a branched cover over p_i, we can look at the same decomposition, in the induced trivialization. We then have : $c_1^\tau(T_{\omega_i}) = \chi(\dot{\Sigma}) + Z(d\varphi_i)$, using the computation (3.16) from [**wendl_automatic_2009**], and because the only critical points of ω_i are the branch points of φ_i. We also have : $c_1^\tau(N_{\omega_i}) = 0$, because there still exist a global non-vanishing section of this bundle. Moreover, by the Riemann-Hurwitz formula, we have that $\chi(\dot{\Sigma}) + Z(d\varphi_i)$ is equal to $k_i \chi(p_i)$, with k_i the degree of φ_i. Finally, we know that $\chi(p_i) = 1 - \#\Gamma_+(p_i)$ as it only has one negative end. Putting all this together, we get :

$$\begin{aligned} \mathrm{ind}(u) &= -2 + \#\Gamma + \#\Gamma_1^+(u) - \#\Gamma_1^-(u) + 2\sum_{i \in I} k_i(1 - \#\Gamma_+(p_i)) \\ &= -2 + 2\#\Gamma_1^+(u) - \#\Gamma_1^-(u) + 2\sum_{i \in I} k_i(1 - \#\Gamma_+(p_i)) \end{aligned}$$

□

Remark 5.16. Unlike lemma 4.9 of [**SOBD2**], this does not directly imply strong bounds on the index as negative rigid page may not be cylinders. For example, consider a negative rigid page which is a pair-of-pants, and a triple cover of it given by a surface with 2 positive ends with branching index 3, and 3 negative ends, and not other branch points. Then the index of this cover is :

$$\mathrm{ind}(u) = -2 + 2 * 2 + 3 + 2 * 3 * (1 - 2) = 5 - 6 = -1$$

hence it has negative Fredholm index and cannot be regular, which is something that cannot happen when all negative rigid pages are cylinders.

Let us now fix u^∞ a holomorphic building in $\overline{M}^{nice}(J)$. We will denote as Γ the set of punctures, γ_z the set of asymptotic orbits for $z \in \Gamma$, Γ_0 and Γ_1 the set of even and odd orbits punctures respectively, Γ^m the set of punctures with covering multiplicity m. We assume that the bottommost level of u^∞ is non-empty, hence that there is no component of u^∞ which is a regular page of $\mathcal{F}_+$. Denote as $u_i = v_i \circ \varphi_i$ the connected components of the main level of u^∞, each u_i being a branched cover over a simple curve v_i, with branch covering φ_i, for i = 1,..., L. Denote as $k_i := deg(\varphi_i)$ the degree of φ_i, and by k_z the branching order of φ_i at z. We have the following relations, which are consequences of covering relations, Riemann-Hurwitz formula and the index formula :

$$\forall \zeta \in \Gamma(v_i),\ k_i = \sum_{z \in \varphi_i^{-1}(\zeta)} k_z$$

$$Z(d\varphi_i) = \sum_{z \in Crit(\varphi_i)} (k_z - 1)$$

$$\forall z \in \Gamma(u_i),\ \ \gamma_z = \gamma_{\varphi_i(z)}^{k_z}$$

$$Z(d\varphi_i) + \sum_{z \in \Gamma(u_i)} (k_z - 1) = 2(k_i - 1)$$

$$\mathrm{ind}(u_i) = -2 + \#\Gamma(u_i) + \#\Gamma_1(u_i) + 2c_1(u_i)$$

$$\mathrm{ind}(v_i) = -2 + \#\Gamma(v_i) + \#\Gamma_1(v_i) + 2c_1(v_i)$$

Putting all of this together, we get the relation :

$$\begin{aligned}
\mathrm{ind}(u_i) &= k_i \mathrm{ind}(v_i) + Z(d\varphi_i) - \sum_{z \in \Gamma_1(u_i)} (k_z - 1) \\
&= k_i[\mathrm{ind}(v_i) + \#\Gamma_0(v_i)] + Z(d\varphi_i) - \sum_{z \in \Gamma_1(u_i)} (k_z - 1) - \sum_{z \in \Gamma_0(u_i)} k_z
\end{aligned}$$

Finally, using the previous lemmas, we have the relations :
$\mathrm{ind}(v_i) \geq 0$ and : $\mathrm{ind}(u^\infty) = 2 - \#\Gamma_0(u^\infty) - 2\#\Gamma^2(u^\infty)$.
The last result of this part is Lemma 6.18 of **[SOBD2]**, which will be the main ingredient to characterize all possible behaviours of limits of sequences of nicely embedded curves. Its proof is mostly the same, noticing that the additional terms that appear cancel out.

Lemma 5.17. If u^∞ has no nodes, then :

$$\begin{aligned}
&\mathrm{ind}(u^\infty) + \Gamma_0(u^\infty) + 2\sum_{m \geq 2}(m-1)\#\Gamma^m(u^\infty) \\
&= 2(L-1) + \sum_{i=1}^{L}(k_i[\mathrm{ind}(v_i) + \#\Gamma_0(v_i)] + Z(d\varphi_i) + \sum_{\zeta \in \Gamma(v_i)} \sum_{z \in \varphi_i^{-1}(\zeta)} [k_z(2m_\zeta - 1) - 1]
\end{aligned}$$

Proof. Start by taking u^+ to be the (potentially unstable) building formed by all the upper levels of u^∞, adding trivial cylinders if they are initially empty. Denote by L_+ the number of connected components of u_+. For the same reason as in **[SOBD2]**, which is just a combinatorial argument based on the fact that u^∞ has arithmetic genus 0, we have the relation :

$$L_+ = -(L-1) + \sum_{i=1}^{L} \#\Gamma(u_i)$$

Combining this with the previous lemma, and using the fact that we have : $\#\Gamma_+(u_+) = \#\Gamma(u^\infty)$, we get the following relation :

$$\begin{aligned}\mathrm{ind}(u^+) &= -2L_+ + \#\Gamma_0(u^\infty) + 2\#\Gamma_1(u^\infty) + \sum_{i=1}^{L} \#\Gamma_0(u_i) + 2\sum_{i\in I} k_i(1-\#\Gamma_+(p_i)) \\ &= 2(L-1) - \sum_{i=1}^{L}[\#\Gamma_0(u_i) + 2\#\Gamma_1(u_i)] + \#\Gamma_0(u^\infty) + 2\#\Gamma_1(u^\infty) + 2\sum_{i\in I} k_i(1-\#\Gamma_+(p_i))\end{aligned}$$

Moreover, denoting by m_z the degree of a puncture, we have the following relation :

$$\sum_{z\in\Gamma(u^\infty)} m_z = \sum_{i=1}^{L}\sum_{z\in\Gamma(u_i)} m_z + \sum_{i\in I} k_i(1-\#\Gamma_+(p_i)) = \sum_{i=1}^{L}\sum_{\zeta\in\Gamma(v_i)}\sum_{z\in\varphi_i^{-1}(\zeta)} k_z m_\zeta + \sum_{i\in I} k_i(\#\Gamma_+(p_i)-1)$$

To see why we can this relation between the sum of multiplicities, observe that as all negative rigid curves and trivial cylinders that are considered have simple ends, this is indeed true for a simple parametrization of these : for a trivial cylinder, the multiplicity is 1 for positive and negative ends, and for a negative rigid page, the multiplicity for positive ends is $\#\Gamma_+(p_i)$ and the multiplicity for negative ends is 1, so the difference is $(1-\#\Gamma_+(p_i))$. Now taking a branched cover of one of these curves amounts to multiplying the sum of multiplicities for positive and negative ends by the degree of the covering map, and this does not depend on the presence or not of a branch point on a puncture. Hence the correction term is now $k_i(1-\#\Gamma_+(p_i))$ for each component which is a branched cover of a negative rigid curve p_i, and this gives the stated equality. Observe that unlike in [**SOBD2**], there is no need of a $\mathbb{S}^1$-symmetry to get these result, as all our rigid curves have simple ends.

Now using this formula, we get the relation :

$$\begin{aligned}\#\Gamma_0(u^\infty) &+ 2\sum_{m\geq 2}(m-1)\#\Gamma^m(u^\infty) \\ &= \#\Gamma_0(u^\infty) + 2\sum_{z\in\Gamma(u^\infty)} m_z - 2\#\Gamma(u^\infty) \\ &= -\#\Gamma_0(u^\infty) - 2\#\Gamma_1(u^\infty) + 2\sum_{i=1}^{L}\sum_{\zeta\in\Gamma(v_i)}\sum_{z\in\varphi_i^{-1}(\zeta)} k_z m_\zeta + 2\sum_{i\in I} k_i(\#\Gamma_+(p_i)-1)\end{aligned}$$

Finally, we combine everything, and by writing $\mathrm{ind}(u^\infty) = \mathrm{ind}(u^+) + \sum_{i=1}^{L}\mathrm{ind}(u_i)$, we get :

$$\begin{aligned}\mathrm{ind}(u^\infty) &+ \#\Gamma_0(u^\infty) + 2\sum_{m\geq 2}(m-1)\#\Gamma^m(u^\infty) \\ &= 2(L-1) - \sum_{i=1}^{L}[\#\Gamma_0(u_i) + 2\#\Gamma_1(u_i)] + \#\Gamma_0(u^\infty) + 2\#\Gamma_1(u^\infty) + 2\sum_{i\in I} k_i(1-\#\Gamma_+(p_i)) \\ &+ k_i[\mathrm{ind}(v_i) + \#\Gamma_0(v_i)] + Z(d\varphi_i) - \sum_{z\in\Gamma_1(u_i)}(k_z-1) - \sum_{z\in\Gamma_0(u_i)} k_z \\ &- \#\Gamma_0(u^\infty) - 2\#\Gamma_1(u^\infty) + 2\sum_{i=1}^{L}\sum_{\zeta\in\Gamma(v_i)}\sum_{z\in\varphi_i^{-1}(\zeta)} k_z m_\zeta + 2\sum_{i\in I} k_i(\#\Gamma_+(p_i)-1)\end{aligned}$$

Now notice that the terms $2\sum_{i\in I} k_i(1-\#\Gamma_+(p_i))$ cancel out, as well as the terms $\#\Gamma_0(u^\infty)+2\#\Gamma_1(u^\infty)$. Putting together the terms inside the sums over i and separating them according to the parity of the asymptotic orbits, we get the result.

□

The last lemma of this section is lemma 6.21 of [**SOBD2**], and allows one to control the self-intersection of the building u^∞.

Lemma 5.18. We have the relation :

$$\sum_{i,j=1}^{L} u_i * u_j = u^\infty * u^\infty$$

where the term on the right is to be understood in the sense of the intersection product for buildings (see [**wendl_contact_2019**] section C.5).

Proof. For the same reason as in a previous lemma, up to choosing well the contact form, we can arrange the relations : $\alpha_-^\tau(\gamma^k) = 0$ for all "admissible" k, all binding orbits γ, and with τ the natural trivialisation. By lemma 2.7 of [**SOBD2**], this implies that :

$$u^\infty * u^\infty = u^\infty \bullet_\tau u^\infty$$

where the right hand term is the actual intersection number computed from a perturbation given by the trivialisation τ. Now take u, v two components of u^∞ leaving in upper levels $\mathbb{R}\times M$. They are covers of trivial cylinders or rigid curves, hence we can compute explicitly $u \bullet_\tau v$: if one of them is a cover of a trivial cylinder, this is 0 as we can remove intersections with the cylinder by pushing a little bit in a direction given by τ. If they are covers of negative rigid pages, then we know that we have the relation $\tilde{u} \bullet_\tau v = d(u \bullet_\tau v)$ if $\tilde{u}$ is a d-fold cover of u. Hence it suffices to compute $u \bullet_\tau v$ for u, v simple parametrizations of negative rigid curves. But by the same argument as before, this is $u * v$. We saw that this number was 0, which concludes. □

5.2.ii Compactness

We got all the desired bounds and control to now state explicitly the possible objects appearing as the limit of sequences of nicely embedded curves of index 0, 1 or 2 in $\mathcal{M}^{\mathcal{F},nice}(J)$. We refer to [**SOBD2**], proposition 6.13 for the complete list of possible degenerations. Morally, what we want to show is the following :

Proposition 5.19. If u^∞ is an element of $\overline{\mathcal{M}}_i^{nice}(J)$, $i = 0, 1, 2$, then the only possible singularities are :

- Lefschetz-type nodal degenerations, which is a transverse intersection of index-0 curves.
- Buildings with 1 upper level, which consists of either : an index-1 curve below, and a simple negative rigid page upstairs ; or of an index-0 curve below and 2 simple negative rigid pages upstairs. The breaking orbit are broken binding components of the broken book.
- Exotic fibers, i.e buildings which consist of an index-0 curve in $\widehat{W}$, and a double cover of a trivial cylinder over an elliptic orbit on the first level.

In every case, all the curves except for the double cover of the trivial cylinder are nicely embedded ; and every other element in $\overline{\mathcal{M}}_i^{nice}(J)$ is also a simple nicely embedded curve. All the curves considered are also Fredholm regular.

In order to show this, we first sum up the consequences of the last part, which correspond to lemmas 6.19, 6.20, 6.23 of [**SOBD2**]. Lemma 6.22 is not necessary as we showed that we always had $u^\infty * u^\infty = 0$. The proofs are mostly "combinatorials", and base on the lemmas and formulas we showed in the previous part, so they work exactly the same in our setting and nothing need to be changed. All of these can be summarized as follow :

Lemma 5.20. Take $u^\infty \in \overline{\mathcal{M}}^{nice}(J)$ as before. Then the main level of u^∞ either has 2 component which are both nicely embedded curves intersecting transversely at a node and with odd asymptotic orbits, or has 1 component which is also a nicely embedded curve, and can be :

- An index 2 curve with only odd ends.
- An index 1 curve with any number of odd ends, and exactly one even end.
- An index 0 curve with any number of odd ends, and exactly two even ends.
- An index 0 curve with only odd ends, but one of them exactly is doubly covered.

Remark 5.21. The first case corresponds to Lefschetz-type singularities, and the different options in the second case correspond respectively to : regular pages, (positive) rigid pages, intersection between "walls" of rigid pages and exotic fibers.

We will refer to the previous possibilities as $1, 2.a, 2.b, 2.c$ and $2.d$ respectively.
In order to conclude on the possible type of curves we get, it remains to show that the upper levels are easy to understand.

Lemma 5.22. The upper levels of u^∞ are the following :

- In case 1 and $2.a$, there are none.
- In case $2.b$ and $2.c$ there is 1 or 2 simple parametrizations of negative rigid curves, "glued" to the main level on the even hyperbolic orbits.
- In case $2.d$, there is a branched double cover of a trivial cylinder over the elliptic orbit the is doubly covered by an end of the main level.

Proof. First notice that if all odd orbits asymptotic to the curves in the main level are simply covered, then for genus reason, all the curves in upper levels are simply covered. Moreover, we saw that they had to be covers of leaves, and we assumed that there was no cover of regular pages or positive rigid pages (in these cases, we know exactly what happens already). Hence they are either covers of trivial cylinders of of negative rigid pages. Hence in cases 1, $2.a$, $2.b$ and $2.c$, the only possible behaviours are : no upper levels in case 1 and $2.a$, 1 upper level in case $2.b$ with a single simply covered negative rigid page in it, and 1 upper level in case $2.c$ with two simply covered negative rigid pages in it.
Now in case $2.d$, as there is no even orbit, the only thing that can appear in an upper level is a double cover of a trivial cylinder over an elliptic binding orbit. The possible double covers are : an unbranched double cover of a trivial cylinder, or a branched double cover of a pair of pants with one negative end and two positive ends over a trivial cylinder. Moreover, we know that u^∞ has at most 1 upper level. In the first case, the index of u^∞ is 0. However, we know that $\mathcal{M}_0^{nice}(J)$ consists of a finite set of points, as these are non-intersecting isolated curve ; hence it coincides with $\overline{\mathcal{M}}_0^{nice}(J)$ and there can be no building with index 0 and an upper level in $\overline{\mathcal{M}}^{nice}(J)$, so this first case cannot happen. In the case of a branched double cover, the index of the upper level is 2, hence the total index of u^∞ is also 2 which is possible, and corresponds to the last case listed above. Finally, note that in this last cast, the upper curve is a branched double cover of a pair-of pants over a trivial cylinder : the curve has index 2, has 1 critical point has the trivial cylinder has none and the covering map has one, and its normal Chern number is, by definition :

$$c_N(u) = \text{ind}(u) - 2 + 2g + \#\Gamma_0(u) = \text{ind}(u) - 2 = 0$$

Hence we have the inequality : $\operatorname{ind}(u) > Z(du) + c_N(u)$, so the automatic transversality criterion of [**wendl_automatic_2009**] is verified and even this double cover is Fredholm regular. □

Remark 5.23. This last precision about the regularity of the double cover is actually important in what will follow, as it allows one apply a gluing theorem to this curve.

This concludes the classification of possible degenerations.

5.2.iii Extension of the foliation

Now that we have seen how nicely embedded curves degenerate in our setting, we are ready to show that the initial curves of the foliation "spread" through the filling and eventually foliate the whole space, with specific singularities that we are able to describe.

Lemma 5.24. If $[u] \in \widehat{\mathcal{M}}_2^{\mathcal{F},nice}(J)$, then there exists an open neighbourhood of $[u]$ inside $\widehat{\mathcal{M}}^{\mathcal{F}}(J)$ such that it forms a 2-parameter family of curves whose main levels are disjoint index-2 nicely embedded curves that foliate an open set of $\widehat{W}$.

Proof. This follows as in [**SOBD2**] from the following facts : let v be the main level, it is nicely embedded and has index 2, so : $2 = \operatorname{ind}(v) + \#\Gamma_0(v) + 2\#\Gamma^2(v)$, so $\#\Gamma_0(v) = 0$. Moreover, one has that $2c_N(v) = \#\Gamma_0(v) = 0$, and v is embedded so : $\operatorname{ind}(v) > c_N(v) + Z(dv)$ which means the automatic transversality criterion is verified. This means that there is an open set around v in $\mathcal{M}_2^{nice}(J)$ which are disjoint curves (because $v * v = 0$ as noticed previously), foliating an open subset of $\widehat{W}$. From this, we can get an open subset in $\widehat{\mathcal{M}}^{\mathcal{F}}(J)$ because u cannot have non-trivial upper levels, as all the ends of v are simple and elliptic, and by the asymptotic conditions imposed on the curves in $\mathcal{M}^{\mathcal{F}_+}(J_+)$. □

Lemma 5.25. If $[u] \in \widehat{\mathcal{M}}_1^{\mathcal{F},nice}(J)$, then there is an open neighbourhood $\mathcal{U}$ of $[u]$ in $\widehat{\mathcal{M}}^{\mathcal{F}}(J)$ and a homeomorphism : $\Psi :]-1,1[^2 \to \mathcal{U}$ such that : $\forall (x,y) \in]-1,1[^2$, $\Psi(0,y) \in \widehat{\mathcal{M}}_1^{\mathcal{F},nice}(J)$ and $\Psi(x,y) \in \widehat{\mathcal{M}}_2^{\mathcal{F},nice}(J)$ if $x \neq 0$. Moreover, we also have that the main levels of these curves are disjoint and foliate an open subset of $\widehat{W}$.

Proof. The argument is the same than in [**SOBD2**]. The only difference is that here, we will glue the pair of negative rigid pages to the main level in order to get the index-2 curves in each side of the "wall". Note that by the condition that the broken book is balanced, this gives curves with the same energy.
For the sake of completeness, let us sum up again the arguments :
First we use the same argument than in the proof of the last lemma. This is, we have : $2 = \operatorname{ind}(v) + \#\Gamma_0(v) + 2\#\Gamma^2(v)$ hence $\#\Gamma_0(v) = 1$, and $2c_N(v) = -1 + 1 = 0$ again. v is also again regular, and we get a 1-parameter family of disjoint curves. Now v has one end at a hyperbolic broken orbit, hence there are 2 negative regular pages in the symplectization that can be glued to v (all the curves are regular). This gives index-2 nicely embedded curves, because the glued curves verify : $w * w = 0$, by doing the same intersection computations as before, and $\delta(w) = \delta_\infty(w)$ (for example because of the adjunction formula). Hence using the previous lemma, a neighbourhood of the 3-dimensional "wall" is foliated by these index 2 curves, which gives the stated result. □

We now look at $\widehat{\mathcal{M}}^{\mathcal{F},nice}(J)$, which is the closure of the union of the previously studied spaces, minus the curves of $\mathcal{M}^{\mathcal{F}_+}(J_+)$ (these are curves or building of the initial foliation, without a main level). Now elements of $\widehat{\mathcal{M}}^{\mathcal{F},nice}(J)$ which are not curves in $\widehat{\mathcal{M}}_1^{\mathcal{F},nice}(J)$ or $\widehat{\mathcal{M}}_2^{\mathcal{F},nice}(J)$ have been studied in the last part as they are limits of nicely embedded curves. Hence these can only be curves of building, which corresponds to cases $1., 2.c, 2.d$ which we studied before. These are exactly the elements inside $\mathcal{M}_0^{\mathcal{F},nice}(J)$.

Lemma 5.26. $\widehat{\mathcal{M}}^{\mathcal{F},nice}(J) = \widehat{M}^{\mathcal{F}}(J) \backslash (\widehat{\mathcal{M}}^{\mathcal{F}_+}(J_+))$ and all buildings in this moduli space satisfy $u * u' = 0$.

Proof. Once again the proof is exactly the same as in [**SOBD2**] and will not be reproduced in full details here. It simply relies on the fact that the only thing remaining to show is that an open neighborhood in $\widehat{\mathcal{M}}^{\mathcal{F}}(J)$ of Lefschetz singularities, rigid nicely embedded curves with 2 even ends, and exotic fibers, is composed exclusively of nicely embedded curves. For Lefschetz fibers, this is true as there is a degenerating 2-parameter family of nicely embedded curves that foliate a neighbourhood of the fiber ; for rigid curves this is also true because the only possible choices for upper levels respecting the asymptotic condition are the 2 negative rigid pages of the corresponding pair, and gluing any of these (or both) gives a nicely embedded curve. Finally, for exotic fibers, we saw that automatic transversality still applied on the double branched cover of a trivial cylinder in the upper level, which is the only possibility of upper level that respects the asymptotic conditions, hence we can glue and once again, and this still gives a nicely embedded curve by invariance of the intersection product. □

Finally, we get the conclusion we wanted :

Lemma 5.27. Every point of $\widehat{W}$ is in the image of a unique curve of $\widehat{\mathcal{M}}^{\mathcal{F},nice}(J)$.

5.2.iv Structure on the filling

Finally, we explain how to get the structure of SNBLF on the filling from what we got in the previous parts. This will also justify the definition that was chosen for a SNBLF, as a generalization of NBLF.

Define the projection $\hat{\pi} : \widehat{W} \to \widehat{\mathcal{M}}^{\mathcal{F}}(J)$ to be the projection which sends each point to the curve that goes through that point. This can be restricted to an extension of W, denoted W_R, see [**SOBD2**] for the construction. We can assume that all nodal singularities are inside the interior of W_R. We denote as $\pi : W_R \to \mathcal{M}_0$ the restriction of $\hat{\pi}$ to this domain.

Lemma 5.28. There exists a smooth structure on $\widehat{\mathcal{M}}^{\mathcal{F}}(J)$ such that π is smooth, and $\mathcal{M}_0$ is a compact oriented surface with boundary. Moreover, we can choose W_R such that its boundary naturally comes with a broken book decomposition, with symplectic fibers ?

Proof. The first part of this lemma follows from what was showed before, for the same reason as in [**SOBD2**]. Namely, one can use the holomorphic foliation obtained before in order to define smooth charts on $\mathcal{M}^{\mathcal{F}}(J)$.

A neighbourhood of nodal singularities then looks like Lefschetz singularities.

Moreover, we can use the model from [**min_spinal_2024**] to study a neighbourhood of exotic fibers (note that here there is no condition such as Lefschetz amenability that could allow one to exclude such fibers from existing). Specifically, propostion 4.9 which describes the local model around an exotic fiber still applies, which allows us to describe locally near the intersection of such an exotic fiber with $\partial_h W$ the holomorphic foliation as the level sets of :

$$\begin{aligned} \pi :& (\mathbb{C}\backslash\{0\}) \times \mathbb{C} \to \mathbb{C} \\ & (z_1, z_2) \mapsto z_2^2 - z_1 \end{aligned}$$

Finally, it remains to describe how the foliation behaves on $\partial_h W$, in particular near the walls. Like for exotic fibers, while initial holomorphic walls consisted of 1-parameter family of index-1 holomorphic curves, here we truncate $\widehat{W}$ which result of a "thickening" of the walls. These are indeed not visible in the case of spinal open books because negative rigid pages are always cylinders, however in our case this creates the apparition of singularities in the horizontal boundary.

□

Remark 5.29. In particular, this implies that $\mathcal{M}_0$ is a connected surface (and so is $\widehat{\mathcal{M}}^{\mathcal{F}}(J)$) which was not obvious from the definition.

Remark 5.30. In fact, one could rule out the existence of *certain* exotic fibers in the previous construction. Suppose there exists a maximal generalized spine component which is simple. Recall this means that all negative rigid pages inside it are cylinders, i.e that we are working with a domain of the form $\Sigma \times \mathbb{S}^1$. Then one can use the "Lefschetz-amenability condition" in order to rule out the existence of exotic fibers with an end in this spine component. Namely, for any strong filling, there cannot be exotic fibers with ends inside this component if the following condition is verified:
"For any compact oriented surface Σ_0 such that $\#\partial\Sigma_0 = \#\pi_0(M_0)$,

Question 6. Can a weakly planar contact 3-manifold (M, ξ) be strongly fillable if one of the following situations happens?

- ξ is not supported by a strongly balanced planar broken book.
- ξ is not supported by a balanced planar broken book.
- ξ is supported by a (strongly balanced) planar broken book, with a cylindrical negative rigid page having a non-trivial characteristic foliation.

Note that for now there does not even exist any explicit example of a contact manifold verifying one of these conditions.

7 The Weinstein conjecture in dimension 3 and 5

Using the results of the previous sections, we are now able to prove the Weinstein conjecture for certain classes of contact structures in dimension 3 and 5, in the spirit of [**abbas_weinstein_2005**]. First let us recall the sketch of the proof for planar contact structures in dimension 3.

Theorem 7.1. **[abbas_weinstein_2005]** *Let (M^3, ξ) be a 3-dimensional contact manifold, supported by a planar open book. Then it verifies the Weinstein conjecture.*

Proof. Consider α_0 the Giroux form for the planar open book, and λ any other contact form for ξ. Construct a cobordism $\mathbb{R}_a \times M$, with a positive end isomorphic to the symplectization of α_0, and a negative end isomorphic to the symplectization of λ. This is possible up to multiplying α_0 by a constant large enough. Take J a compatible almost complex structure cylindrical on the ends, such that the pages of the planar open book lift as index 2 curves in the positive end. Consider u_0 such a curve. u_0 is a nicely embedded regular index 2 curve, with self-intersection 0.
Consider $\mathcal{M}^{u_0}$ the connected component of u_0 inside the moduli space of asymptotically cylindrical finite energy maps with the right asymptotics. For $u = (a, f) \in \mathcal{M}^{u_0}$, define $m(u) := \min\{a(z) \,|\, z \in \dot{\Sigma}\}$ where $\dot{\Sigma}$ is the domain of u. Define $M = \inf\{m(u) \,|\, u \in \mathcal{M}^{u_0}\}$. Assume $M > +\infty$. Now $\mathcal{M}^{u_0}$ is a 2-dimensional smooth manifold and each curve inside it is a nicely embedded regular index 2 curve, and two of these never intersect. Take a sequence u_n such that $m(u_n) \to M$. Up to a subsequence, u_n convergences to some nodal holomorphic building u_∞. If u_∞ has an underground level, it breaks on a Reeb orbit for λ and we get what we want. If not, suppose it has an upper level. Then by intersection theory, it has to be a page of the open book in the symplectization of α_0, which is not possible by definition of u_n. Hence it only has one level. It is either a curve of $\mathcal{M}^{u_0}$, or a nodal curve with one nodal point, and in any case it verifies $m(u_\infty) = M$. In either case, as we saw in the previous part, a neighborhood of this curve is foliated by elements of $\mathcal{M}^{u_0}$, so there exists $u \in \mathcal{M}^{u_0}$ such that $m(u) < M$, which is a contradiction. Hence $M = -\infty$, and taking again the limit of a sequence u_n such that $m(u_n) \to -\infty$ yields a Reeb orbit for λ. ☐

By a short adaptation of this proof, we get the same result for broken books.

Theorem 7.2. *Let (M^3,ξ) be a contact structure nicely supported by a liftable planar balanced broken book. Then it verifies the Weinstein conjecture.*

Remark 7.3. Note that in particular contact structures supported by planar spinal open books are included in this theorem.

Proof. Consider the same cobordism as before, with α_0 the Giroux form for the planar balanced broken book, and λ any contact form for ξ. Consider $\widehat{\mathcal{M}}^{\mathcal{F}}(J)$ the same moduli space as before, i.e the quotiented compactified moduli space of holomorphic curves in the same connected component as the curves of the initial foliation living "at infinity". Now, as in the last proof, define : $M := \inf\{m(u)\,|\,u \in \widehat{M}^{\mathcal{F}}(J)\}$, where $m := \min\{a(z)\,|\,z \in \dot{\Sigma}\}$. Suppose M is not $-\infty$, otherwise there is necessarily a closed Reeb orbit for λ. Then because we know that $\widehat{M}^{\mathcal{F}}(J)$ is compact, and $m : \widehat{\mathcal{M}}^{\mathcal{F}}(J)$ is continuous, M is in its image. Now take u such that $m(u) = M$. By the compactness results shown in the last part, the main level of u is a nicely embedded curve, and a neighbourhood of the image of u is foliated by holomorphic curves of $\widehat{\mathcal{M}}^{\mathcal{F}}(J)$ (these results did not depend on the compactness of the symplectic filling). Hence there exists v such that $m(v) < m(u)$, which is a contradiction. □

Now we want to focus on 5-dimensional contact manifolds. We start by giving the details of the proof, filling in some missing details from [**acu_weinstein_2017**]. The idea is to perform some sort of "neck-stretching" in the symplectization, and to show that a family of curves will persist throughout the deformation, which shows the existence of a periodic orbit in the end.

We start by giving some details on the explicit model we use. This was done in [**roux_cas_nodate**], base on [**wendl_open_2009**] for the 3-dimensional case. See also cite(papier variétés suturées)
First start with (M,ξ) a 5-dimensional contact manifold whose contact structure is supported by an open book decomposition (Γ,p) such that $(\Gamma,\xi_{|\Gamma})$ is a 3-dimensional planar contact manifold. We will see it as the realization of an abstract open book (here we use the letter Γ by analogy with the sutured case).

Take W a Weinstein domain corresponding to a page, with symplectic form $\omega = d\beta$, and $\beta = e^{-\rho}\beta_0$ near the boundary (with ρ the coordinate given by Liouville flow in negative time), where β_0 is contact form on Γ such that the pages of the planar open book for ξ_Γ lifts as J_0-holomorphic curves, with $J_0 \in \mathcal{J}(\beta_0)$. Now our open book also depends on a choice of exact symplectomorphism W_Ψ equal to the identity near the boundary (in fact, the exactness assumption is not necessary as up to isotopy one can always arrange for such a symplectomorphism to be exact). Define the mapping torus W_Ψ, and the contact form $d\phi + \beta$ where $d\phi$ is the angular coordinate. We can then glue back $\mathbb{D}^2_{(\rho,\phi)} \times \Gamma$, and extend the contact form with the following construction :

$$\alpha = \begin{cases} d\phi + \beta & \text{for} \\ f(\rho)\beta_0 + g(\rho)d\phi & \text{for } \rho < 1+\delta \end{cases}$$

where $\delta > 0$ is small, and $f = f(\rho)$ and $g = g(\rho)$ are smooth maps $[0,1] \to \mathbb{R}$ verifying the following properties :

1. f is decreasing, takes value $b > 1$ at 0 and coincides with $e^{1-\rho}$ near 1.
2. g is increasing, takes value 0 at 0 and 1 in a neighbourhood of 1.
3. $f(\rho)$ and $\frac{g(\rho)}{\rho^2}$ are smooth at $\rho = 0$.
4. Writing $D(\rho) = fg' - gf'$, then $D > 0$ everywhere.
5. $\exists \eta > 0$ such that $-\frac{g'}{f'} \leq \frac{1}{b} + \eta \in \mathbb{R}\backslash\mathbb{Q}$, with equality for $\rho \leq \frac{2}{3}$.
6. For $\rho \geq \frac{2}{3}$, we have $g(\rho) \geq \frac{2}{3}$ and $f(\rho) \leq 1 + \frac{b}{3}$.

Such maps exist, see [**roux_cas_nodate**],[**wendl_open_2009**],[**wendl_holomorphic_2010**]. In particular, we have : $D(1) = e^{1-\rho} > 0$, so condition 4) is compatible with the others. For another reference of this construction, see [**geiges_introduction_nodate**], theorem 7.3.3.

From this construction, it is possible to construct an α-compatible almost complex structure J which allows one to "lift" pages of the open book as complex hypersurfaces, isomorphic to the completion of the these pages seen as Weinstein domains (see article variétés suturées et [**roux_cas_nodate**]). Hence, the finite energy foliation living inside $\mathbb{R} \times \Gamma$ from the planar open book decomposition extends inside each page as a Lefschetz fibration. Moreover, using higher dimensional intersection theory and a little bit of normal bundle analysis encore les mêmes articles, one can show the following results :

Lemma 7.4. The curves appearing in the above foliation are regular, and their index is 4 if they come from regular pages of the initial foliation, 2 if they come from components of nodal curves (i.e singular fibers of the Lefschetz fibration), and 0 for those coming from trivial cylinders.

Lemma 7.5. Any J-holomorphic curve in $\mathbb{R} \times M$ whose positive ends are all binding components inside Γ_0 is a curve of the previous foliation.

Hence define $\mathcal{M}_+(J)$ the moduli space of all these unparametrized curves modulo $\mathbb{R}$-translation ; it is a smooth manifold whose compactification is diffeomorphic to $\mathbb{S}^3$ ($\mathcal{M}_+(J)$ itself is $\mathbb{S}^3$ with a finite set of disjoint unlinked unknots removed).

Let us now define the defomration of almost complex structure that we will use. Take λ a contact form for ξ the same contact structure as before. Take a smooth generic path of almost complex structures $\{J_\tau\}$ on $\mathbb{R}_a \times M$ verifying the following conditions.

First, J_1 is a generic λ-compatible almost complex structure. Then, choose an increasing function $h : [0, 1[\to \mathbb{R}$ such that $h(0) < -2$ and $\lim\limits_{\tau \to 1} h(\tau) = +\infty$. Now for $\tau < 1$, we ask J_τ to coincide with J_1 for $a < h(\tau)$ and with J for $a > h(\tau) + 1$. This is possible up to shinking λ enough by a multipicative constant, which does not change the existence or not of a closed Reeb orbit.

Now take $x_0 \in M$ such that x_0 belongs to the image of a regular curve of index 4, and consider consider u_0 the unique curve such that $(0, x_0) \in \mathrm{Im}(u_0)$ in the symplectization (we have uniqueness dues to the previous lemma). The strategy of proof is the following : show that for each time $\tau < 1$, there exists a J_τ-holomorphic curve such that (ϵ, x_0) is in the image of this curve for ϵ arbitrarily small. By SFT compactness, this will be enough to conclude as it will show the existence of a J_1-holomorphic building with non-empty main level, which implies the existence of a Reeb orbit for λ. Note that we can apply the SFT compactness theorem as this setting is close enough from to a stretching of the neck : consider the hypersurface $\{h(0)\} \times M$. Then stretching the neck along this hypersurface yields an equivalent deformation of the almost complex structure in the upper part of the symplectization ; the lower part is invariant and λ-compatible so there is no issue there.

Let $(\tau_n)_{n\in\mathbb{N}}$ be a sequence of times such that $\lim\limits_{n\to+\infty} \tau_n = 1$.

Proposition 7.6. There exist a small generic perturbation x of x_0 and $\epsilon > 0$ arbitrarily small, such that for all n, there exists a J_{τ_n}-holomorphic curve which contains (ϵ, x) in its image.

Proof. Define $\mathcal{M}(\{J_\tau\}, \gamma)$ to be the parametric moduli space of tuples (τ, u) where $\tau \in [0, 1]$ and u is J_τ-holomorphic, with positive asymptotic ends at binding orbits of the open book. We will simply write $\mathcal{M}$ for shorthand, and $\mathcal{M}_1$ for the space of curves with a marked point. Chosing our path of almost complex structures generically allows to consider that these moduli spaces are smooth manifolds, as all the curves have simple ends. Moreover, $\mathcal{M}$ is 5 dimensional and $\mathcal{M}_1$ is 7-dimensional.
Fix $n \geq 0$ and define : $T = \{\tau \in [0, 1] \mid \exists(\tau, u_\tau) \in \mathcal{M},\ s.t\ (0, x) \in \mathrm{Im}(u_\tau)\}$. We will show that up to

a generic perturbation of $(0, x)$, we have that : $\sup T = 1$, which is sufficient in order to conclude (the supremum is well defined as $0 \in T$ as $(0, x)$ belongs to the initial foliation).

Consider the following map :

$$\begin{aligned} F : \mathcal{M}_1 \times \mathcal{U} &\to W \times W \\ (\tau, [u, z], x) &\mapsto (u(z), x) \end{aligned}$$

where $\mathcal{U}$ is a small neighbourhood around $(0, x_0)$ in W. F is obviously transverse to the diagonal, hence $\tilde{M}_1 := F^{-1}(\Delta)$ is a smooth manifold of dimension 7.
Define :

$$\begin{aligned} \pi : \ \tilde{M}_1 &\to W \\ (\tau, [u, z], x) &\mapsto x \end{aligned}$$

Then π is smooth, and using Sard theorem, we get that its regular values are generic. Moreover, $\mathcal{U}$ is in the image of π and is an open set of W hence there is a regular value of π inside $\mathcal{U}$ (chosen as a "generic small perturbation of x_0"). Indeed, for all x in $\mathcal{U}$, there is a J_0-holomorphic curve of $\mathcal{M}$ whose image contain x. Choose such an element x. Then $\pi^{-1}(x)$ is a smooth 1-dimensional manifold, which we will denote by I.

Take the compactification $\overline{\mathcal{M}_1}$ of $\mathcal{M}_1$, and denote by S the set $\overline{\mathcal{M}_1} \backslash \mathcal{M}_1$. If we can show that curves in I cannot converge in the SFT topology to a curve in S, then we get the result : indeed, in that case I is a compact one-dimensional manifold of $\mathcal{M}_1$, with exactly one boundary component which is a J_0-holomorphic curve. this means there is another boundary component on the "other side" of $\mathcal{M}_1$, i.e a J_1-holomorphic curve which is in I, which is what we want. However, in general this might not be the case as some nodal degeneracy can appear.

Consider a sequence (τ_n, u_n) of curves in I, which converges for the SFT topology to a curve (J_τ, u_∞).

Lemma 7.7. u_∞ is a nodal curve with an arbitrary number of nodes (possibly 0) and whose components are either one index-4 curve, or two index-2 curves, and an arbitrary number of index-0 curves and ghost bubbles.

Proof. The main level of u_∞ is non-empty by definition, as $x \in \mathrm{Im}(u_\infty)$. However, because of the asymptotic conditions and intersection properties, no upper level of u_∞ can have a negative end, except if it were a branched cover of a trivial cylinder over a binding orbit, which is also not possible as all ends are simple. Hence because u_∞ is connected, it has no upper level. Now if the main level has nodes, we can assume that all components are regular by the genericity assumption (this is true if we consider them as part of parametric moduli spaces). In this case, nodes have a contribution of 0 to the index. Moreover, because all components are regular "parametrically", they have index at least -1. However, the index formula is :

$$\mathrm{ind}(u) = (n - 3)\chi(\dot{\Sigma}) + 2c_1^\tau(u) + \sum_{\gamma \in \boldsymbol{\gamma}} \mu_{CZ}^\tau(\gamma)$$

where τ is the trivialization naturally induced by the open book construction defined above (in particular, it splits as $\tau_1 \oplus \tau_2$ where τ_1 corresponds to the trivialization coming from the 3-dimensional open book, and τ_2 comes from the "5-dimensional open book"). Here, n is the complex dimension so it is 3, and all Conley-Zehnder indices are even by construction (in fact it is exactly 2). Hence the index of all components is always even, and is either $0, 2$ or 4. We know that the sum of these indices is 4, so we get the result. $\square$

We can now finish the proof of the proposition.

$\square$

Remark 7.8. What makes the proof work here is that the space of singular curves in the compactification has codimension at least 2, which roughly means that we can ignore them generically. We will see that in the case of broken books, this does not hold anymore because of index-1 curves (the "walls" mentioned in the last part).

Let us now consider balanced planar broken books. Suppose there is a contact structure on M which is supported by an open book (Γ, p) with $(\Gamma, \xi_{|\Gamma})$ weakly planar. We apply the same construction as before and get a foliation by holomorphic curves of $W := \mathbb{R} \times M$.

Lemma 7.9. The different type of lifted pages have the following indices :

- $\operatorname{ind}(u) = 4$ if u is a regular page.
- $\operatorname{ind}(u) = 3$ if u is a positive rigid page.
- $\operatorname{ind}(u) = 1$ if u is a negative ritid page.

Proof. The computation for regular pages is the same than in the previous situation.
If u is a positive rigid page, we have :

$$\begin{aligned}\operatorname{ind}(u) &= (n-3)\chi(\dot{\Sigma}) + 2c_1^\tau(u) + \sum_{\gamma \in \Gamma_e^+} \mu_{CZ}^\tau(\gamma) + \mu_{CZ}^\tau(\gamma_h) \\ &= 2\chi(\dot{\Sigma}) + 2c_1^\tau(N_u) + \sum_{\gamma \in \Gamma_e^+} \mu_{CZ}^\tau(\gamma) + \mu_{CZ}^\tau(\gamma_h) \\ &= 2(2 - k_+ - 1) + 2k_+ + 1 \\ &= 3\end{aligned}$$

where $k_+ = \#\Gamma_e^+$ is the number of positive ends asymptotic to elliptic binding components in Γ_0, and because the Conley-Zehnder index of the only hyperbolic asymptotic orbit γ_h is 1 in the same trivialization as before.
Similarly, for a lift of a negative rigid page :

$$\begin{aligned}\operatorname{ind}(u) &= 2\chi(\dot{\Sigma}) + 2c_1^\tau(N_u) + \sum_{\gamma \in \Gamma_e^+} \mu_{CZ}^\tau(\gamma) - \mu_{CZ}^\tau(\gamma_h) \\ &= 2(2 - k_+ - 1) + 2k_+ - 1 \\ &= 1\end{aligned}$$

□

Let us now take the same setup as before.

Proposition 7.10.

We

Remark 7.11. As a final remark, it should be possible to extend these results to higher dimensions, with a suitable notion of "iterated weakly planar manifolds". However, this has not yet been fully worked out.

8 Examples

In this section we construct several explicit examples of 3-manifolds with contact structures supported by planar balanced broken books, and deduce explicit contact structures on 5-manifolds for which the Weinstein conjecture holds.

8.1 Mapping tori and circle bundles over surfaces

One main source of examples that was already studied in [**SOBD1**] is the one of circle bundles over surfaces.
The data of a circle bundle over a compact surface with non-empty boudary corresponds exactly to an open book decomposition.

8.2 Branched coverings

Take a broken book decomposition, and a transverse knot embedded inside a regular page. Then the branched double cover of the manifold with branching locus this knot has a natural induced broken book decomposition structure. Moreover, taking the knot transverse to the characteristic foliation ensures that the contact structure lifts to the cover, and is supported by this new broken book decomposition. This gives a whole new class of examples which may be not planar in the usual sense, or in the sense of spinal open books.

Let us give a little bit more details on this construction.

We first recall the standard construction of a contact structure on the double branched covering of $\mathbb{S}^3$ along a transverse knot.

Lemma 8.1. Let $K \subset \mathbb{S}^3$ be a transverse knot, and let $p : M \to \mathbb{S}^3$ be a branched double cover over K, with branching locus the knot K_1. Then there exists a contact structure on M which coincides with $p^*\xi$ on $M \backslash K_1$.

Proof. □

In order to show that this contact structure is actually supported by the broken book decomposition, we will need another expression of a defining contact form.
Consider a planar open book decomposition on $\mathbb{S}^3$ and α a Giroux form for this open book, and take an embedded transverse loop inside a page. Then a neighbourhood of this loop can be described using coordinates (t, τ, θ), where the contact form α can be written under the form :

$$\alpha = dt + e^{\tau} d\theta$$

Using polar coordinates (r, ϕ) for the plane (t, τ), we can also write α under the form :

$$\alpha = \cos(\phi)dr - r\sin(\phi)d\phi + e^{r\sin(\phi)}d\theta$$

Now locally the pages can be arranged so that they are the level sets of t, with the page where the loop is being $\{t = 0\}$. Take the branched double cover M over $\mathbb{S}^3$, where the branched covering p can locally be described around the branching locus K_1 as : $p(r, \phi, \theta) = (r^2, 2\phi, \theta)$. This amounts to : $p(t, \tau, \theta) = (t^2 - \tau^2, 2t\tau, \theta)$. Hence level sets for t become level sets for $t^2 - \tau^2$, which are hyperbola branches, except for $t = \pm\tau$ which is to the lift of the initial page, and now corresponds to the four rigid pages asymptotic to K_1. Hence we get a broken book as expected. It remains to show that the contact structure lifted on M is supported by this broken book.

Lemma 8.2. There exists a contact form $\tilde{\alpha}$ on M such that $\tilde{\alpha}$ coincides with $p^*\alpha$ away from a neighbourhood of K_1, it defines the same contact structure as before, and its Reeb vector field is supported by the broken book decomposition.

Proof. First we compute the pullback of the form α away from K_1 in coordinates (t, τ) :

$$\alpha_0 := p^*\alpha = 2tdt - 2\tau d\tau + e^{2t\tau}d\theta$$

where we used coordinates (t, τ, θ) on M near K_1. We have : $d\alpha_0 = e^{2t\tau}(2td\tau \wedge d\theta + 2\tau dt \wedge d\theta)$. This converges to 0 as r goes to 0, hence it does not define a contact form.
Now, on $M \backslash K_1$, α_0 is supported by the broken book decomposition by definition. In particular, in the neighbourhood of K_1 that we consider, we have : $R_{\alpha_0} = \frac{1}{2r^2}(t\partial_t - \tau\partial_\tau)$ which corresponds to a "standard hyperbolic model" for a vector field.
Take $\delta < \delta' < 1$, $\beta := f(r)d\phi$ on $\mathbb{D}^2$, where f is a map such that $f = r^2$ for $r < \delta'$, $f = 0$ near 1, and f is increasing for $r < \delta < 1$.
Now, define :

$$\tilde{\alpha} := \begin{cases} p^*\alpha & \text{for } r \geq 1 \\ \alpha_0 + \epsilon\beta & \text{for } r \leq 1 \end{cases}$$

where ϵ is chosen small enough.
First notice that $\tilde{\alpha}$ is well defined, by definition of β. Then, up to choosing ϵ small enough, we can ask for $\tilde{\alpha}$ to be a contact form for $\delta \leq r$. Moreover, we have :

$$\tilde{\alpha} \wedge d\tilde{\alpha} = \alpha_0 \wedge d\alpha_0 + \epsilon(\alpha_0 \wedge d\beta + \beta \wedge d\alpha_0)$$

We know that away from $r = 0$, we have $\alpha_0 \wedge d\alpha_0 > 0$. We also have, for $r < \delta$:

$$\alpha_0 \wedge d\beta = e^{2t\tau} f'(r) d\theta \wedge dr \wedge d\phi > 0$$

Finally, $\beta \wedge d\alpha_0$ happens to be negative sometimes, but we know that $d\alpha_0$ converges to 0 when $r \to 0$, and $\beta = r^2 d\phi$ on this domain, so it also converges to 0 at speed "$2r$", hence up to choosing δ small enough, one can arrange that $\alpha_0 \wedge d\beta + \beta \wedge d\alpha_0$ is positive for $r < \delta$ (in fact this can also be seen easily by writing down the explicit formula for this). Note that here we choose first δ adapted to β and α_0, and then fix ϵ small enough.

This proves that $\tilde{\alpha}$ is a contact form on M. To show that the associated Reeb vector field is supported by the broken book decomposition, it suffices to notice that $R_{\tilde{\alpha}} = \partial_\theta$ on $\{r = 0\}$. Finally, the Reeb vector field for $r < \delta$ can be written as : $g(t, \tau, \theta)(t\partial_t - \tau\partial_\tau) + h(t, \tau, \theta)\partial_\theta$ where g and h are non-zero functions, which prove that it is always transverse to the pages, hence positively transverse.

We now prove that the induced contact structure is the same than the one we built in the first proposition. In order to do this, consider α_0 on $M \backslash K_1$. The first construction shows that even though α_0 is not a contact form on K_1, $\mathrm{Ker}(\alpha_0)$ is a contact structure on all of M. Hence, take $\alpha_t = \alpha_0 + t\beta$ for $t \in [0, \epsilon]$ and (ξ_t) is then a homotopy of contact structures, which gives an isotopy by Gray's theorem.

□

This gives a whole class of contact structures on 3-manifolds supported by broken book decompositions which are not spinal open book decompositions in general. Moreover, these broken books are planar and balanced in the sense mentioned in the last parts, hence are well suited to study symplectic fillings. We can classify all manifolds arising from such a construction with the help of the following lemma.

Proposition 8.3. The contact 3-manifold one can obtain with this construction are exactly : - - - -

Proof. □

Remark 8.4. In certain cases we can actually understand the structure of SNLF induced on such fillings. For instance, take...

Corollary 8.4.1. *Take W to be one of the compact symplectic manifolds with boundary cited in the proposition above, and ψ a symplectomorphism equal to the identity near the boundary. Let M be the contact manifold created by the procedure described in the last part, namely one take the mapping torus W_ψ, and "fills" it with $\Gamma \times \mathbb{D}^2$ where $\Gamma \simeq \partial W$, with the natural contact structure coming from this* *procedure.*

Then M verifies the Weinstein conjecture.

Corollary 8.4.2. *The Weinstein conjecture holds for the following class of contact* 5*-manifolds : 5 dimensional tori ? would be cool. You would need to show that the contact structure is supported by a cool open book with balanced planar boundary.*

9 Conclusion